Cell–Cell Communication Analysis (CellChat v2) of Intestinal IEL and LPL Compartments in Crohn's Disease

Bioinformatics Analysis Report

Davide Maccarrone

2026-08-11

Contents

1 Executive Summary **2**

2 Analytical Rationale **3**

2.1 Why cell–cell communication inference as a complementary method . . . 3

2.2 Pipeline architecture . . . 3

2.3 Scope . . . 4

3 Methodological Framework: Cell Type Comparison Validity **4**

3.1 IEL compartment . . . 4

3.2 LPL compartment . . . 5

4 Global Network Architecture **6**

5 IEL Compartment: Pathway-Level Findings **11**

5.1 MHC-I . . . 11

5.2 MHC-II — *receiver population remodeling rather than simple amplification* . . . 14

5.2.1 Bubble plot . . . 17

5.2.2 Hierarchy plots . . . 17

5.3 LIGHT (TNFSF14) — *within-CD characterization* . . . 18

5.4 MIF — *mixed, with CD-associated network remodeling* . . . 20

5.5 GALECTIN — *CD-emergent, epithelial-centered communication pattern* . . . 22

5.6 CDH1 (E-cadherin) — *epithelial-associated CDH1 communication pattern* . . . 24

5.7 Biological interpretation . . . 27

5.8 Signaling-role scatter plots and communication patterns . . . 27

6 LPL Compartment: Pathway-Level Findings **42**
6.1 MHC-I . . . 43
6.2 MHC-II — *population substitution rather than simple intensification* . . . 46
6.3 SIRP (SIRPG–CD47) — *network reorganization* . . . 49
6.4 GALECTIN — *CD-associated, within-CD characterized* . . . 53
6.5 IL2 — *CD-associated, receptor-context dependent* . . . 54
6.6 LIGHT — *fragile, population-limited* . . . 55
6.7 Communication patterns and cell-type signaling changes . . . 57

7 Cross-Validation with GSEA **64**

8 Known Technical Issues (CellChat v2) — Methods Note **64**

9 Comparison with External Literature **65**

10 Cross-Compartment Synthesis **65**

11 Limitations **66**

12 Reproducibility **66**
12.1 Environment . . . 66
12.2 Reference scripts . . . 66
12.3 Reproducibility settings . . . 66

13 Deliverables **66**
13.1 Recommended manuscript figures . . . 67

14 Conclusion **67**

15 Appendix A: Glossary of Abbreviations **67**

16 Appendix B: References **68**

17 Appendix C: Figure Index **68**

1 Executive Summary

This report presents the cell–cell communication analysis of intraepithelial (IEL) and lamina propria (LPL) lymphocyte compartments in Crohn's disease (CD) versus control intestinal tissue,using CellChat v2 (Jin et al., 2024) as an orthogonal, network-level complement to the pseudobulk differential expression and GSEA analyses reported separately. Where GSEA characterizes *which programs* are active within a cell type, CellChat characterizes *who is signaling to whom*, and through which ligand–receptor pairs — a distinct and complementary axis of biological information.

Two independent CellChat objects were built per compartment (Control and CD), never merged prior to inference, and subsequently aligned (`liftCellChat()`) for differential comparison. Across both compartments, the analysis converges on a small number of well-supported, literature-concordant findings:

- **A denser, structurally more complex communication network in CD** in both compartments, consistent with active mucosal inflammation, though the mechanism differs: in IEL the CD network is *diffuse* (more pathways, lower average edge strength — partly a function of more cell populations clearing the detection threshold in CD), whereas in LPL the CD network is *proportionally more intense* (more interactions and more total strength, ~1.6–1.7× Control).
- **MHC-II signaling shows a disease-associated reorganization involving regulatory T-cell populations in both compartments.**, but through population-specific rather than generic mechanisms: in IEL, CD-emergent Treg/macrophage/TH17-like populations dominate the CD-side receiver profile; in LPL, the receiving Treg subset is *replaced* between conditions (Activated effector Treg in Control → FOXP3 activated Treg in CD), not merely intensified.
- **LIGHT (TNFSF14) signaling toward FOXP3 IL2RA Treg via LT R, with a broader TNFRSF14 (HVEM) axis toward multiple populations (IEL)**, and **SIRPG–CD47 T–T costimulatory signaling (LPL)** are identified as compartment-specific, biologically coherent, literature-supported axes.
- **CD103 (ITGAE/ITGB7)–E-cadherin** emerges as the most specific ligand–receptor pair recovered in the IEL compartment, mechanistically consistent with the established role of this axis in intraepithelial lymphocyte retention.
- The single most important methodological caveat governing the entire analysis is **severe, condition-dependent cell type imbalance**, formalized here as an explicit validity classification per compartment. All findings should be interpreted through this framework.

This report consolidates and reframes findings already produced and internally reviewed in the IEL and LPL CellChat pipelines (scripts `13-16` and `17-20` respectively).

2 Analytical Rationale

2.1 Why cell–cell communication inference as a complementary method

Pseudobulk DESeq2 and GSEA describe transcriptional programs *within* a cell type. They cannot, by construction, describe the *directionality* of intercellular signaling — which population is the likely source of a ligand, which population expresses the cognate receptor, and how that sender–receiver structure reorganizes with disease. CellChat v2 infers this structure from ligand–receptor co-expression using a curated database of literature-supported interactions and provides both quantitative summaries (aggregated network strength/count, information flow per pathway) and cell-type-resolved output.

2.2 Pipeline architecture

- Two fully independent CellChat objects per compartment (`createCellChat` → `subsetData` → `identifyOverExpressedGenes/Interactions` → `computeCommunProb` (trimean method, package default) → `filterCommunication(min.cells = 10)` → `computeCommunProbPathway` → `aggregateNet`), never merged prior to inference.
- `liftCellChat()` was used to align cell type levels across conditions for comparison; `netAnalysis_computeCentrality()` was re-run after lifting because lifted matrices differ in dimension from the originals.
- Reproducibility: `netEmbedding(..., umap.method = "uwot")` with `set_seed(1234)` immediately before each stochastic block (functional similarity embedding, `selectK`, `identifyCommunicationPatterns`).

- IFN-II, IL17, and TGFb signaling are absent from CellChat results across both compartments and both conditions. TNF signaling is absent in IEL (both conditions) but is detected in LPL (both Control and CD). This partial absence should not be interpreted as biological absence of these cytokine programs — it most plausibly reflects the detection threshold of the trimean-based default method, combined with compartment-specific differences in which cell populations express the relevant ligand-receptor pairs at sufficient frequency to pass filtering.

2.3 Scope

Unlike the GSEA analysis, CellChat operates directly on single-cell expression and therefore includes every annotated cell type in each compartment, subject to the `min.cells = 10` per-condition filter applied independently to each condition's network.

Interpretation rule

Cell types that fail the `min.cells = 10` threshold in one condition cannot support a standard quantitative CD-vs-Control comparison for that population. The resulting outputs must be labeled as within-condition characterization, imbalanced comparison, or fully comparable comparison.

3 Methodological Framework: Cell Type Comparison Validity

`filterCommunication(min.cells = 10)` excludes any cell type with fewer than 10 cells in a given condition from that condition's network; `liftCellChat()` subsequently re-inserts it as an all-zero row/column when aligning objects for comparison. Both compartments show substantial, condition-dependent imbalance in cell type composition. This imbalance is the single most important lens through which every result in this report must be read.

Tier boundaries are set on the absolute cell count on the weaker side of each population (not on the CD:Control ratio): fewer than 10 cells on the weaker side aligns with the `filterCommunication(min.cells = 10)` threshold and places a population in the emergent/dominant tier (within-condition characterization only); 10–~30 cells on the weaker side is treated as imbalanced (comparison possible but indicative); above ~30 cells on both sides is treated as fully comparable.

3.1 IEL compartment

Tier	Cell types (n CD / n Control)	Interpretation
Fully comparable	GZMK effector memory CD8 T cells (183/686); Effector T cell mixed state (213/356); CXCL13 TFH-like T cells (75/1394); Cycling T cells (207/35)	Fully interpretable CD-vs-Control differential comparison
Imbalanced	CD39 tissue-resident CD8 T cells (2551/20); Cytotoxic T cells (2113/11); NK-like cytotoxic T cells (249/10)	Comparison possible but Control side is statistically weak; treat as indicative

Tier	Cell types (n CD / n Control)	Interpretation
CD-emergent	FOXP3 IL2RA Treg (79/0); CST3 LYZ macrophages (35/1); TH17-like CD4 T cells (1619/2); TH17-like innate-associated CD4 T cells (832/6); Terminal effector CD8 T cells (529/0); Naïve CD4 (678/8); Epithelial cells/enterocytes (20/6)	**Within-CD characterization only**
Control-dominant	/NK-like cytotoxic T cells (1/2287); Naive CD4 T cells (4/1935); TH17 effector (4/1508); IL7R memory-like CD8 T cells (0/1481); Naïve/central memory CD4 (5/1238); Activated memory CD4 (4/983)	**Within-Control characterization only**

3.2 LPL compartment

Tier	Cell types (n CD / n Control)	Interpretation
Fully comparable	Activated cytotoxic CD8 T cells (5650/2002); Naive CD4 T cells (2284/3756); Activated effector Treg (873/617); CD4 TRM-like activated memory T cells (753/249)	Fully interpretable CD-vs-Control differential comparison
Imbalanced (moderate)	/ innate cytotoxic T cells (486/57); Cycling T cells (230/23)	Comparison possible; one side weaker — indicative estimates
CD-heavy	Th17-like activated CD4 T cells (4118/53); Innate-like/TRM CD8 (2194/16); IFN-activated CD8 cytotoxic T cells (1474/47); Activated CD4 memory T cells (5578/1)	Characterization predominantly in CD
Control-heavy	Activated CD4 T cells stress response (15/2628); T follicular helper (24/2000)	Characterization predominantly in Control
CD-exclusive	FOXP3 activated Treg (2205/0); NK-like cytotoxic T cells (296/0); TRM-like CD8 cytotoxic T cells (204/0)	**Within-CD characterization only**
Excluded	Epithelial/stromal contamination (36/0)	Declared contamination; excluded from pathway deep-dives and hierarchy plots

Structural note

Unlike IEL, the LPL compartment contains **no myeloid or epithelial populations** — it is entirely T-cell-composed. This is a structural property of the compartment and explains why

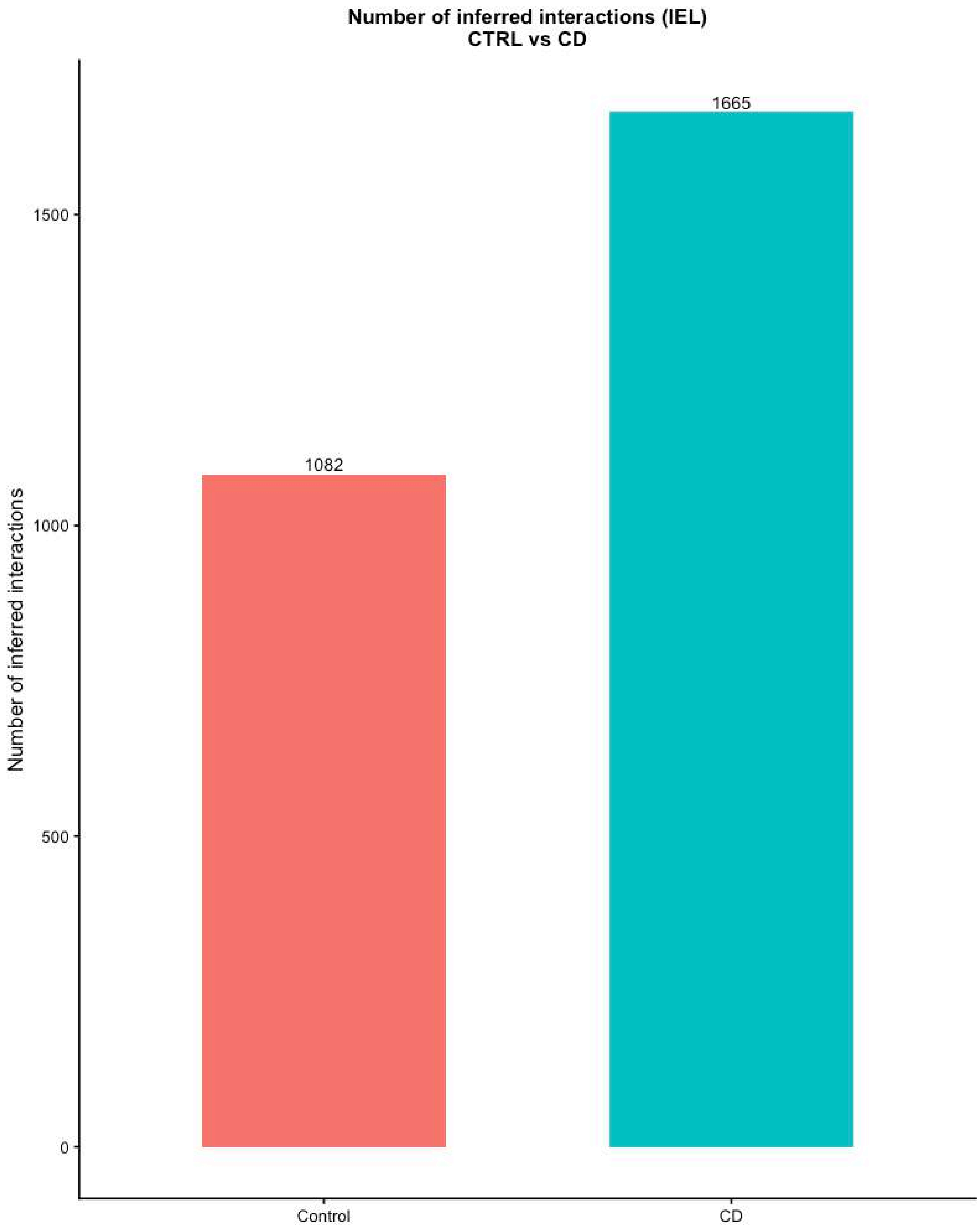
LPL lacks direct analogues to the IEL macrophage- or epithelial-centered pathways MIF and CDH1.

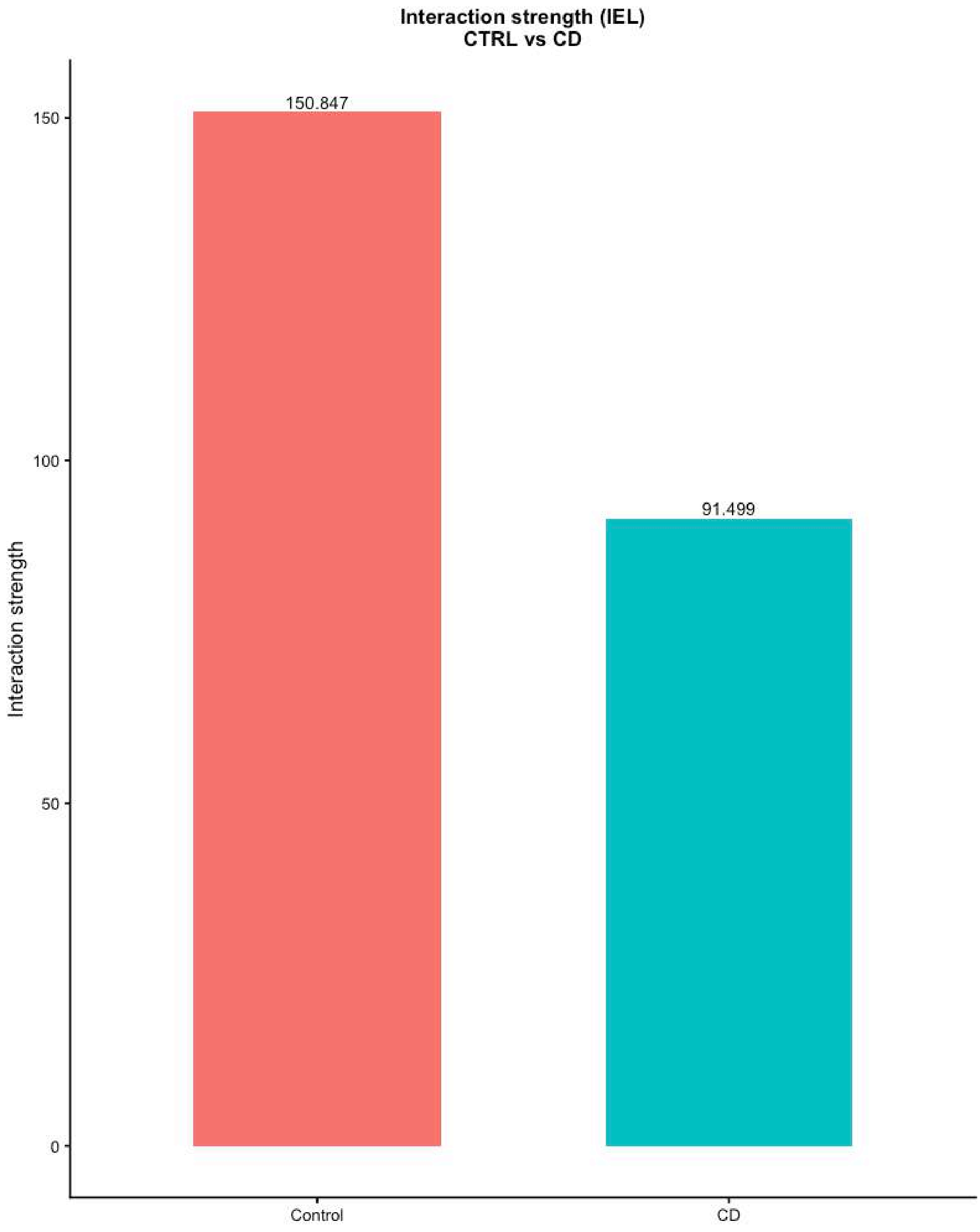
4 Global Network Architecture

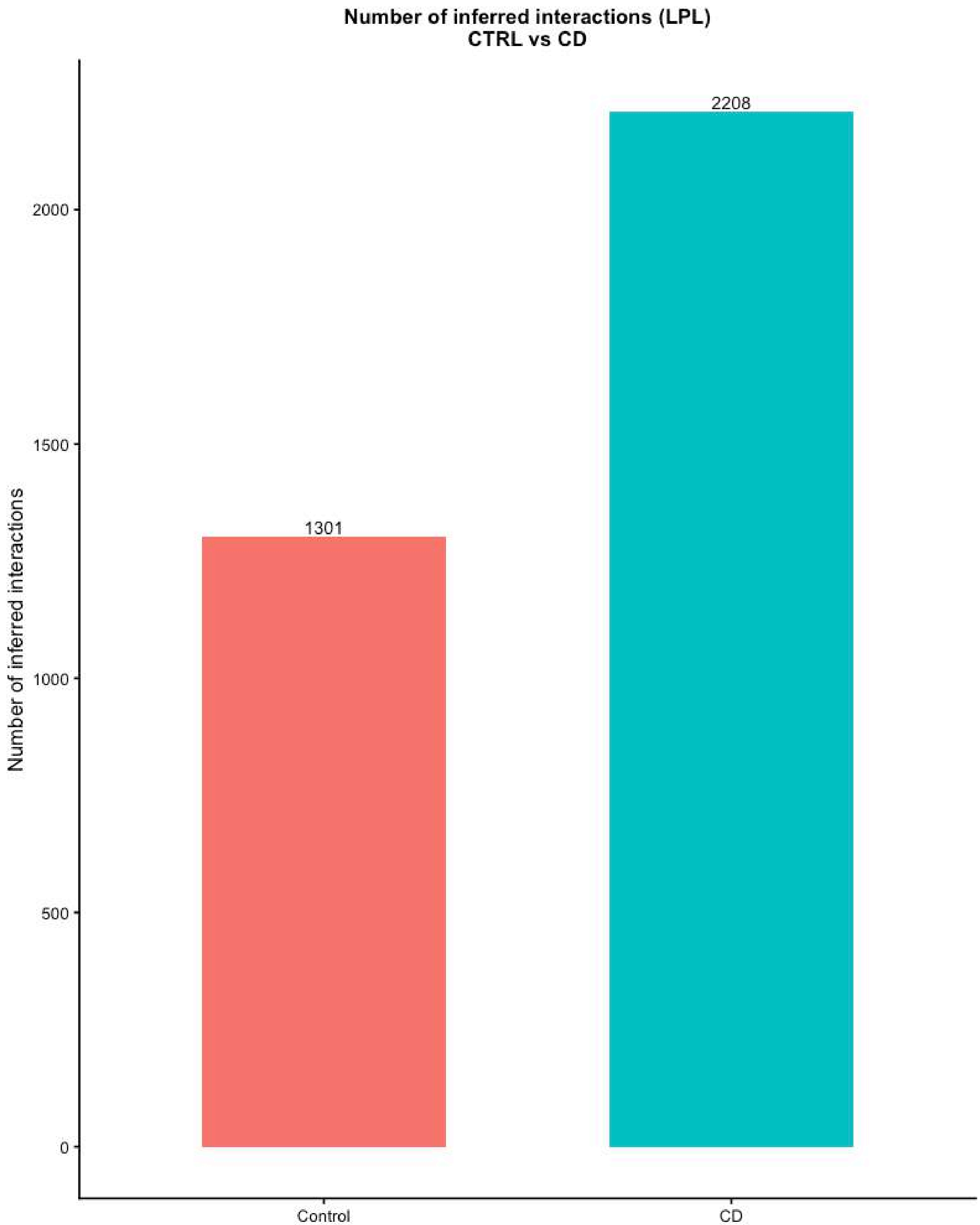
Metric	IEL (CD vs Control)	LPL (CD vs Control)
Interaction count	~3× more pathways detected in CD	2208 vs 1301 (~1.7×)
Interaction strength	Lower average strength per edge in CD (diffuse network)	161.0 vs 101.4 (~1.6×)
Communication-pattern complexity	Shared K across conditions	Outgoing K=4 Control vs K=8 CD; Incoming K=7 vs K=7
Overall pattern	More diffuse, less concentrated network in CD	Proportionally more intense network in CD

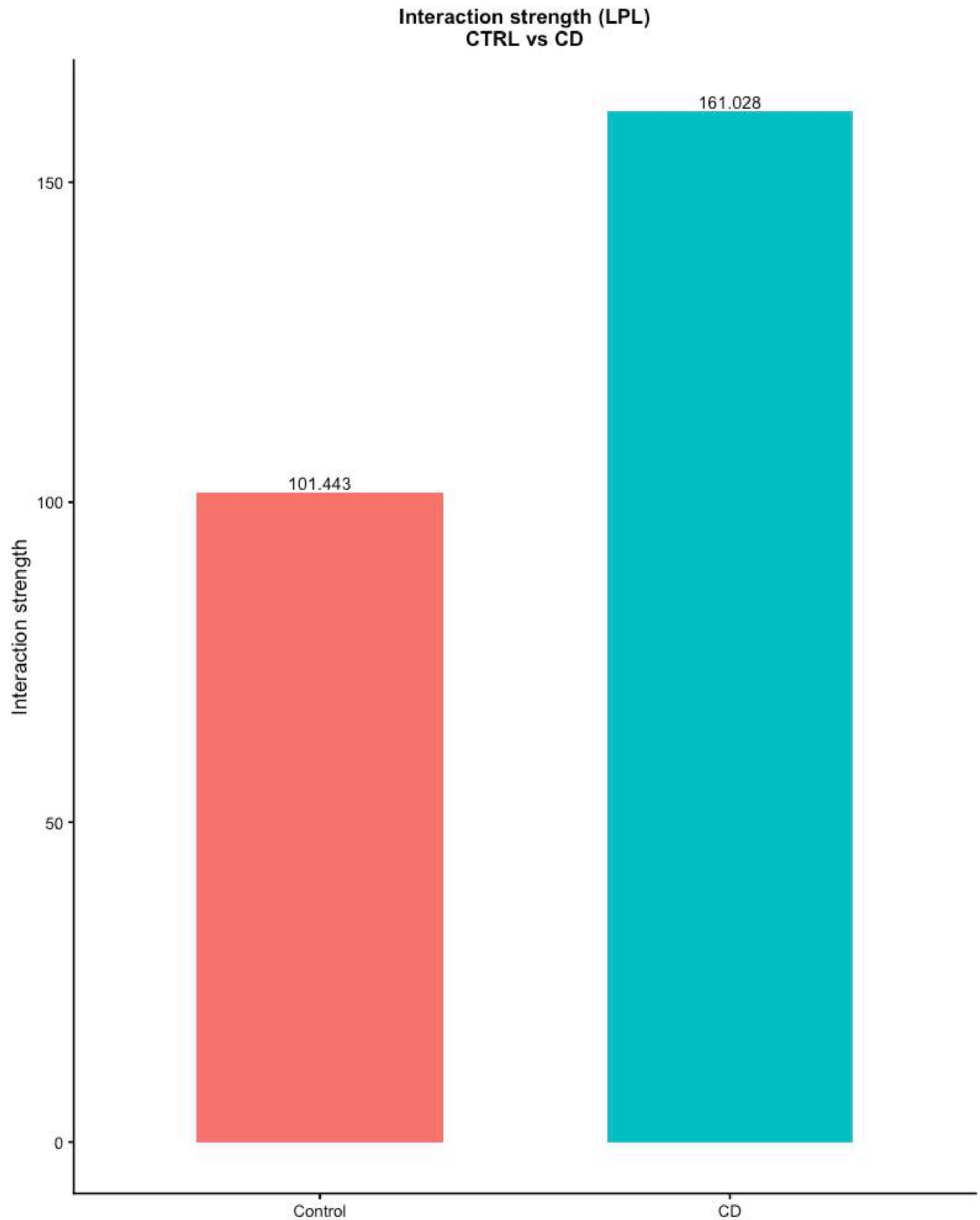
`compareInteractions()`, `netVisual_diffInteraction()`, and `netVisual_heatmap()` operate on the full aggregated network and are therefore valid at the network level without depending on the statistical power of a single cell type. Functional similarity embedding and `rankSimilarity` likewise describe pathway-level reorganization.

Figure 1. Global interaction count and strength, Control versus CD, for both compartments.


Number of inferred interactions (IEL)
CTRL vs CD
Number of inferred interactions
1665
1082
1500
1000
500
0
Control
CD


Interaction strength (IEL)
CTRL vs CD
150.847
91.499
Interaction strength
150
100
50
0
Control
CD


Number of inferred interactions (LPL)
CTRL vs CD
2208
1301
Number of inferred interactions
2000
1500
1000
500
0
Control
CD



A compartment-level asymmetry is worth noting explicitly: the larger apparent pathway count in CD IEL reflects a combination of genuine transcriptional/functional heterogeneity and the near-mechanical consequence of more distinct cell populations clearing the `min.cells = 10` threshold in CD.

5 IEL Compartment: Pathway-Level Findings

Six pathways were selected for deep-dive interpretation: **MHC-I, MHC-II, LIGHT, MIF, GALECTIN, CDH1**.

The `celltypes_of_interest` set comprised seven populations: FOXP3 IL2RA Treg, CST3 LYZ macrophages, TH17-like CD4 T cells, CD39 tissue-resident CD8 T cells, Epithelial cells / enterocytes, GZMK effector memory CD8 T cells, and Effector T cell mixed state.

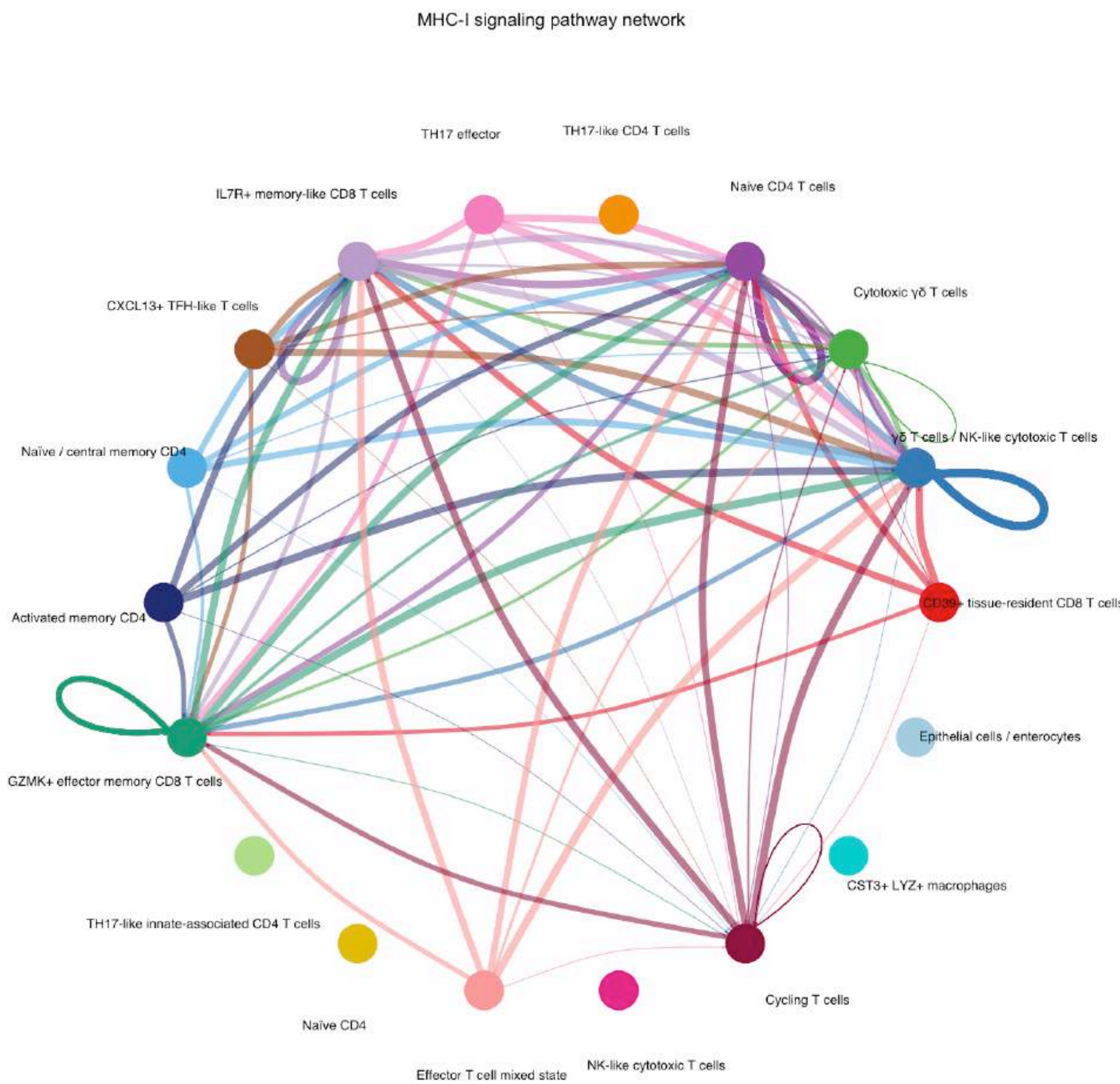
By cell-count validity tier (Control vs. CD; see Section 3.1): FOXP3 IL2RA Treg (0/79), TH17-like CD4 T cells (2/1619), and CST3 LYZ macrophages (1/35) are condition-emergent and are interpreted within-condition only. CD39 tissue-resident CD8 T cells (20/2551) is severely imbalanced toward CD and is not treated as a genuine shared population despite being nominally present in both conditions. Epithelial cells / enterocytes (6/20) have low absolute counts in both conditions, limiting statistical power for either within- or cross-condition comparison. Only GZMK effector memory CD8 T cells (686/183) and Effector T cell mixed state (356/213) have cell numbers in both conditions sufficient to support a genuine quantitative cross-condition comparison.

5.1 MHC-I

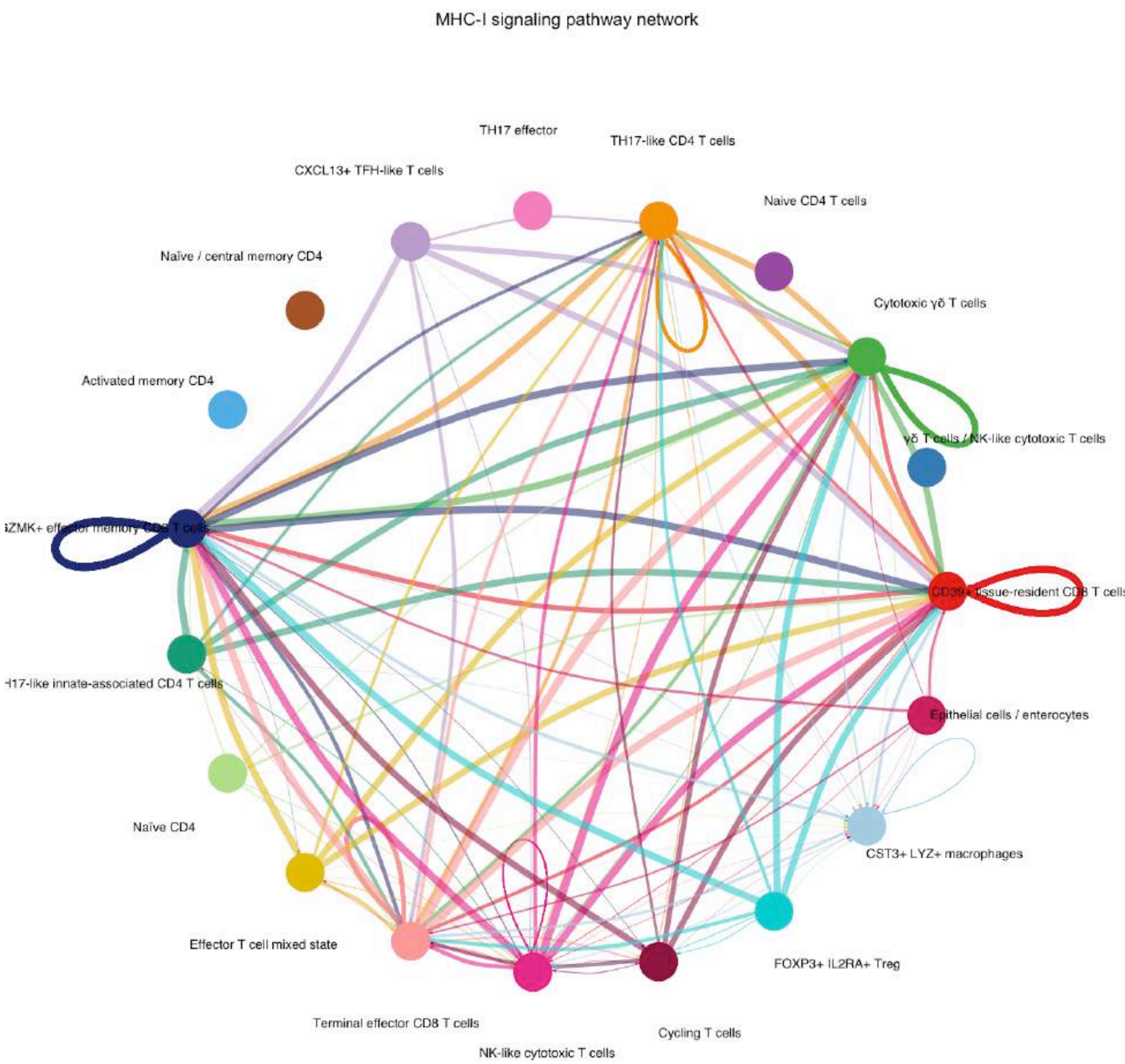
Ubiquitous signaling pattern in both conditions, as expected for a broadly expressed pathway. Beyond simple homogenization, the receiver/sender structure itself shifts between conditions: CST3 LYZ macrophages and Epithelial cells/enterocytes were not detected as active participants in the inferred MHC-I network in Control (zero interactions as either source or target) and emerge as active senders only in CD. Conversely, CD39 tissue-resident CD8 T cells displays a strong increase in inferred MHC-I incoming signaling in CD, consistent with its expansion and increased representation within the CD network. This pattern is compatible with inflammatory IFN-driven remodeling described in intestinal inflammation, and should be read as a structural reorganization rather than a simple intensification of a shared axis. The hierarchy plots below use CD39 tissue-resident CD8 T cells and Cytotoxic T cells as receivers: Cytotoxic T cells is the one target consistently populated in both conditions, while CD39 TRM CD8 becomes substantial only in CD (near-zero in Control).

Figure 6. MHC-I circle plots (Control and CD) and hierarchy plots (Control and CD).

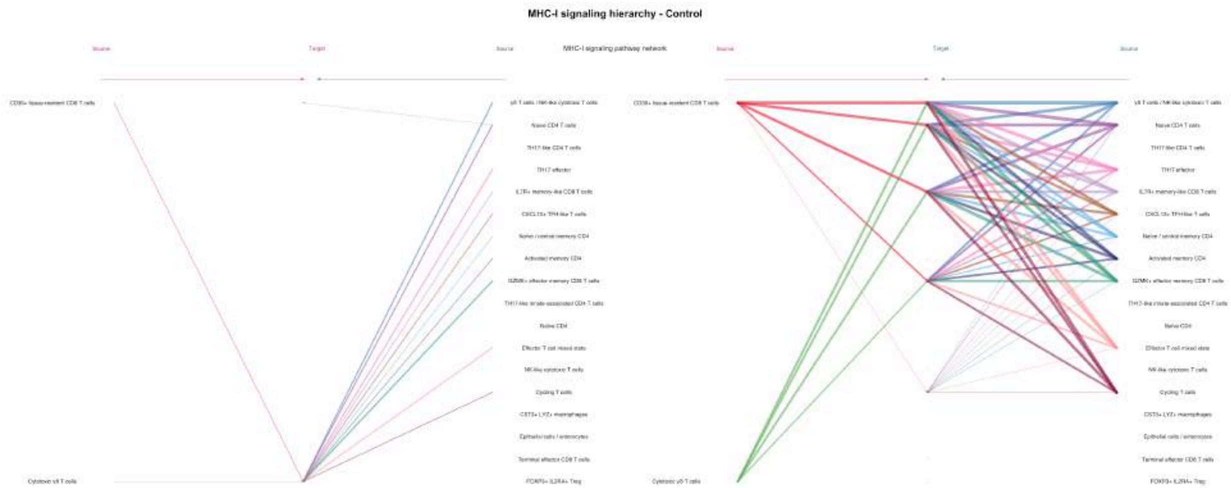
5.1.0.1 Circle plots Control



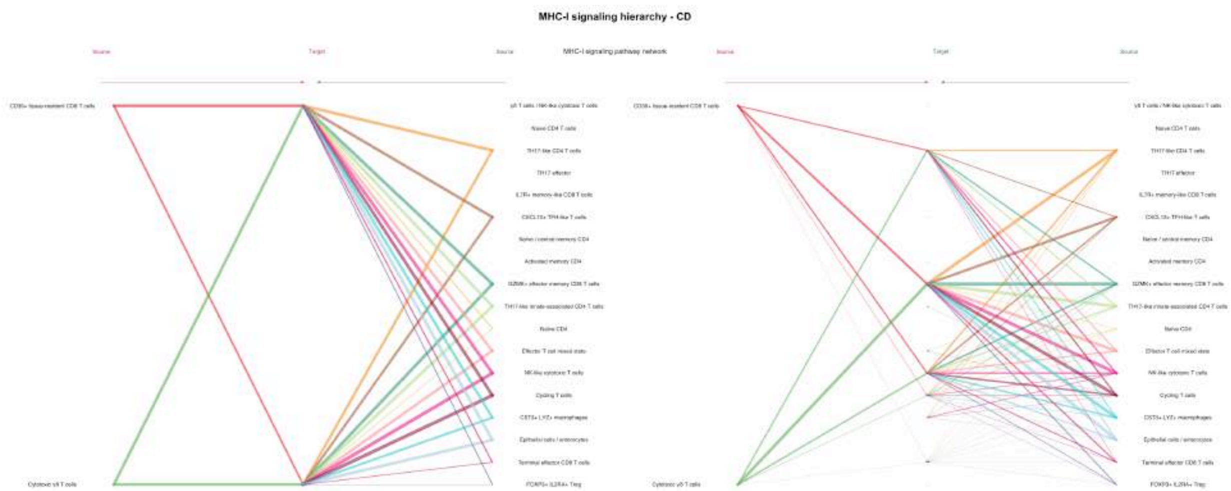
Crohn's disease (CD)



5.1.0.2 Hierarchy plots Control



Crohn's disease (CD)



5.2 MHC-II — *receiver population remodeling rather than simple amplification*

The inferred MHC-II communication network shows distinct receiver profiles between Control and CD.

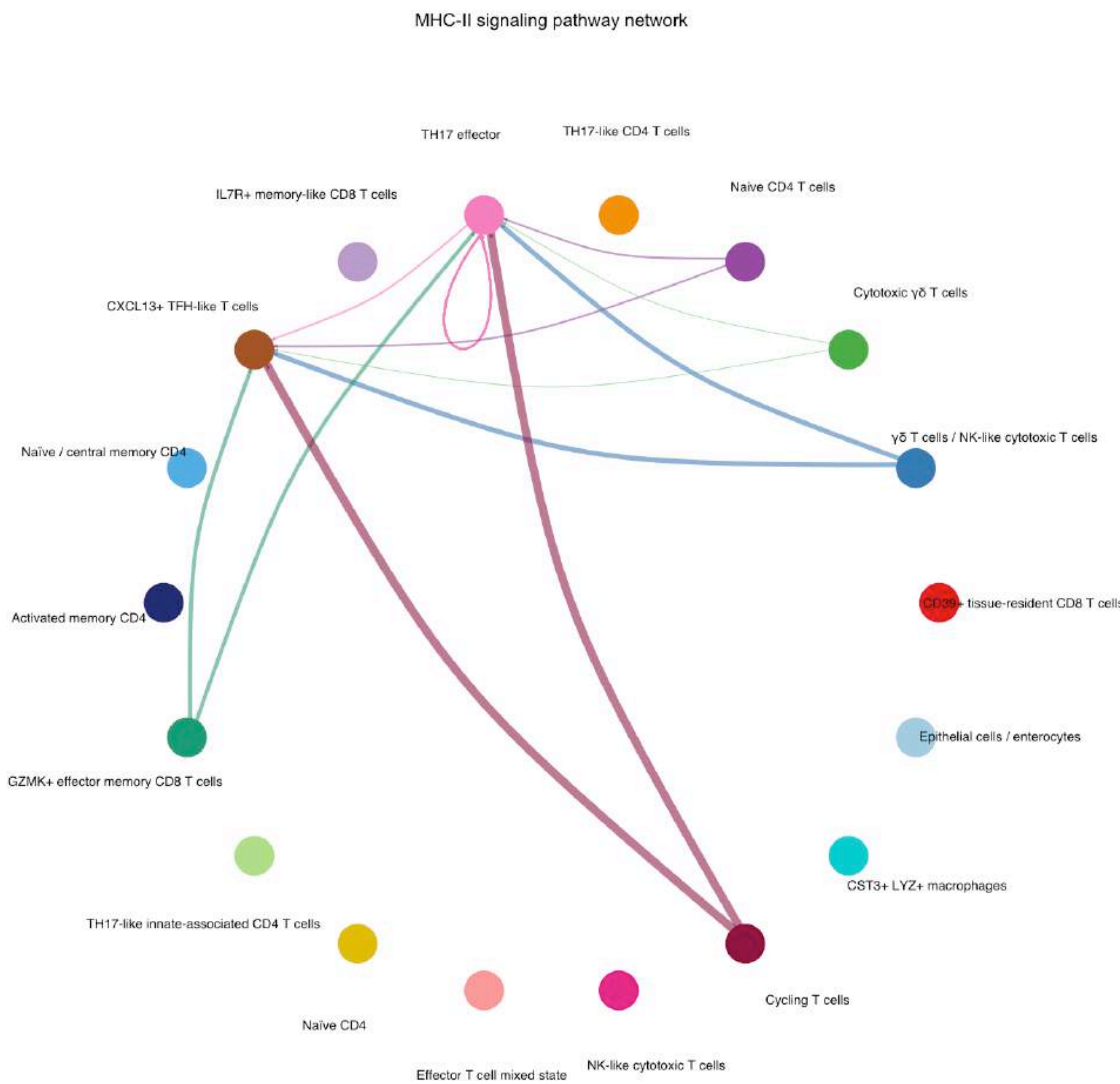
Control: MHC-II signaling is predominantly associated with TH17 effector cells, a strongly Control-enriched population (1508 Control / 4 CD).

CD: the inferred MHC-II network involves FOXP3 IL2RA Treg, TH17-like innate-associated CD4 T cells, and CST3 LYZ macrophages, reflecting the emergence of CD-associated receiver populations. Because several of these populations are strongly condition-skewed or absent in one condition, this result should not be interpreted as quantitative amplification of a conserved Treg-centered circuit. Instead, the data support a remodeling of the cellular context in which MHC-II communication is detected during CD.

Literature context. MHC-II presentation toward regulatory and TH17-like CD4 T cell populations is consistent with CD40-axis biology described in intestinal inflammation, where CD103 dendritic cells and macrophage-lineage antigen-presenting cells shape the local Treg/TH17 balance (Barthels et al. 2017; Wang et al. 2026). The population substitution observed here — a different receiving Treg-associated population in CD versus Control, rather than intensification of one shared population — should be read as a shift in which cell type occupies this signaling role under inflammation, not as direct evidence of the CD40 mechanism itself, which CellChat's ligand-receptor database does not model explicitly for this pathway.

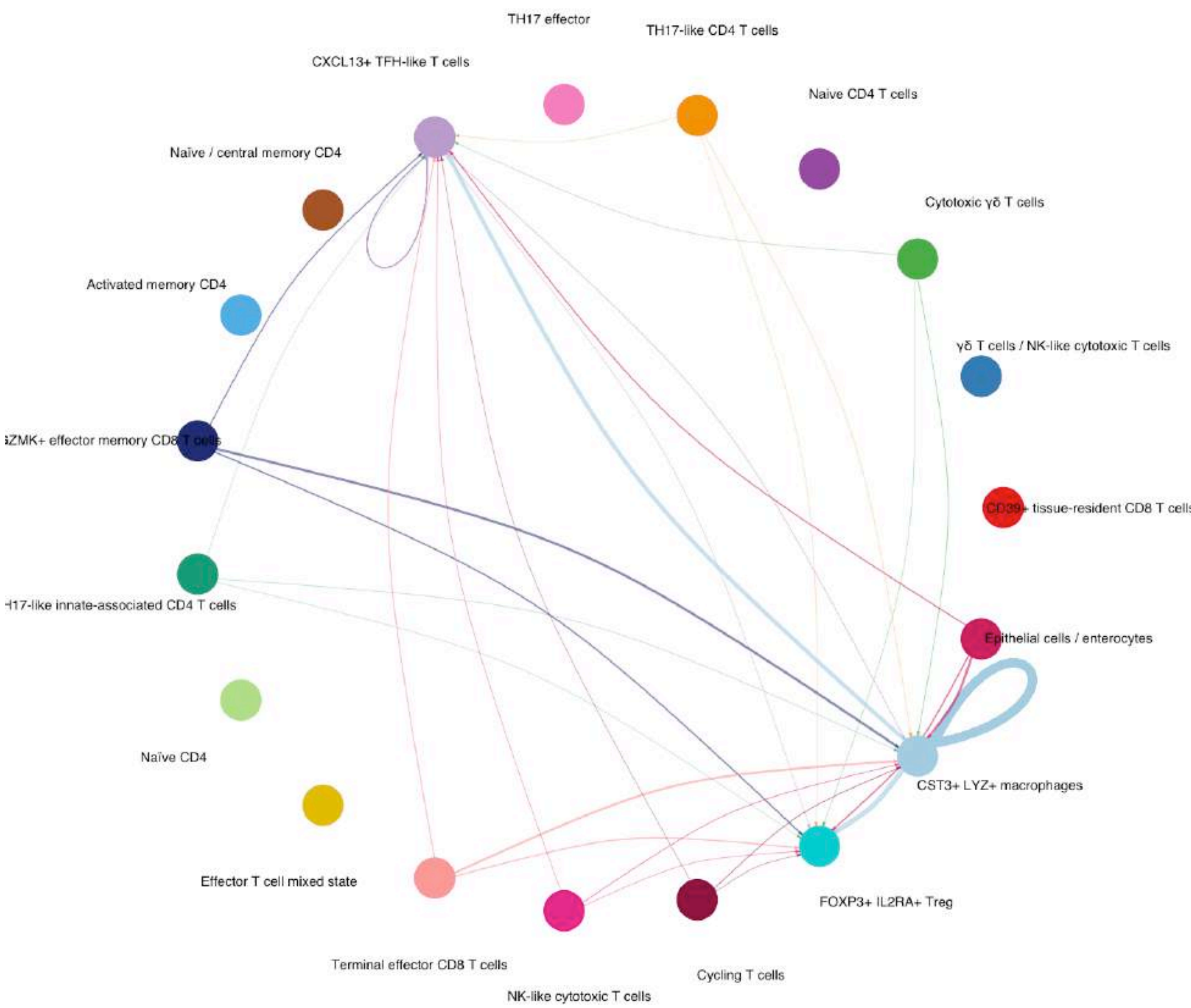
Figure 7. MHC-II circle, bubble, and hierarchy plots.

5.2.0.1 Circle plots Control

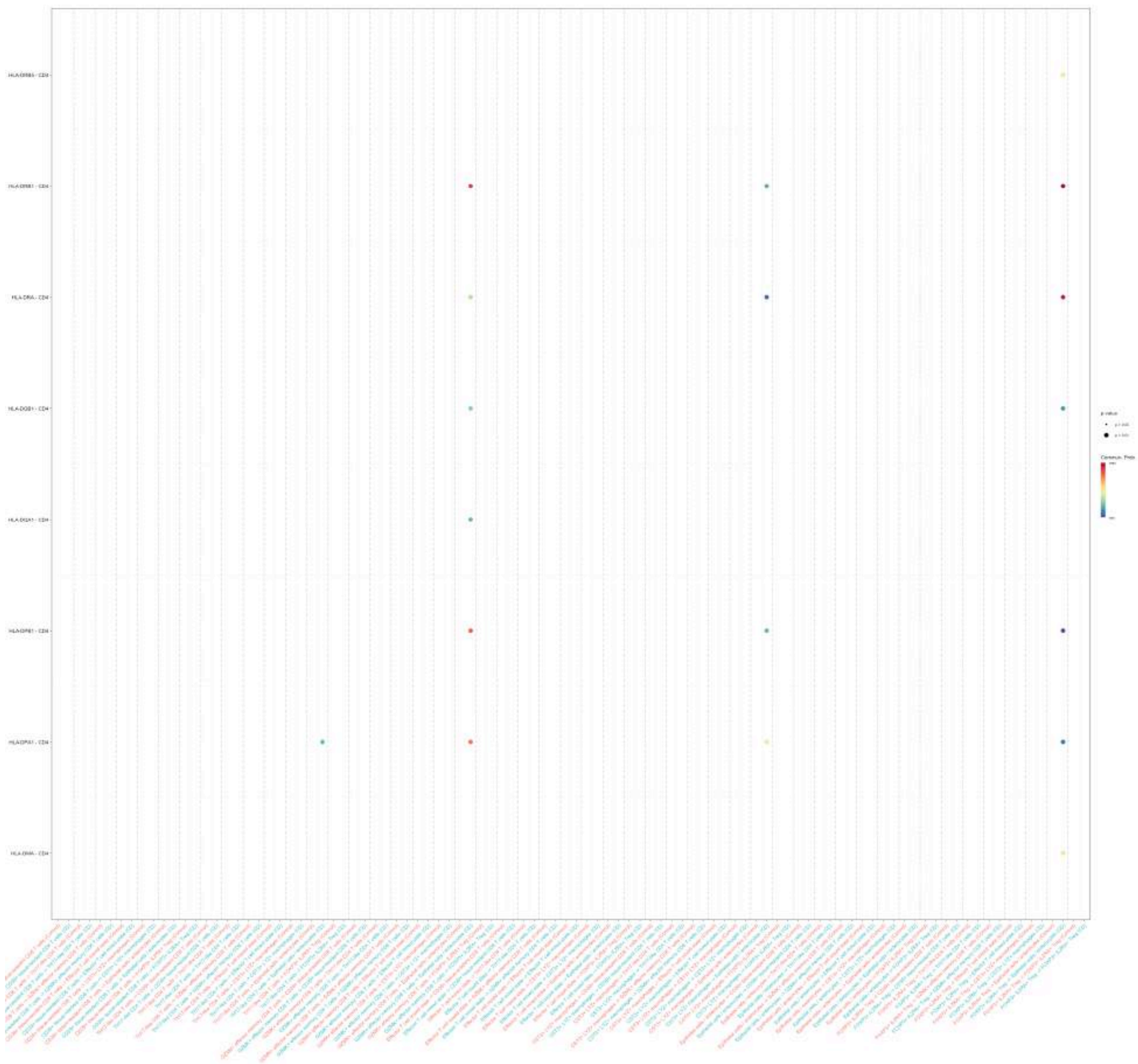


Crohn's disease (CD)

MHC-II signaling pathway network

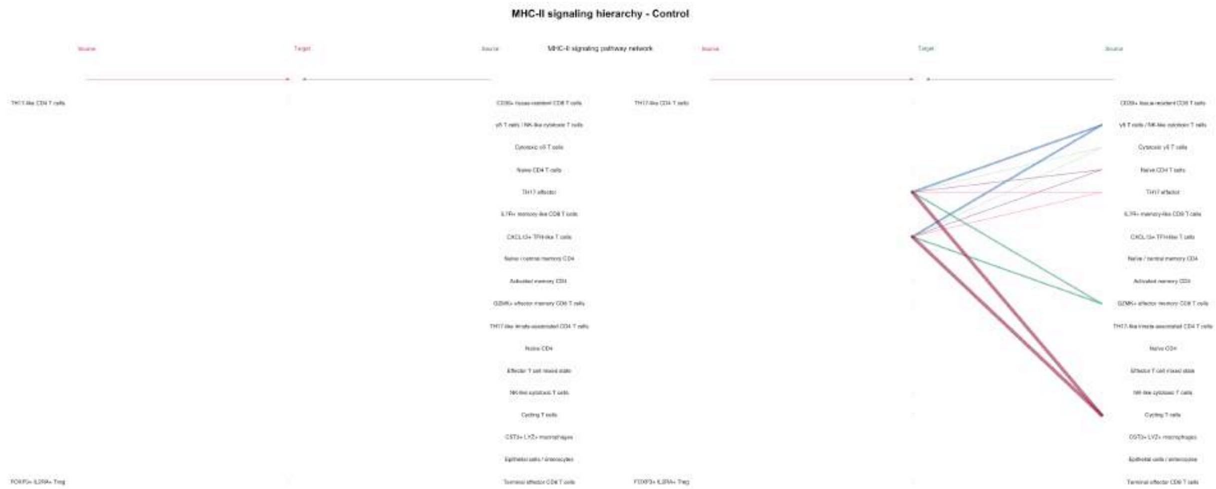


5.2.1 Bubble plot



5.2.2 Hierarchy plots

Control



Crohn's disease (CD)

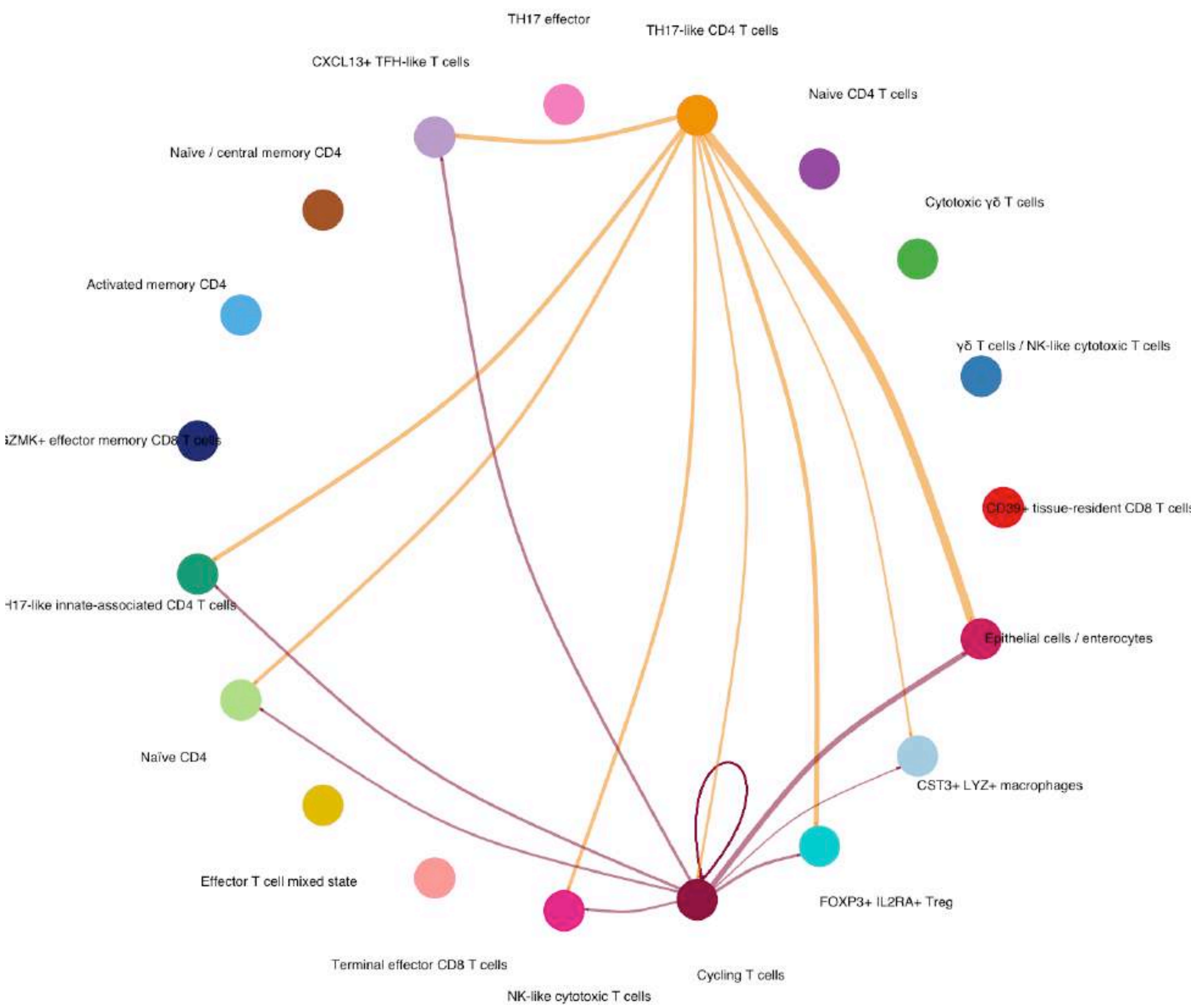
5.3 LIGHT (TNFSF14) — *within-CD characterization*

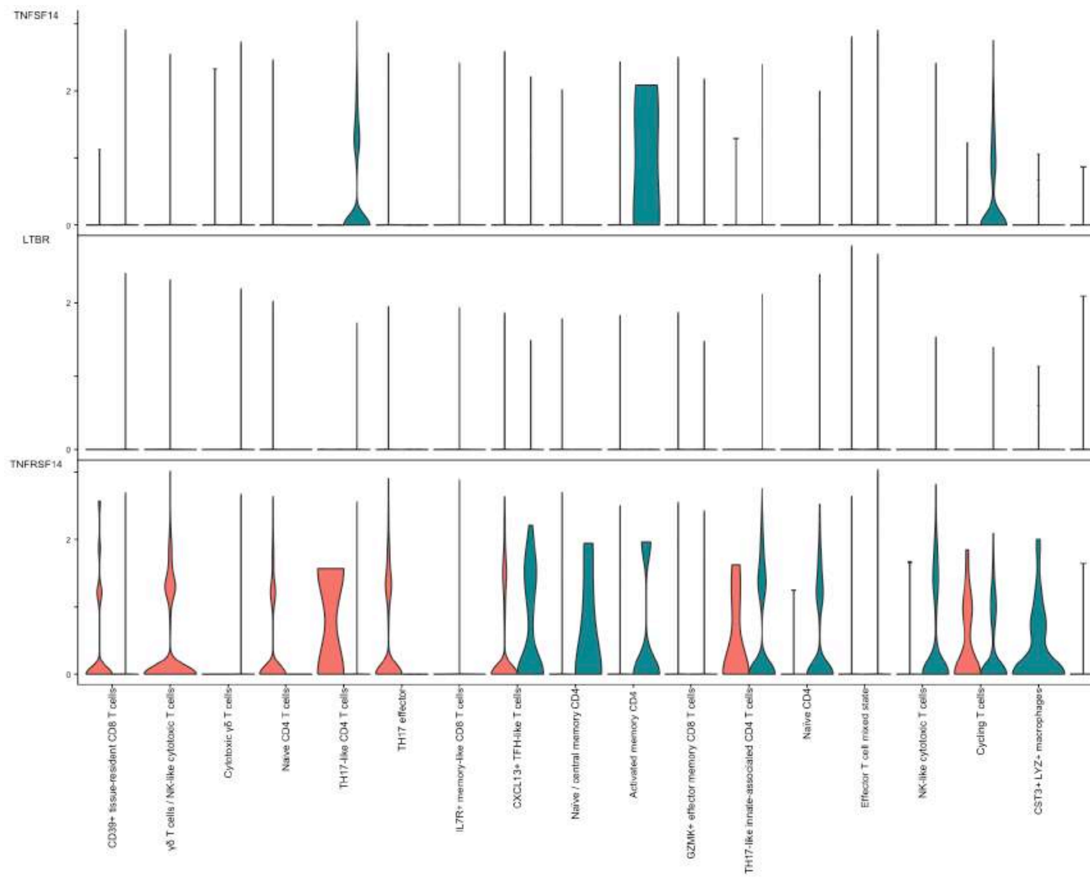
Within the CD LIGHT network, TH17-like CD4 T cells show the highest inferred outgoing contribution, followed by CST3 LYZ macrophages. The LIGHT–LTBR axis includes FOXP3 IL2RA Treg among the inferred receiver populations, while the broader LIGHT–TNFRSF14 (HVEM) axis reaches a wide set of receiver populations including CXCL13 TFH-like T cells, TH17-like innate-associated CD4 T cells, Naïve CD4, Cycling T cells, epithelial cells, and CST3 LYZ macrophages themselves (self-signaling). This should be presented as a within-CD characterization rather than as a confirmatory differential pathway, given the CD-emergent status of the Treg receiver population.

Figure 8. LIGHT circle plot and expression plots.

5.3.0.1 Circle plot Crohn's disease (CD)

LIGHT signaling pathway network





5.3.0.2 Gene expression

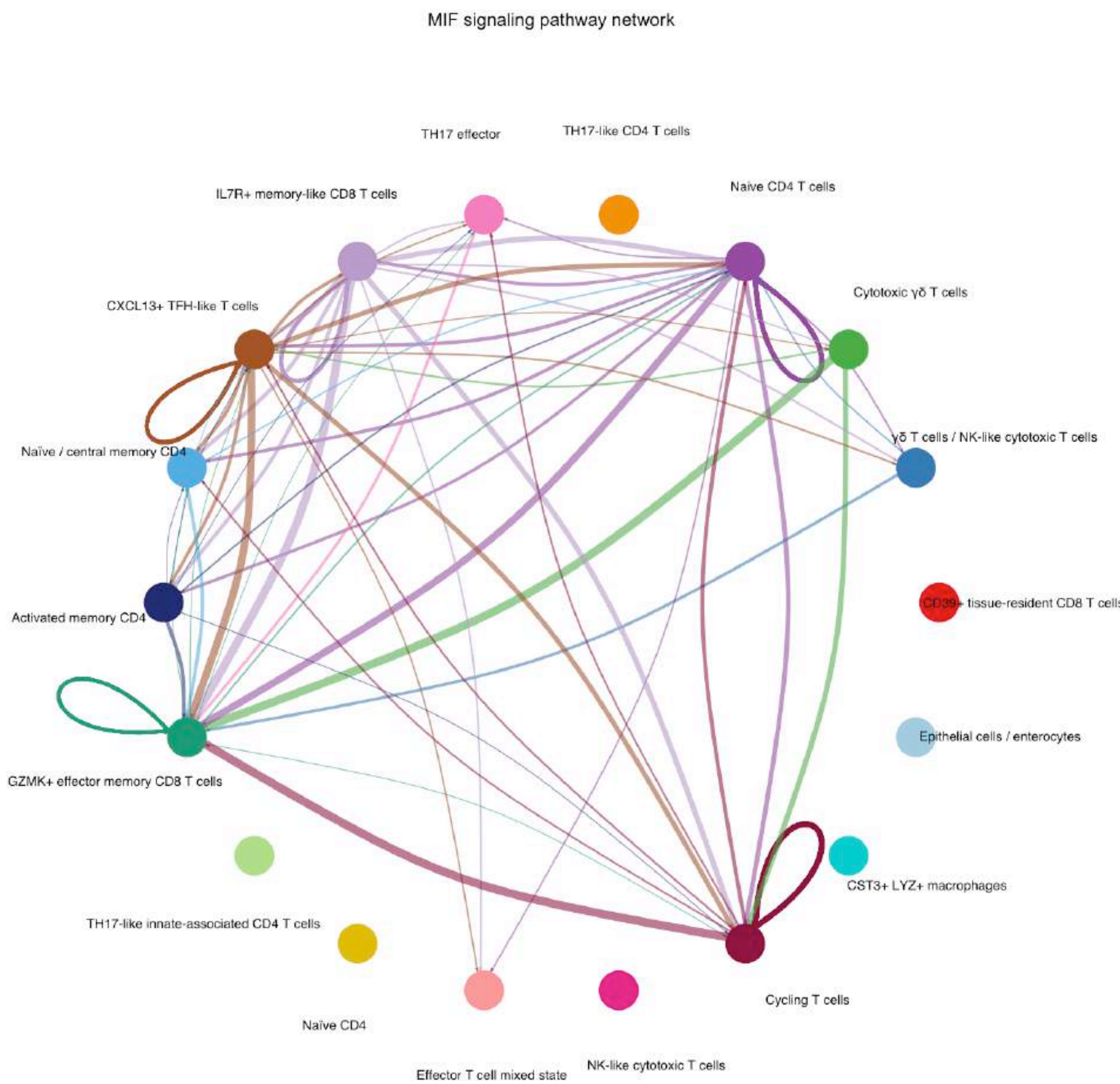
Literature context. LIGHT/TNFSF14 is associated with IBD susceptibility and has been reported as elevated in Crohn's disease. Its receptors include HVEM and LT R, with potentially divergent biological consequences. The detected LIGHT–LT R signaling involving macrophage-associated populations in IEL should therefore not be described simply as pro-inflammatory.

5.4 MIF — *mixed, with CD-associated network remodeling*

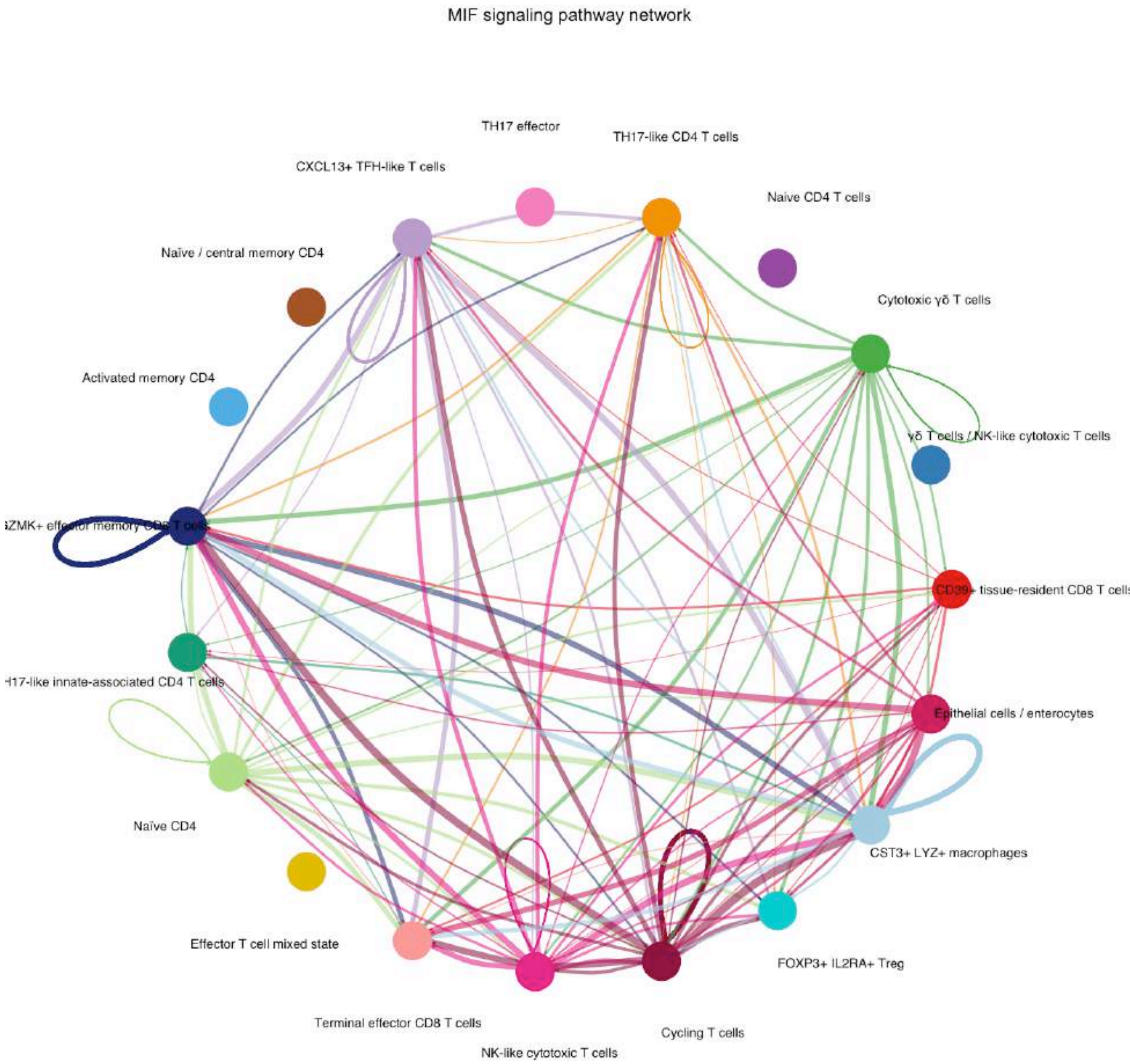
The inferred MIF communication pattern differs substantially between Control and CD. In Control, MIF involves mainly Control-enriched populations, limiting direct interpretation of a quantitative cross-condition comparison. In CD, the inferred MIF network becomes broader, with increased participation of CST3 LYZ macrophages and additional contribution from GZMK effector memory CD8 T cells. CST3 LYZ macrophages represent a prominent node within the CD-associated MIF network; however, this population is strongly CD-enriched (35 CD / 1 Control) and should therefore be interpreted primarily as a within-CD characterization. In contrast, GZMK effector memory CD8 T cells are present in both conditions (183 CD / 686 Control) and represent the most suitable population for direct CD-versus-Control interpretation of MIF-associated communication changes. Overall, MIF should be interpreted as a condition-associated network remodeling event rather than a simple pathway activation comparison.

Figure 9. MIF circle plots.

Control



Crohn's disease (CD)

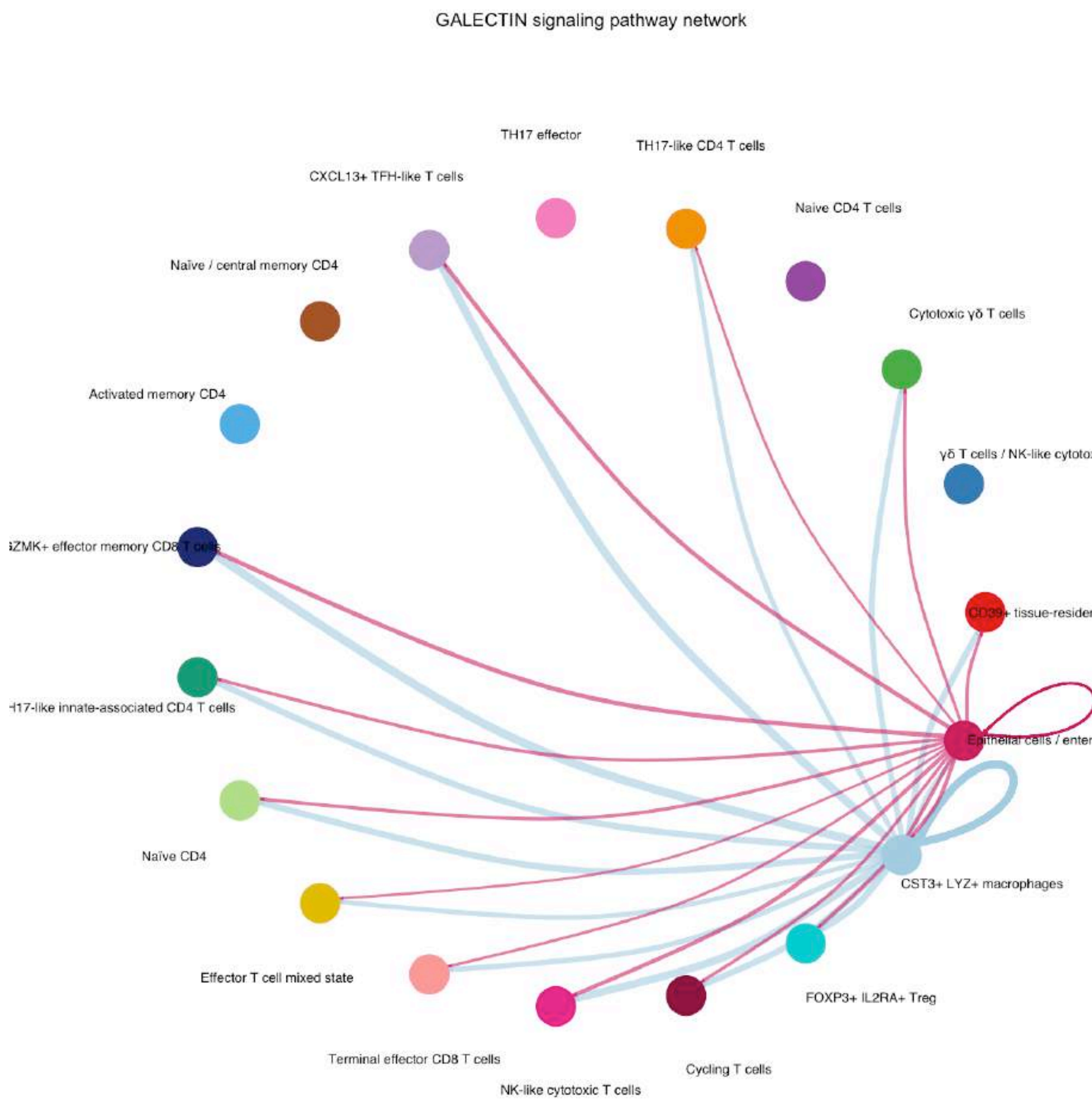


5.5 GALECTIN — *CD-emergent, epithelial-centered communication pattern*

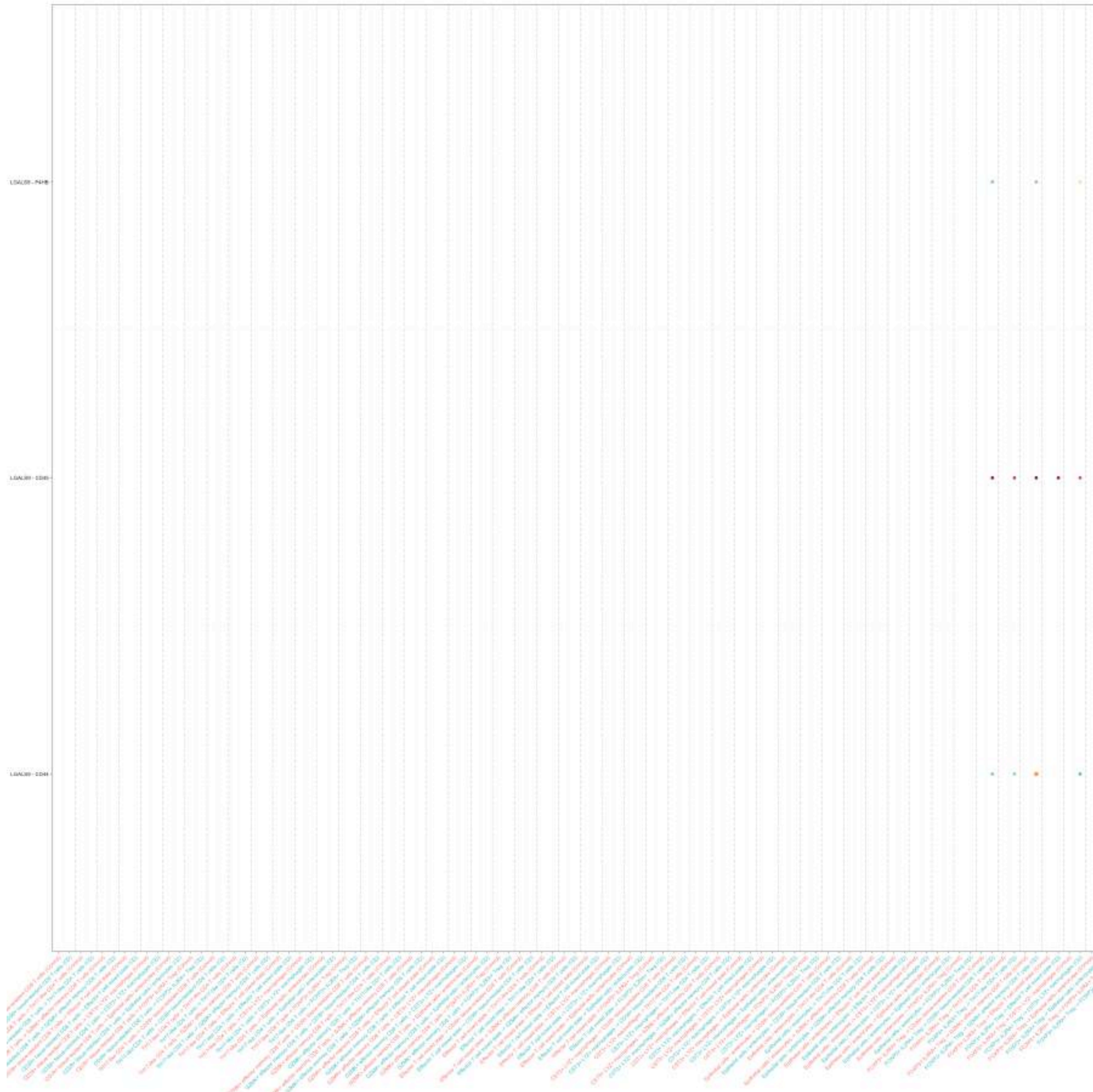
GALECTIN signaling was detected in the CD IEL network and showed an epithelial-centered communication structure. In the circle plot, Epithelial cells / enterocytes represent the dominant visual hub, including a prominent self-signaling component, indicating autocrine epithelial GALECTIN communication within the inferred network.

FOXP3 IL2RA Treg cells participate in the GALECTIN network but do not represent the dominant signaling hub based on the network visualization. Given that both epithelial and Treg populations are strongly condition-dependent in this dataset, this result should be interpreted as a within-CD communication pattern rather than a differential pathway activation claim.

Figure 10. GALECTIN signaling in IEL under Crohn's disease (CD).



5.5.0.1 Circle plot



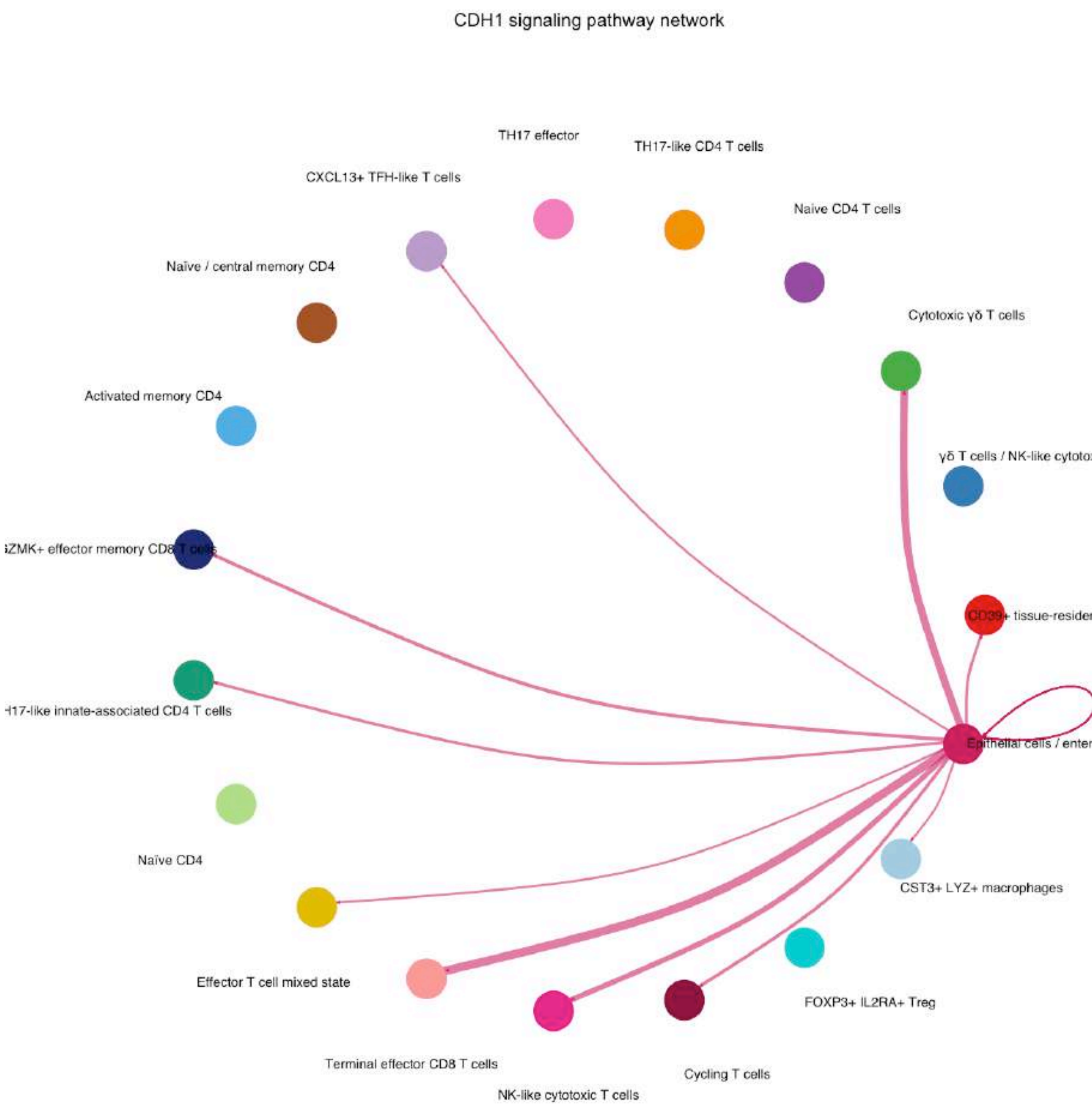
5.5.0.2 Bubble plot

5.6 CDH1 (E-cadherin) — *epithelial-associated CDH1 communication pattern*

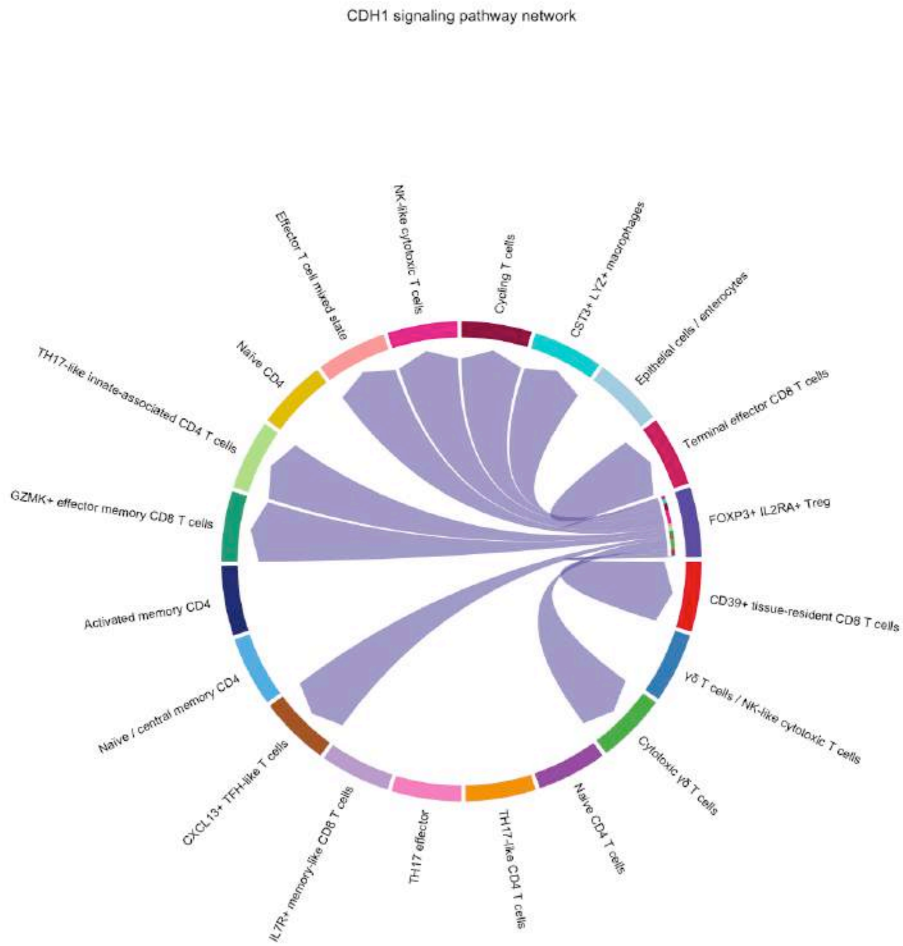
CDH1 signaling in the IEL CD network shows a prominent epithelial-centered structure, with Epithelial cells / enterocytes displaying the strongest visual hub and autocrine component in the circle plot.

Although FOXP3 IL2RA Treg cells contribute to the inferred CDH1 communication network, the current visualization does not support describing Treg as the exclusive or dominant sender population. The canonical epithelial CDH1–CD103 (ITGAE/ITGB7) interaction remains biologically plausible, but sender assignment should follow the complete CellChat communication table rather than visual interpretation alone.

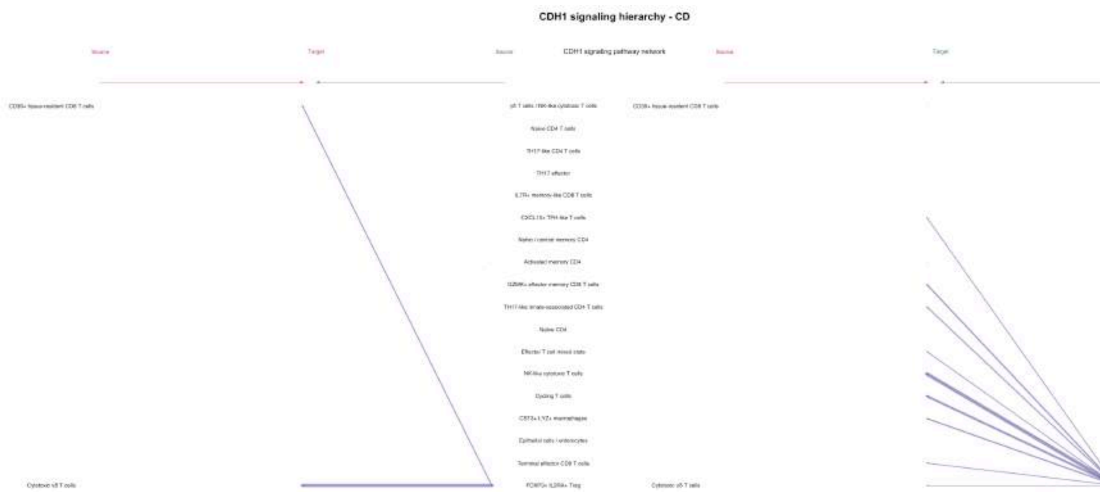
Figure 11. CDH1 (E-cadherin) signaling in IEL under Crohn's disease (CD).



5.6.0.1 Circle plot



5.6.0.2 Chord plot



5.6.0.3 Hierarchy plot

5.7 Biological interpretation

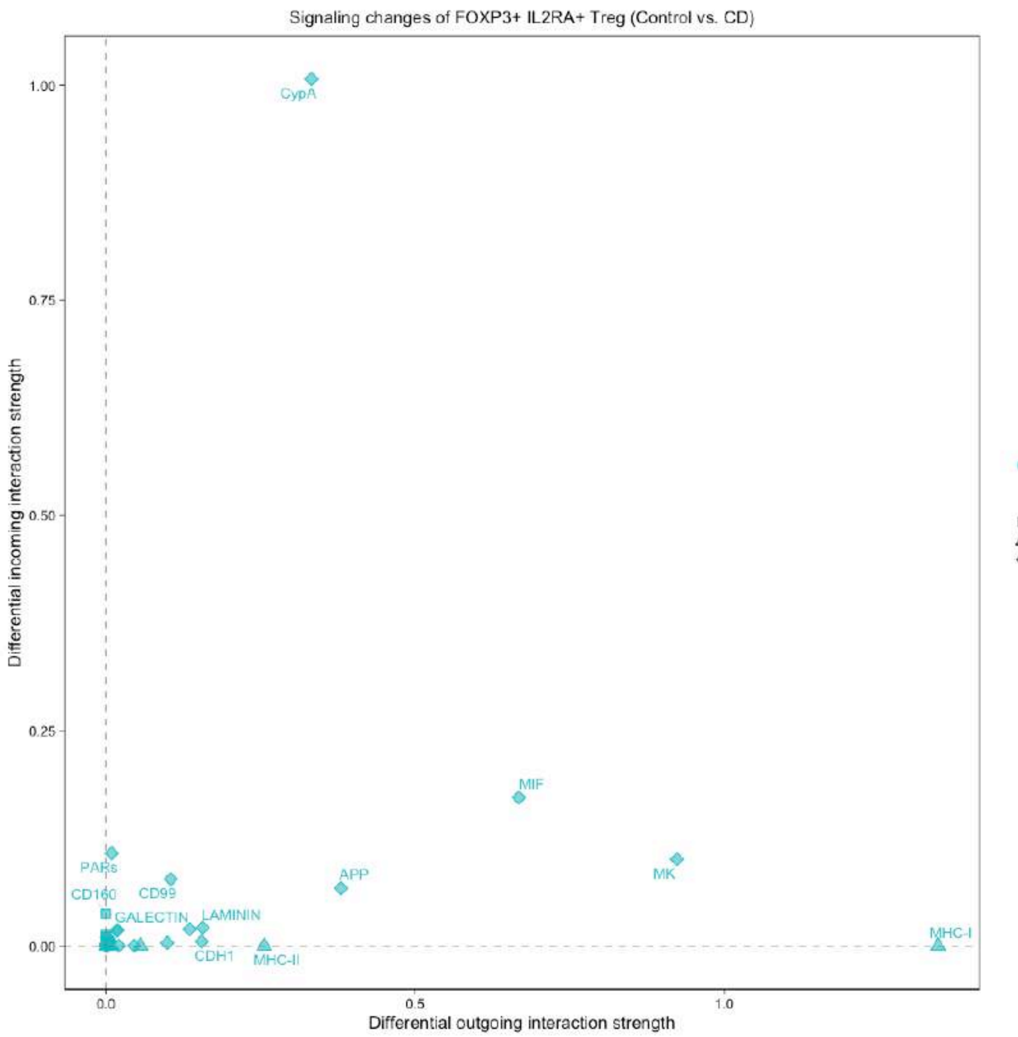
CD103–E-cadherin is an established retention mechanism for intraepithelial lymphocytes and CD103 Tregs. In this dataset it should be presented as a strong, mechanistically anchored **CD-population characterization**, not as a differential CD-vs-Control claim.

5.8 Signaling-role scatter plots and communication patterns

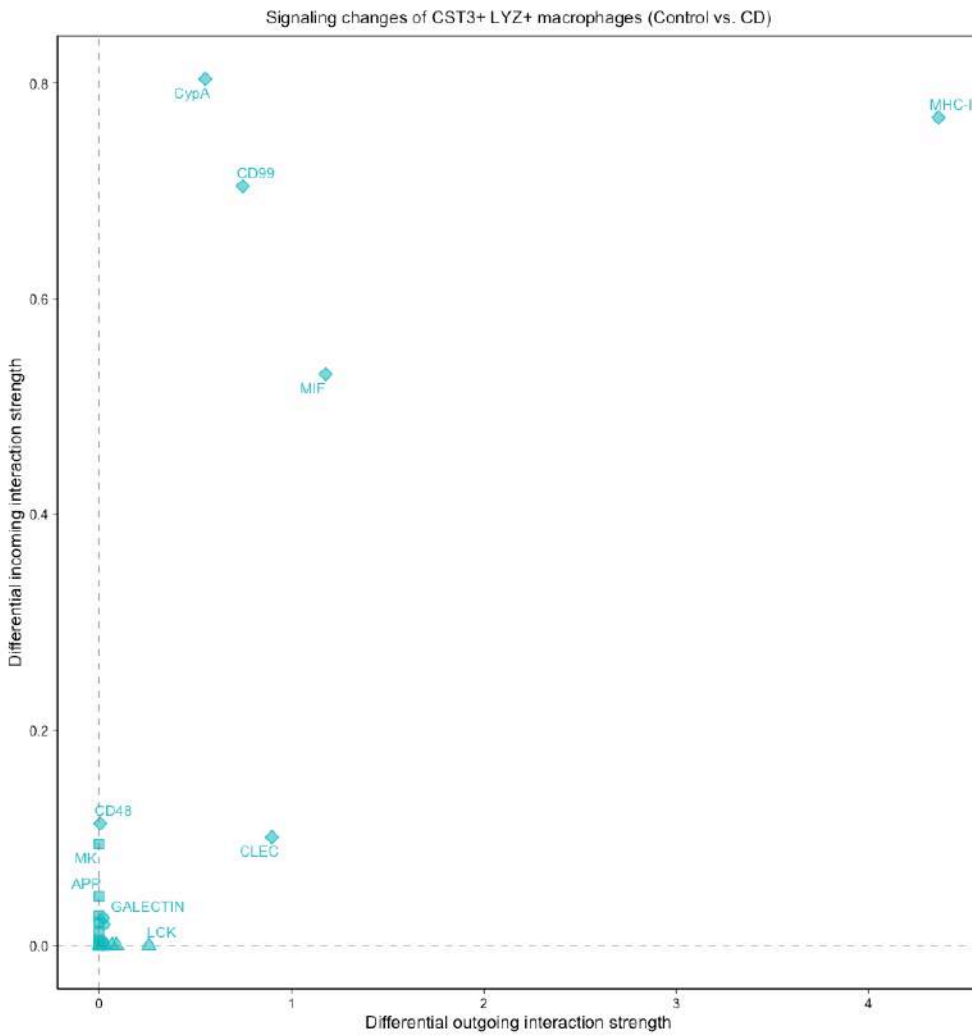
The signaling-role scatter plots summarize CellChat-inferred changes in incoming and outgoing communication strength for selected IEL populations. These outputs provide a network-level overview of pathway redistribution between conditions but should not be interpreted as direct measurements of biological activation.

Interpretation is constrained by the strong cell-type imbalance described in Section 3.1. CD-emergent populations (including FOXP3 IL2RA Treg, CST3 LYZ macrophages, and TH17-like CD4 T cells) primarily represent CD-network characterization rather than quantitative CD-versus-Control comparisons. More balanced populations, such as GZMK effector memory CD8 T cells and effector T-cell mixed states, provide more interpretable comparative information.

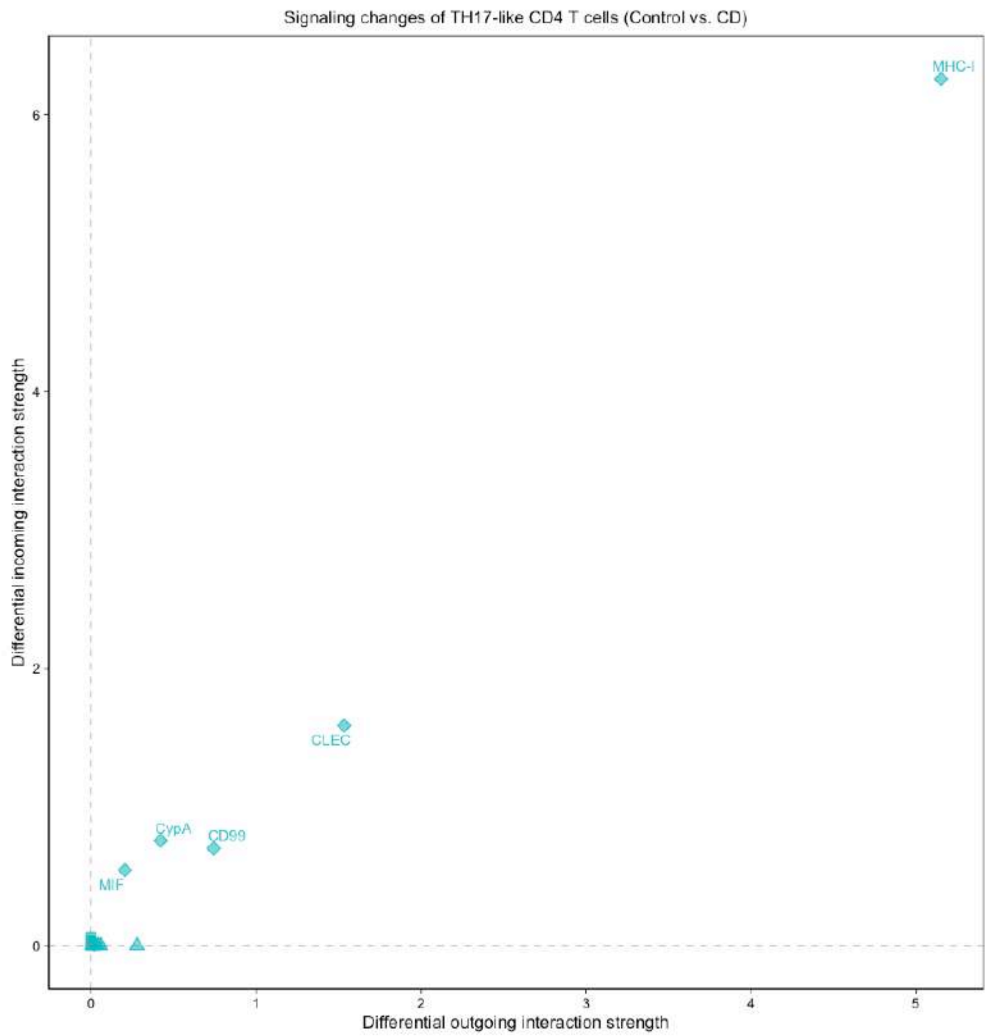
Figure 12. CellChat signaling-role scatter plots for selected IEL populations.



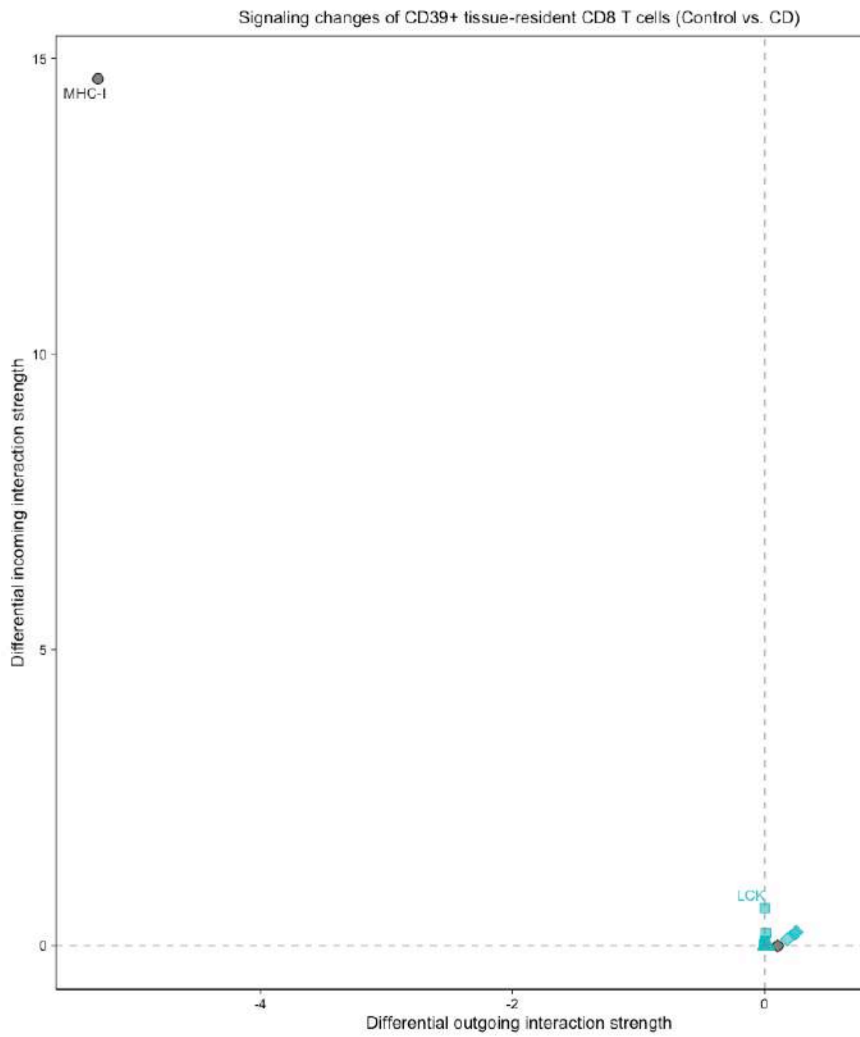
5.8.0.1 FOXP3 IL2RA Treg



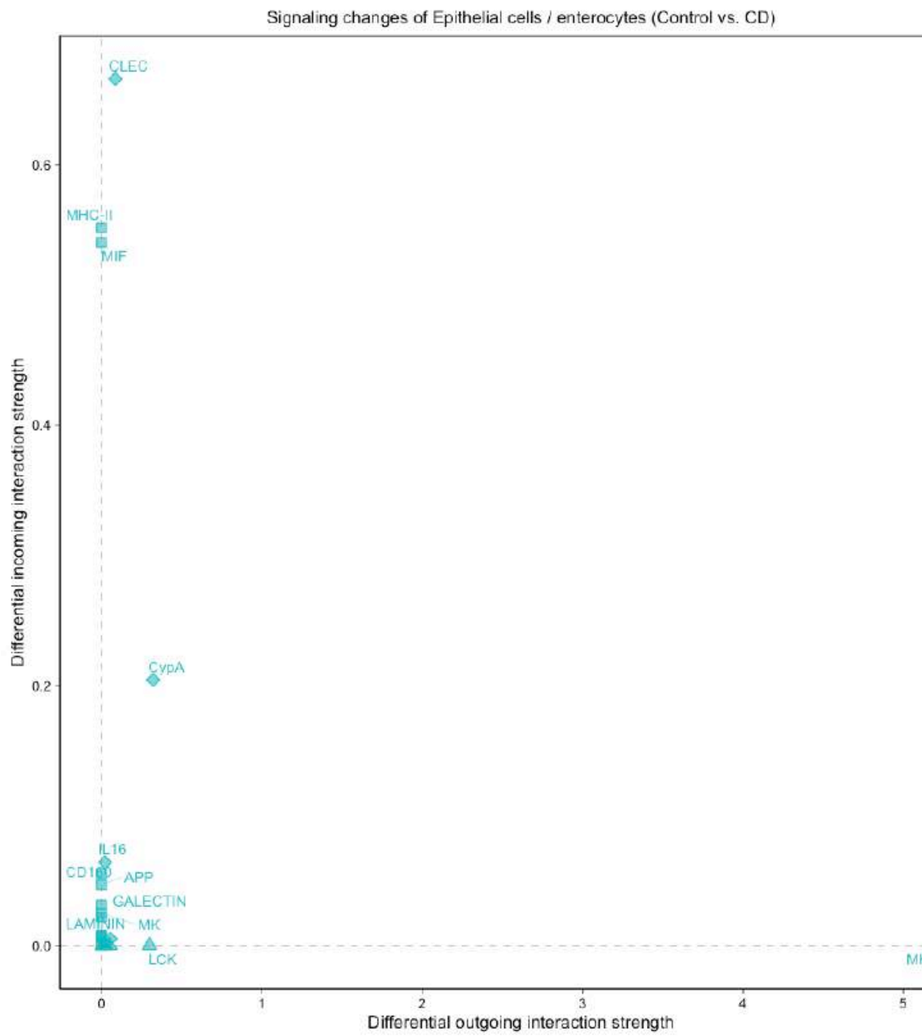
5.8.0.2 CST3 LYZ macrophages



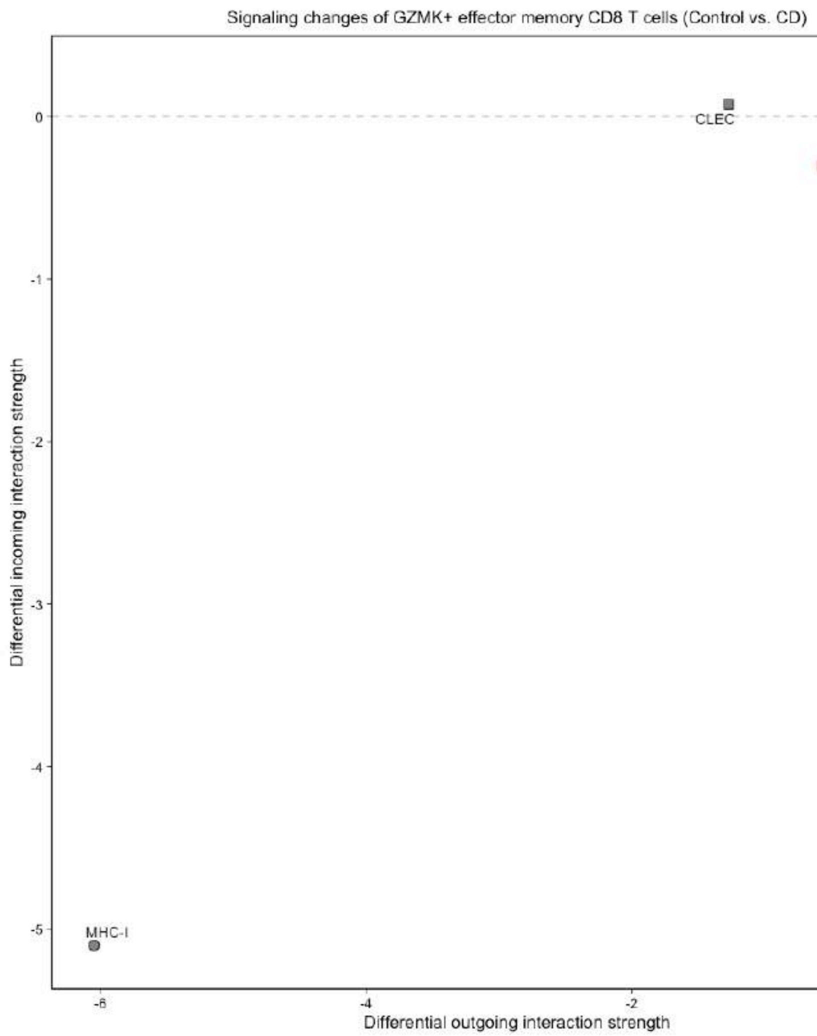
5.8.0.3 TH17-like CD4 T cells



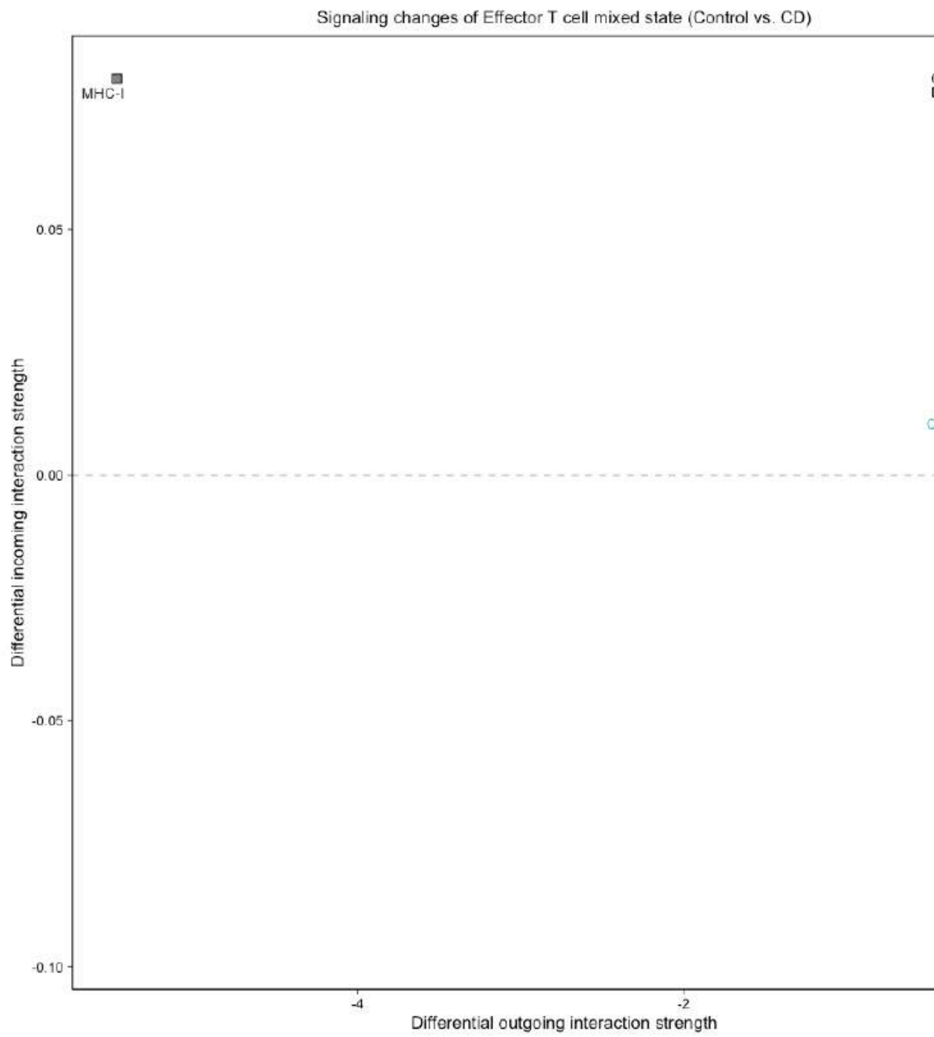
5.8.0.4 CD39 tissue-resident CD8 T cells



5.8.0.5 Epithelial cells / Enterocytes



5.8.0.6 GZMK effector memory CD8 T cells



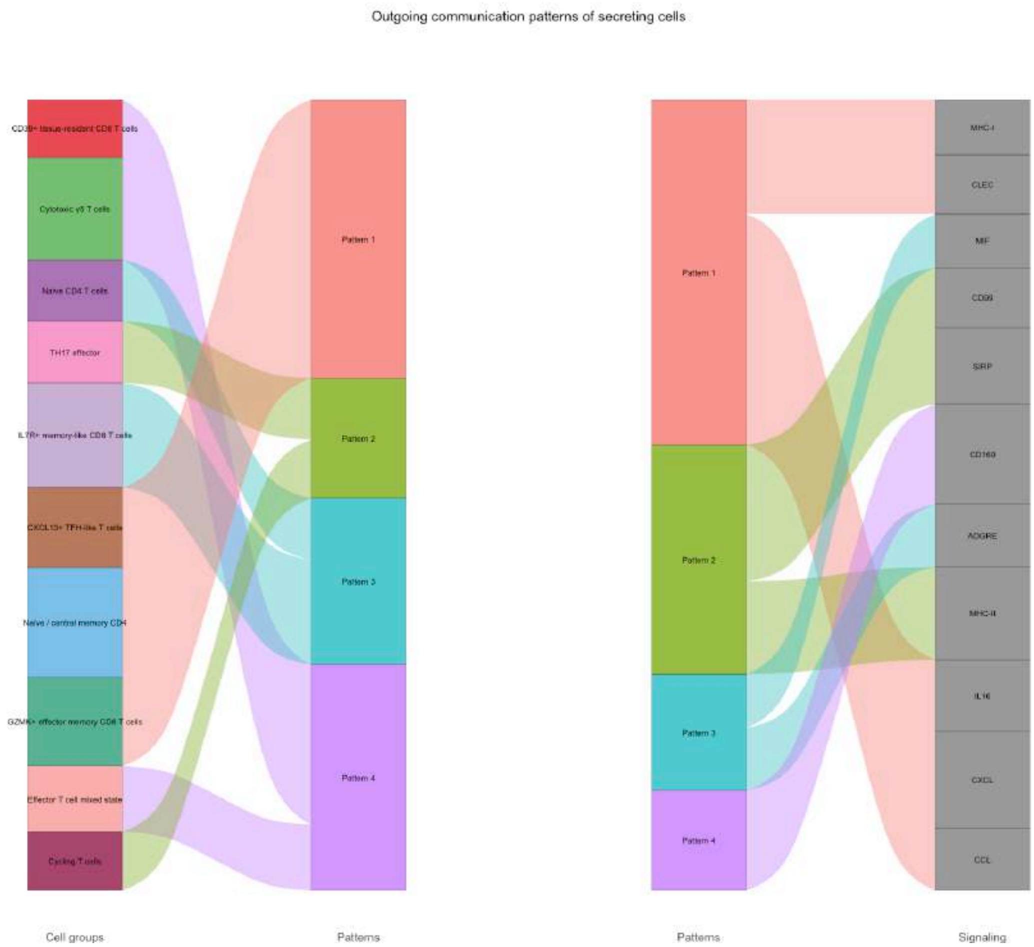
5.8.0.7 Effector T cell (mixed state)

Figure 13. NMF-derived communication patterns in the IEL network.

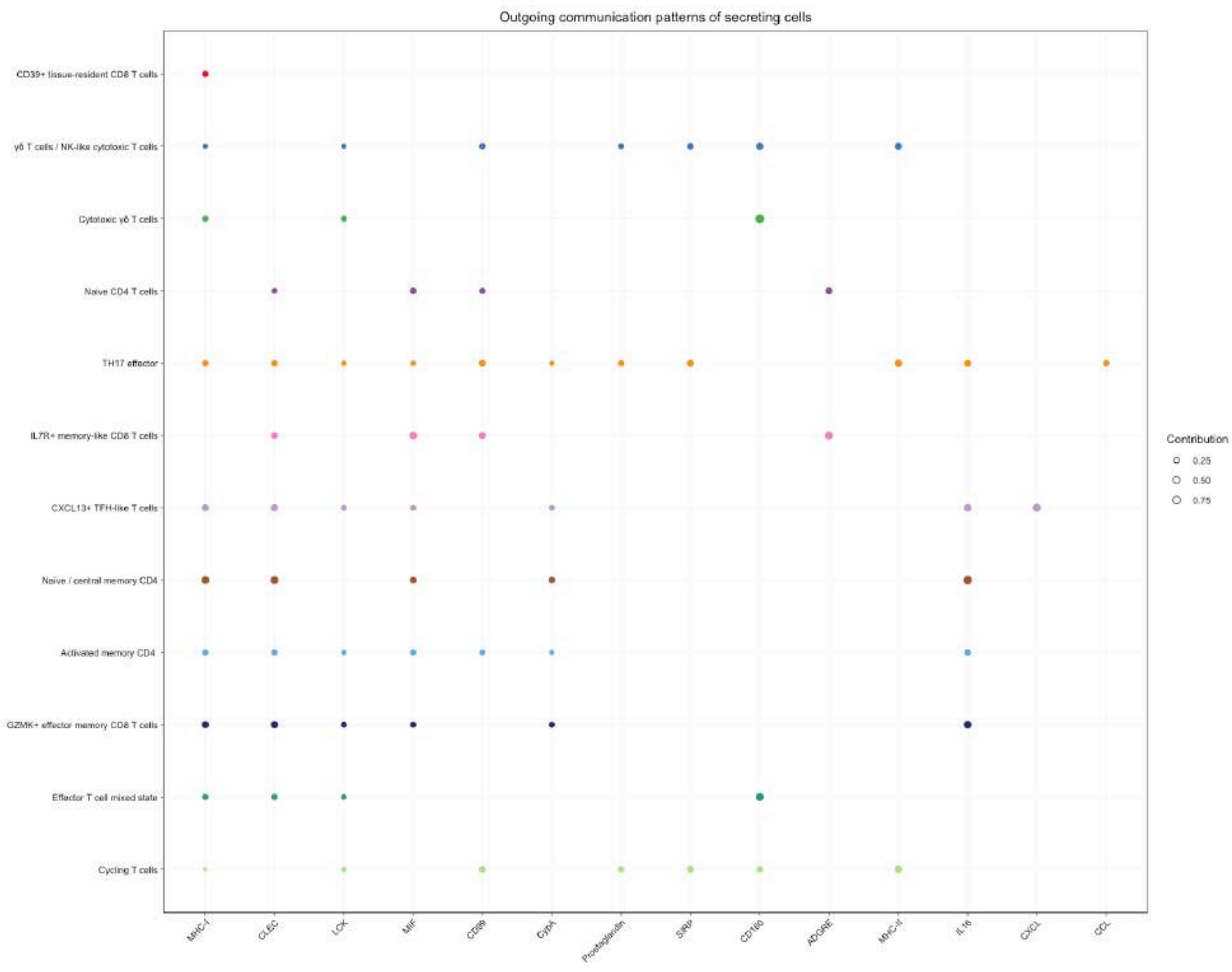
NMF decomposition was performed independently for outgoing and incoming communication programs. Four outgoing patterns (K = 4) and six incoming patterns (K = 6) were identified. Pattern identities are condition-specific and should not be interpreted as directly corresponding between Control and CD. Differences in pattern composition therefore reflect global communication-network reorganization rather than a direct comparison of individual latent factors.

5.8.0.8 Outgoing communication patterns

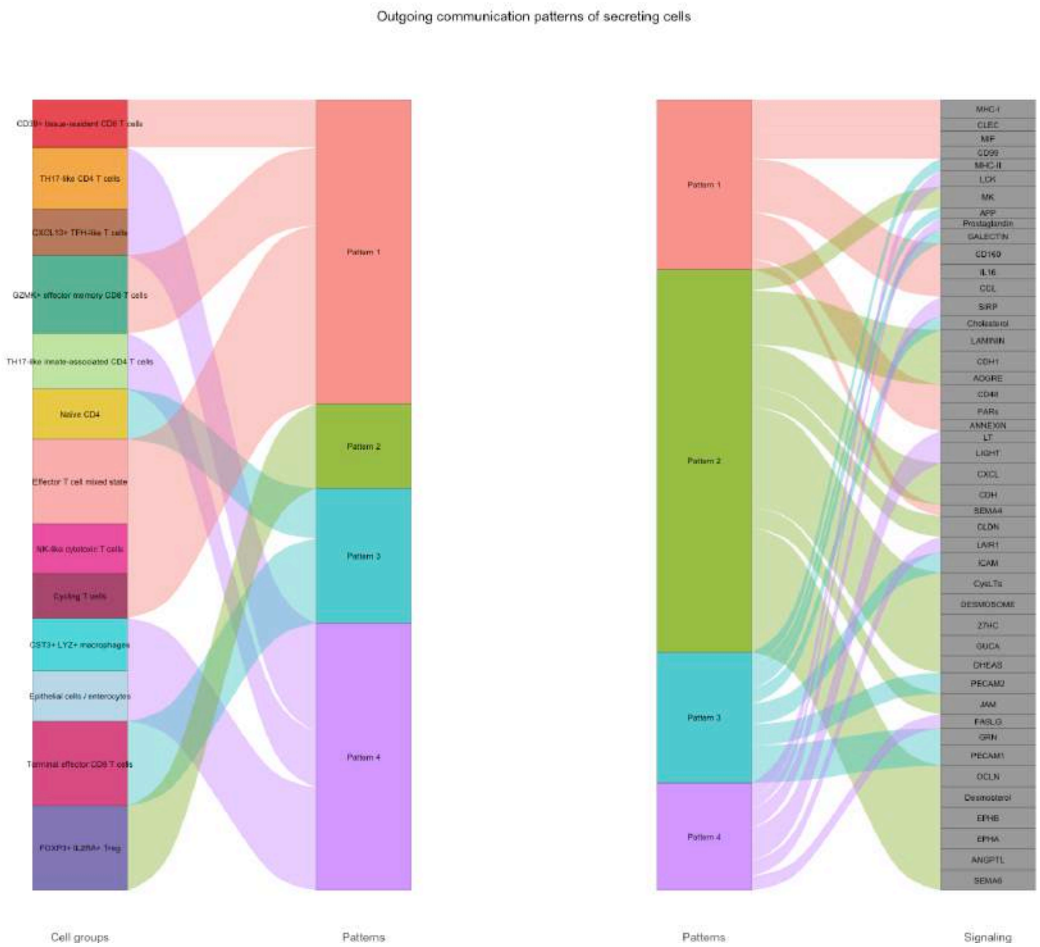
5.8.0.8.1 Control River plot


Outgoing communication patterns of secreting cells
Cytotoxic γδ T cells
Naive CD4 T cells
TH17 effector
CXCL13+ TFH-like T cells
Naive / central memory CD4
GZMK+ effector memory CD8 T cells
Effector T cell mixed state
Cycling T cells
Pattern 1
Pattern 2
Pattern 3
Pattern 4
MHC-I
CLEC
MIF
CD99
SIRP
CD160
ADGRE
MHC-II
IL16
CXCL
CCL
Cell groups
Patterns
Patterns
Signaling

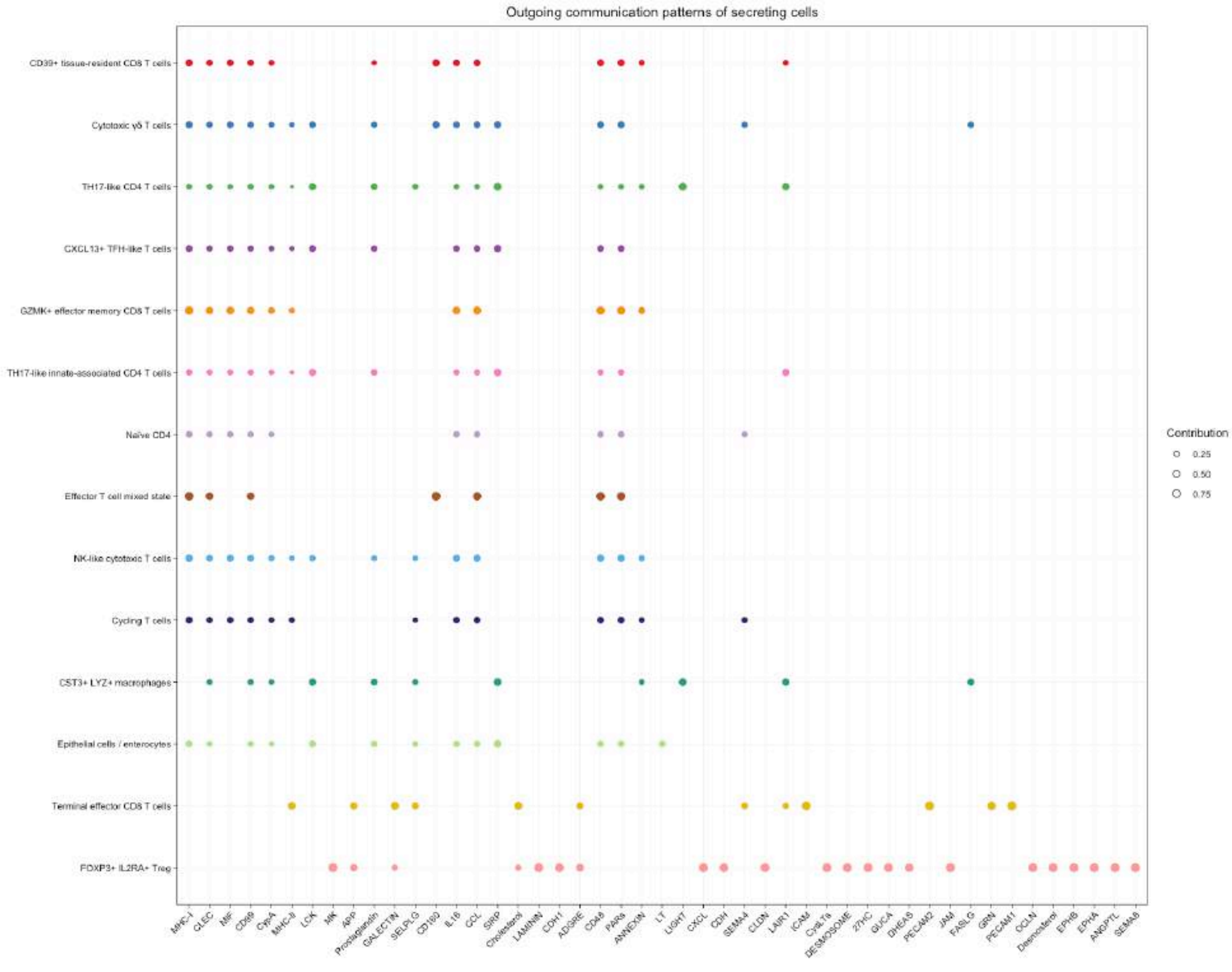
Dot plot



5.8.0.8.2 Crohn's disease (CD) River plot

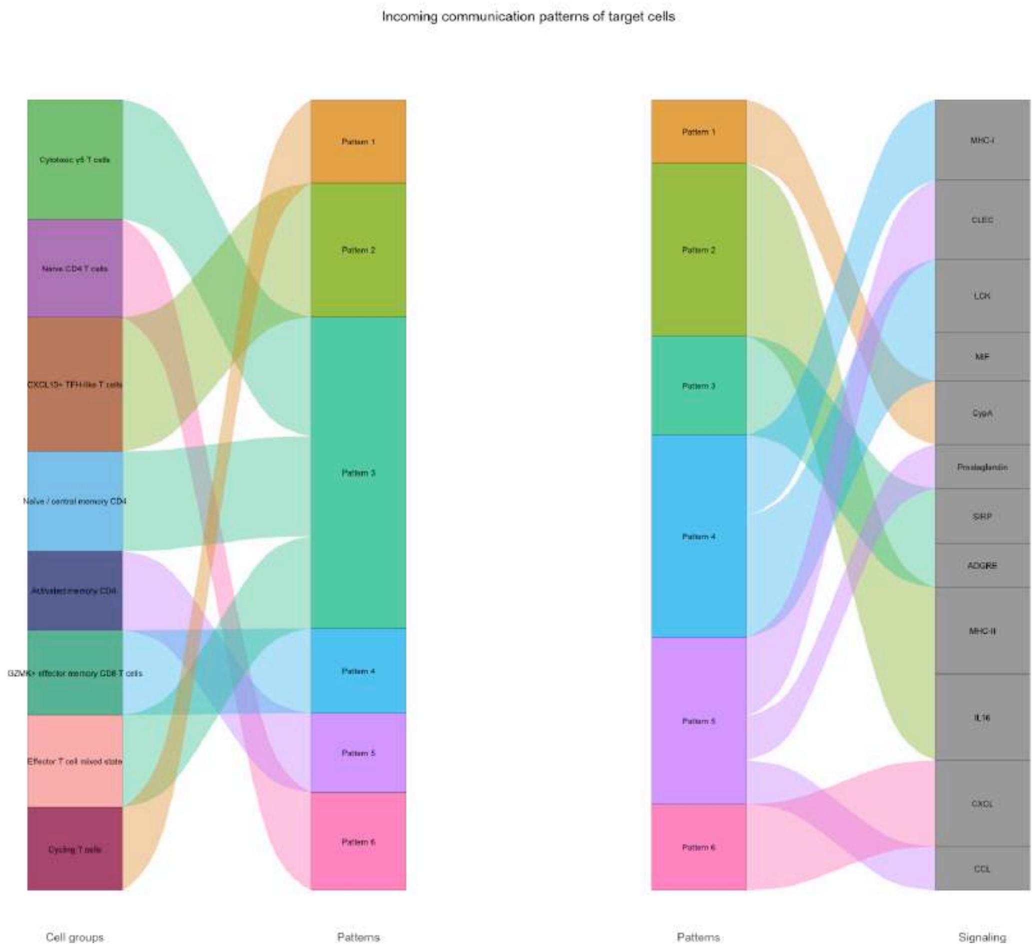

Outgoing communication patterns of secreting cells
CD39+ tissue-resident CD8 T cells
TH17-like CD4 T cells
CXCL13+ TFH-like T cells
GZMK+ effector memory CD8 T cells
TH17-like innate-associated CD4 T cells
Naive CD4
Effector T cell mixed state
NK-like cytotoxic T cells
Cycling T cells
CST3+ LYZ+ macrophages
Epithelial cells / enterocytes
Terminal effector CD8 T cells
FOXP3+ IL2RA+ Treg
Pattern 1
Pattern 2
Pattern 3
Pattern 4
Cell groups
Patterns
Pattern 1
Pattern 2
Pattern 3
Pattern 4
MHC-I
CLEC
MIF
CD99
MHC-II
LCK
MK
APP
Prostaglandin
GALECTIN
CD160
IL16
CCL
SIRP
Cholesterol
LAMININ
CDH1
ADGRE
CD48
PARs
ANNEXIN
LT
LIGHT
CXCL
CDH
SEMA4
CLDN
LAIR1
ICAM
CysLTs
DESMOSOME
27HC
GUCA
DHEAS
PECAM2
JAM
FASLG
GRN
PECAM1
OCLN
Desmosterol
EPHB
EPHA
ANGPTL
SEMA6
Patterns
Signaling

Dot plot

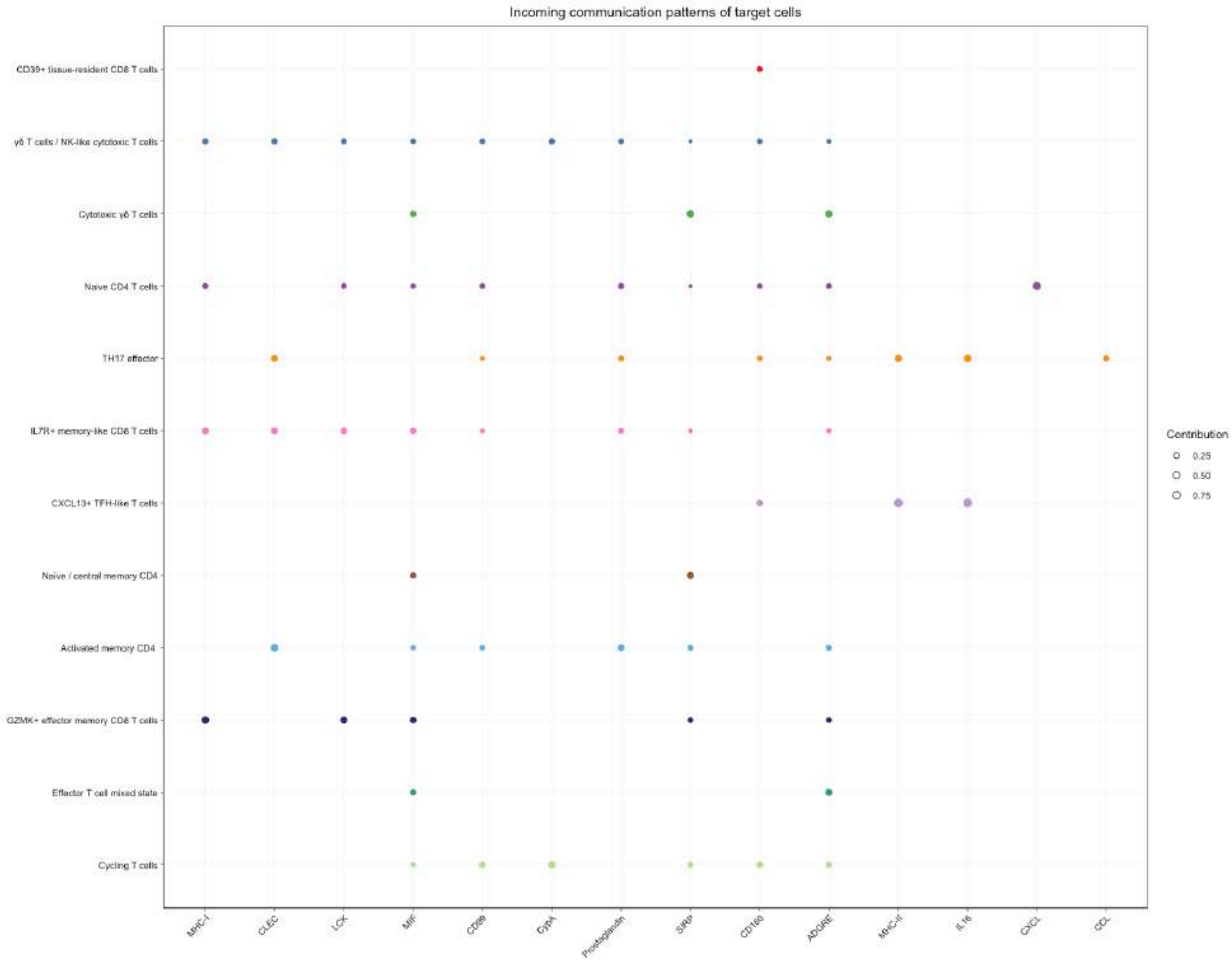


5.8.0.9 Incoming communication patterns

5.8.0.9.1 Control River plot

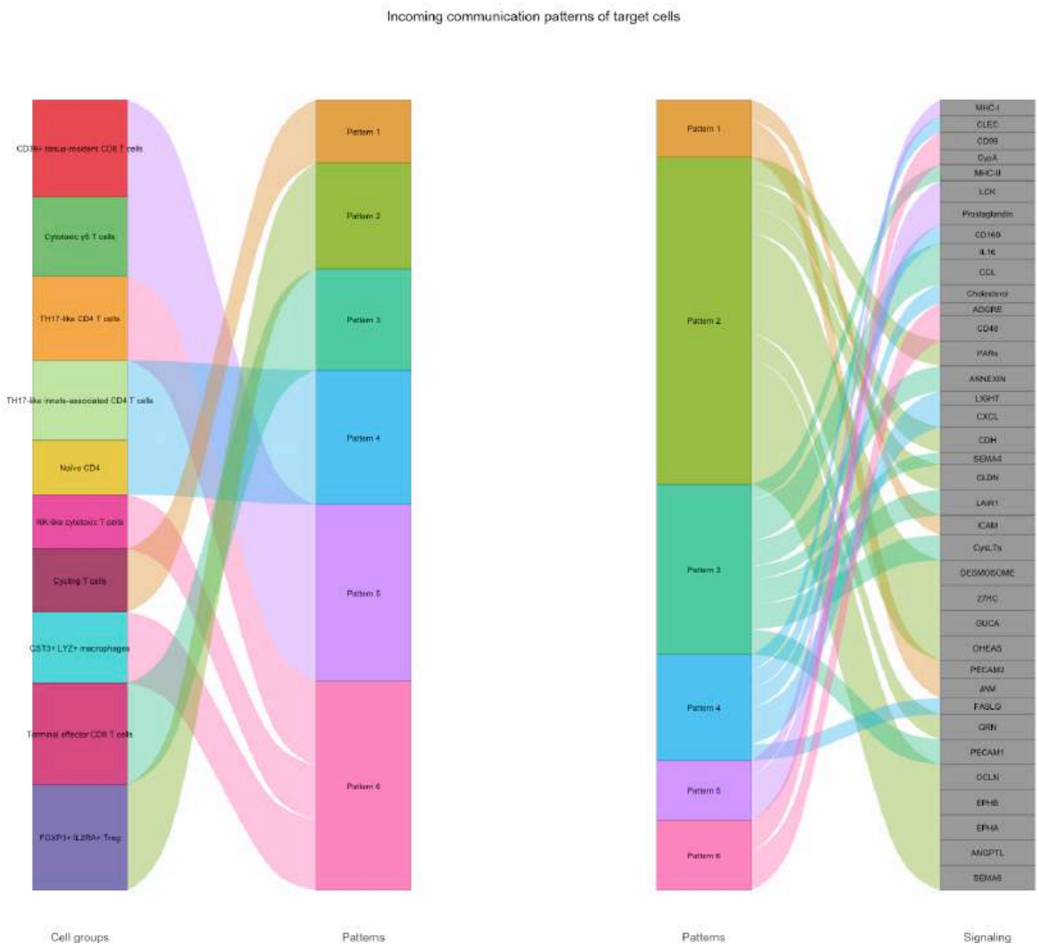

Incoming communication patterns of target cells
Naive / central memory CD4
Activated memory CD4
GZMK+ effector memory CD8 T cells
Effector T cell mixed state
Cycling T cells
Pattern 1
Pattern 2
Pattern 3
Pattern 4
Pattern 5
Pattern 6
MHC-I
CLEC
LCK
MIF
CypA
SIRP
ADGRE
MHC-II
IL16
CXCL
CCL
Cell groups
Patterns
Patterns
Signaling

Dot plot



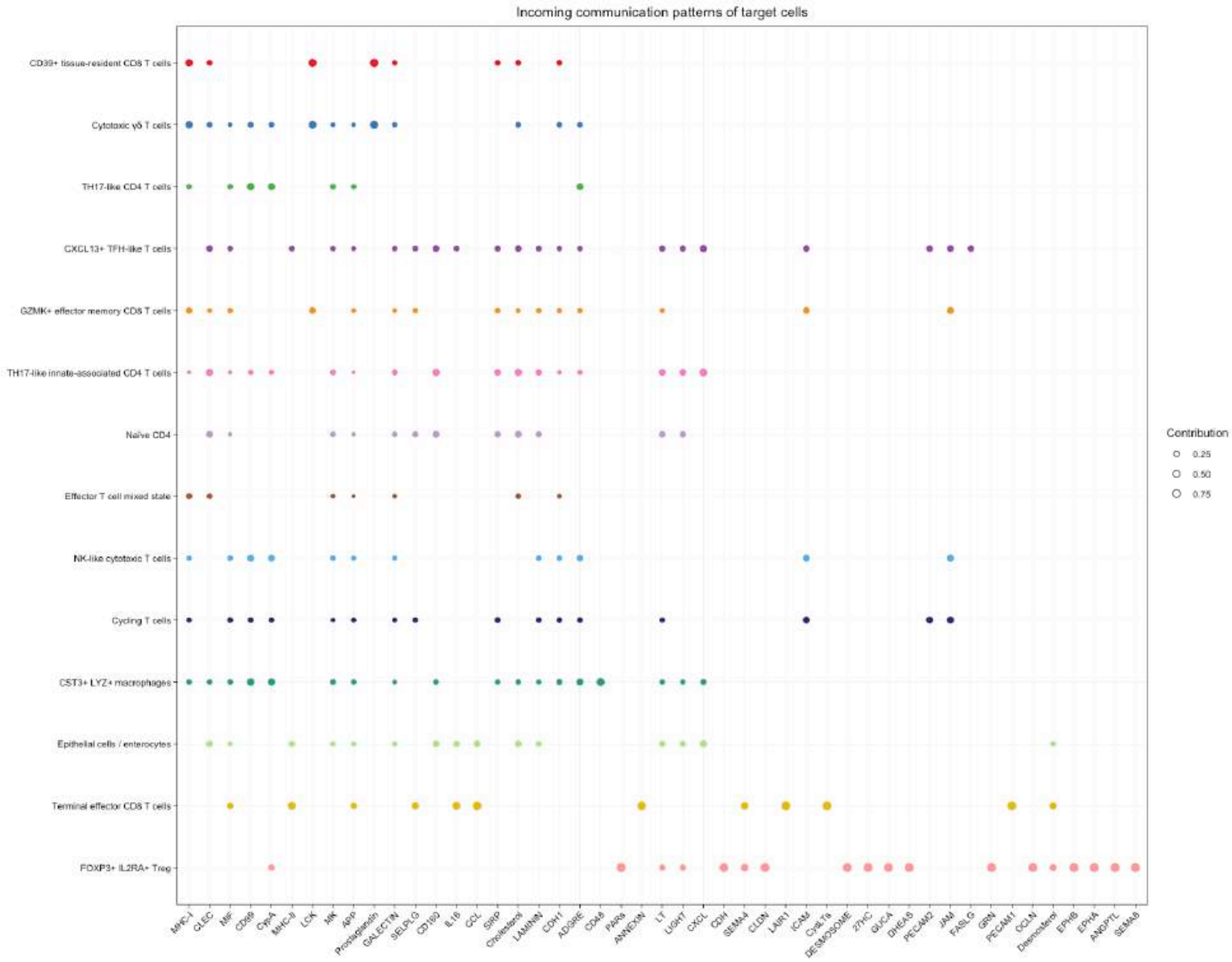
Crohn’s disease (CD)

River plot



Incoming communication patterns of target cells
CD69+ tissue-resident CD8 T cells
Cytotoxic γδ T cells
TH17-like CD4 T cells
TH17-like innate-associated CD4 T cells
Naive CD4
NK-like cytotoxic T cells
Cycling T cells
CST3+ LYZ+ macrophages
Terminal effector CD8 T cells
FOXP3+ IL2RA+ Treg
Pattern 1
Pattern 2
Pattern 3
Pattern 4
Pattern 5
Pattern 6
MHC-I
CLEC
CD99
CypA
MHC-II
LCK
Prostaglandin
CD160
IL16
CCL
Cholesterol
ADGRE
CD48
PARs
ANNEXIN
LIGHT
CXCL
CDH
SEMA4
CLDN
LAIR1
ICAM
CysLTs
DESMOSOME
27HC
GUCA
DHEAS
PECAM2
JAM
FASLG
GRN
PECAM1
OCLN
EPHB
EPHA
ANGPTL
SEMA6
Cell groups
Patterns
Patterns
Signaling

Dot plot



In the CD incoming dot plot, FOXP3 Treg is the row with the broadest pathway association. This is a robust description of the Treg-CD profile, but not a differential claim because Treg is not measurable in Control.

6 LPL Compartment: Pathway-Level Findings

Six pathways were selected: **MHC-I, MHC-II, SIRP, GALECTIN, IL2, LIGHT**. ADGRE and IL10 were discarded after direct verification via `subsetCommunication()` showed too few interactions to justify inclusion.

Pathway	Real sender	Interactions (Control / CD)	Robustness
MHC-I	Broad, near-ubiquitous	>600 rows/condition	Solid
MHC-II	Broad in CD, restricted in Control	83 / 146	Solid for T follicular helper; descriptive for Treg axis
SIRP	Reorganized sender/receiver set	17 / 24	Solid; narrate as reorganization
GALECTIN	IFN-activated CD8 cytotoxic T cells	0 / 38	Moderate, CD-exclusive

Pathway	Real sender	Interactions (Control / CD)	Robustness
IL2	Innate-like/TRM CD8	0 / 11	Weak, receptor-nuance dependent
LIGHT	Activated CD4 T cells stress response	0 / 8	Fragile; n=15 CD sender cells

6.1 MHC-I

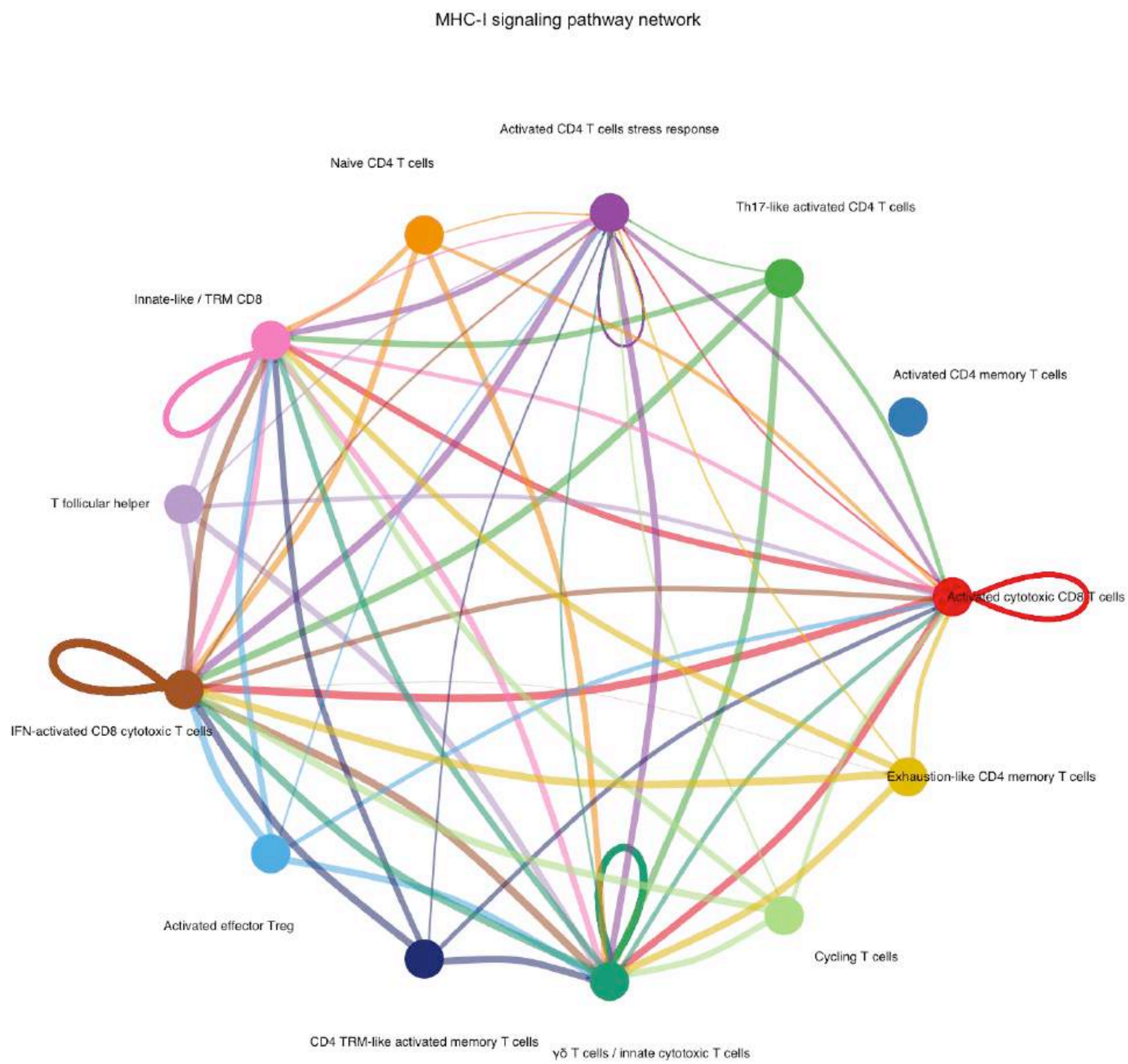
MHC-I represents a broadly detected signaling pathway in both LPL conditions. The dominant inferred receiver population is **Activated cytotoxic CD8 T cells**, a fully comparable population between CD and Control (5650 and 2002 cells, respectively).

Because this population is present at high abundance in both conditions, the MHC-I communication pattern represents one of the more robust pathway-level comparisons in the LPL compartment. The signaling-role scatter highlights a marked redistribution of incoming MHC-I-associated signaling toward **Activated effector Treg**, another population that is quantitatively represented in both conditions (873 CD / 617 Control).

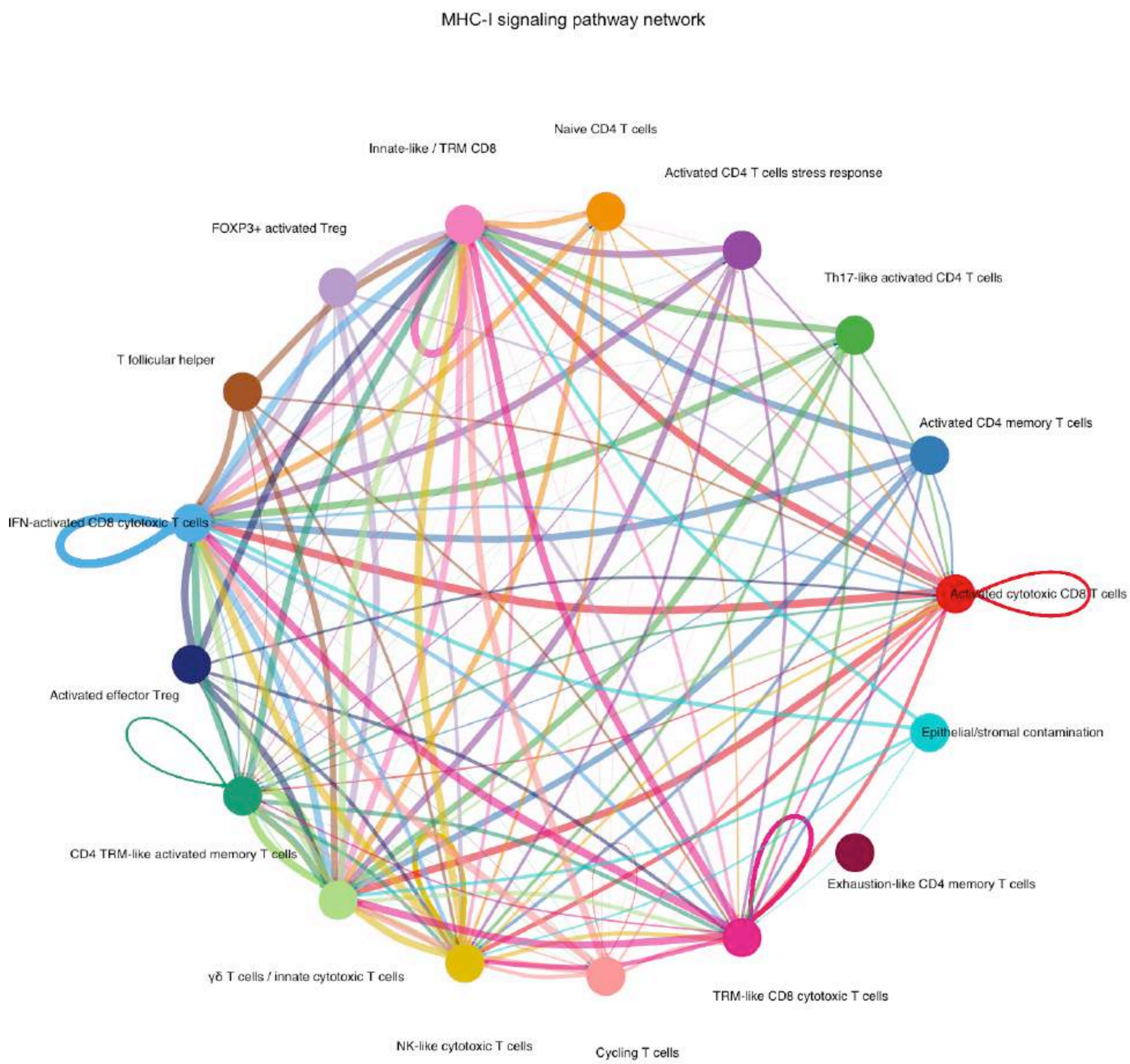
These results indicate a reorganization of MHC-I-associated communication within the LPL immune network, rather than direct evidence of altered MHC-I biological activity. CellChat infers ligand–receptor communication potential and does not measure antigen presentation efficiency or functional T-cell activation.

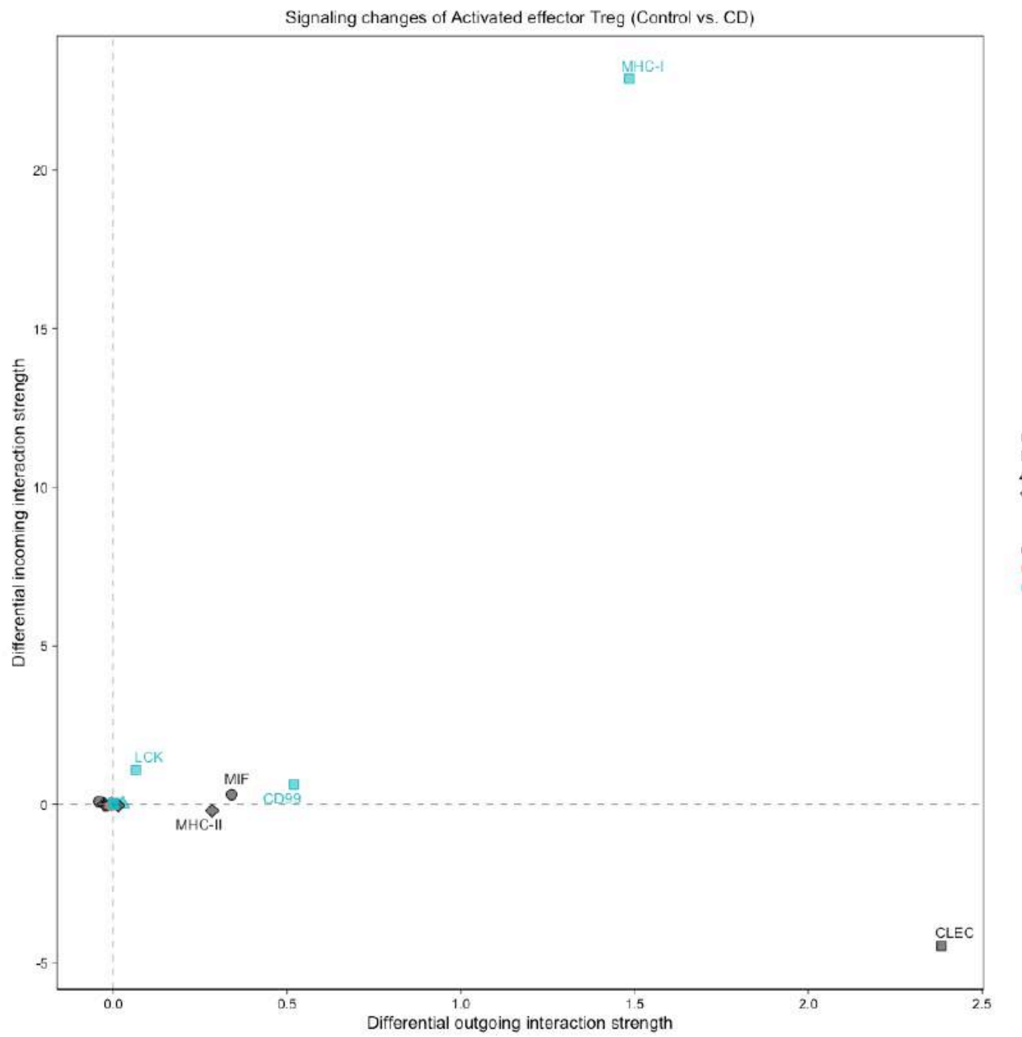
Figure 14. MHC-I signaling in LPL under Control and Crohn's disease (CD), together with the signaling-role change observed for Activated effector Treg.

6.1.0.1 Circle plots Control



Crohn's disease (CD)


MHC-I signaling pathway network
Naive CD4 T cells
Innate-like / TRM CD8
Activated CD4 T cells stress response
FOXP3+ activated Treg
Th17-like activated CD4 T cells
T follicular helper
Activated CD4 memory T cells
IFN-activated CD8 cytotoxic T cells
Activated cytotoxic CD8 T cells
Activated effector Treg
Epithelial/stromal contamination
CD4 TRM-like activated memory T cells
Exhaustion-like CD4 memory T cells
γδ T cells / innate cytotoxic T cells
TRM-like CD8 cytotoxic T cells
NK-like cytotoxic T cells
Cycling T cells



6.1.0.2 Signaling-role change

6.2 MHC-II — *population substitution rather than simple intensification*

MHC-II signaling shows a condition-dependent redistribution of inferred receiver populations rather than a simple increase of a shared Treg-associated circuit.

In Control, MHC-II communication is primarily associated with **T follicular helper cells**, with additional contribution from **Activated effector Treg** and **Cycling T cells**.

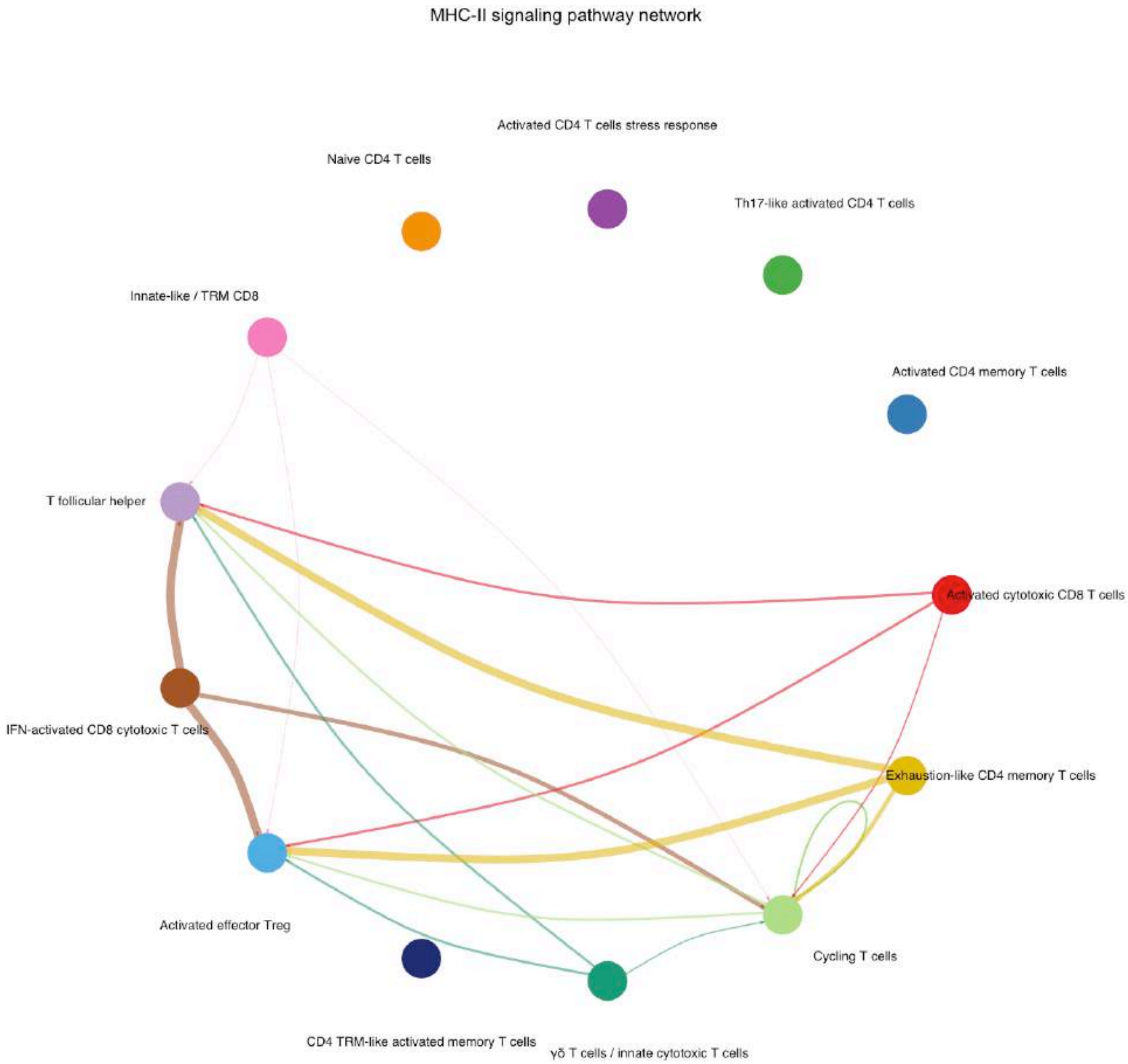
In CD, the receiving profile is reorganized, with **FOXP3 activated Treg** emerging as a major inferred receiver population, together with contribution from **IFN-activated CD8 cytotoxic T cells**. Because FOXP3 activated Treg cells are absent in Control (2205 CD / 0 Control), this represents a CD-specific network characterization rather than a quantitative increase of a conserved Treg pathway.

Activated effector Treg, although present in both conditions (873 CD / 617 Control), is no longer among the dominant inferred MHC-II receiver populations in CD.

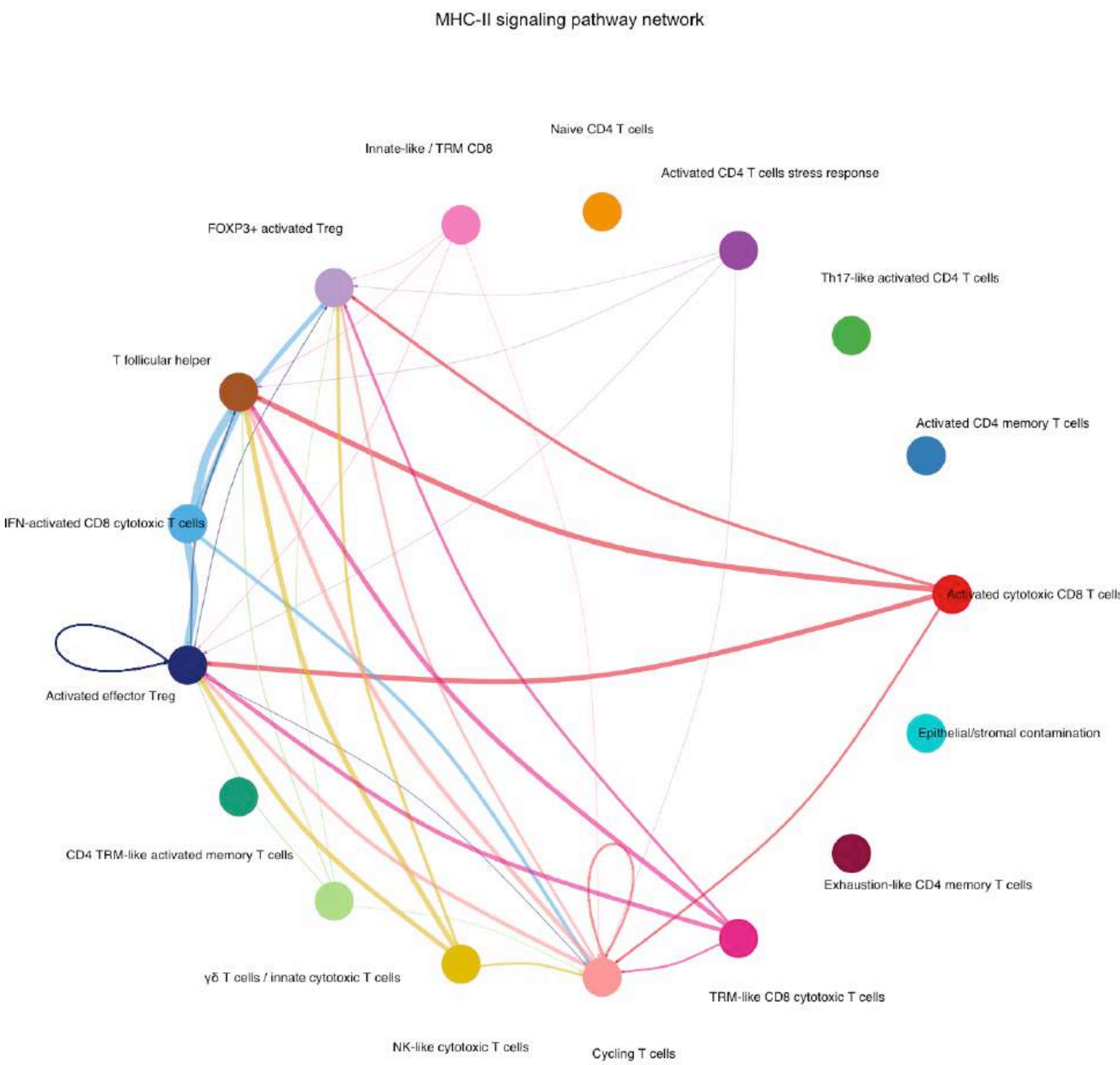
Therefore, the main interpretation is a **change in the cellular identity of the MHC-II receiving compartment**, rather than amplification of the same Treg signaling circuit across conditions.

Figure 15. MHC-II signaling in LPL under Control and Crohn's disease (CD), showing the condition-specific receiving profiles and the inferred signaling hierarchy.

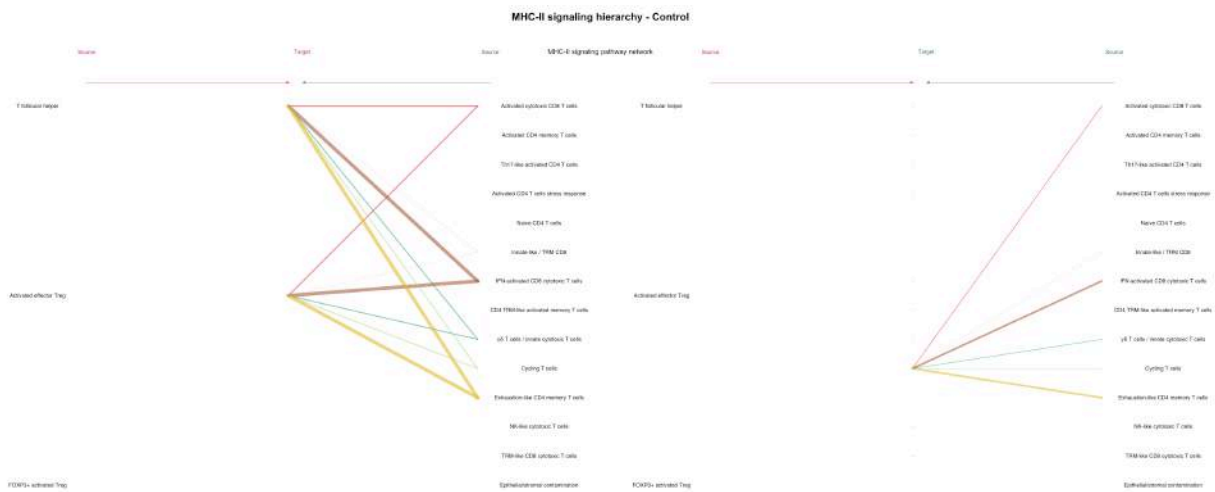
6.2.0.1 Circle plots Control



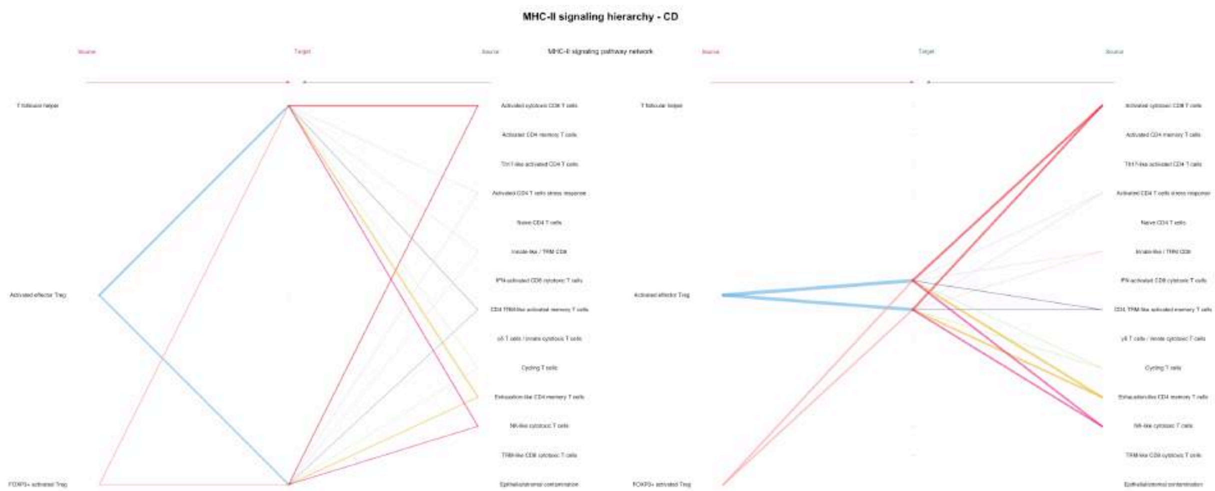
Crohn's disease (CD)



6.2.0.2 Signaling hierarchy Control



Crohn's disease (CD)



6.3 SIRP (SIRPG–CD47) — *network reorganization*

SIRP signaling is detected in both LPL conditions (17 interactions in Control and 24 in CD) but shows one of the strongest changes in pathway-level organization between conditions.

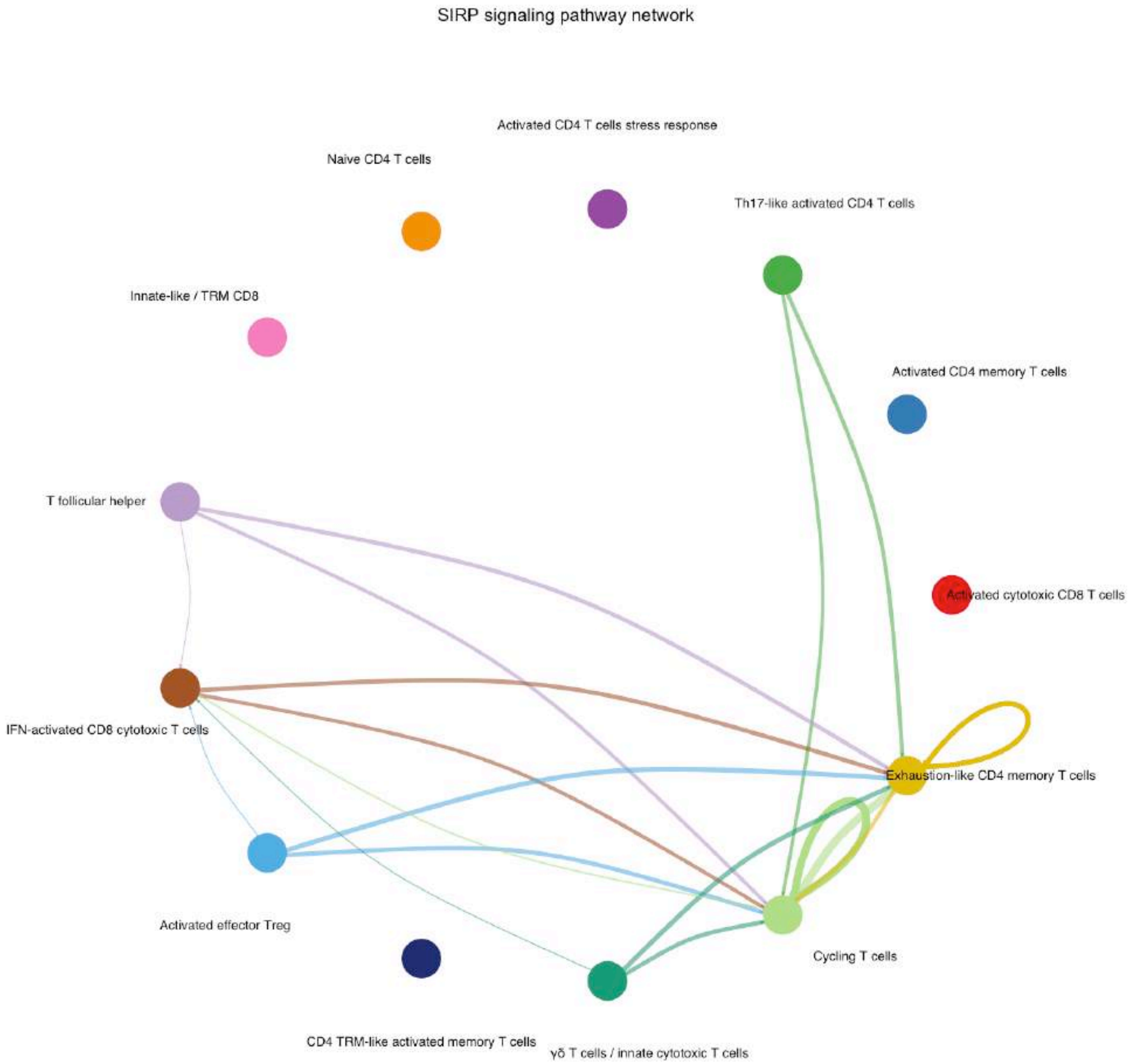
Functional similarity analysis indicates that shared pathways occupy a relatively coherent region of the inferred communication landscape, whereas pathways with condition-specific representation are positioned separately within the embedding space. This should be interpreted as a redistribution of communication-network structure rather than evidence of a newly activated biological pathway.

Among evaluated pathways, SIRP displays the largest rank-based displacement between Control and CD, indicating a marked change in its relative communication pattern across conditions.

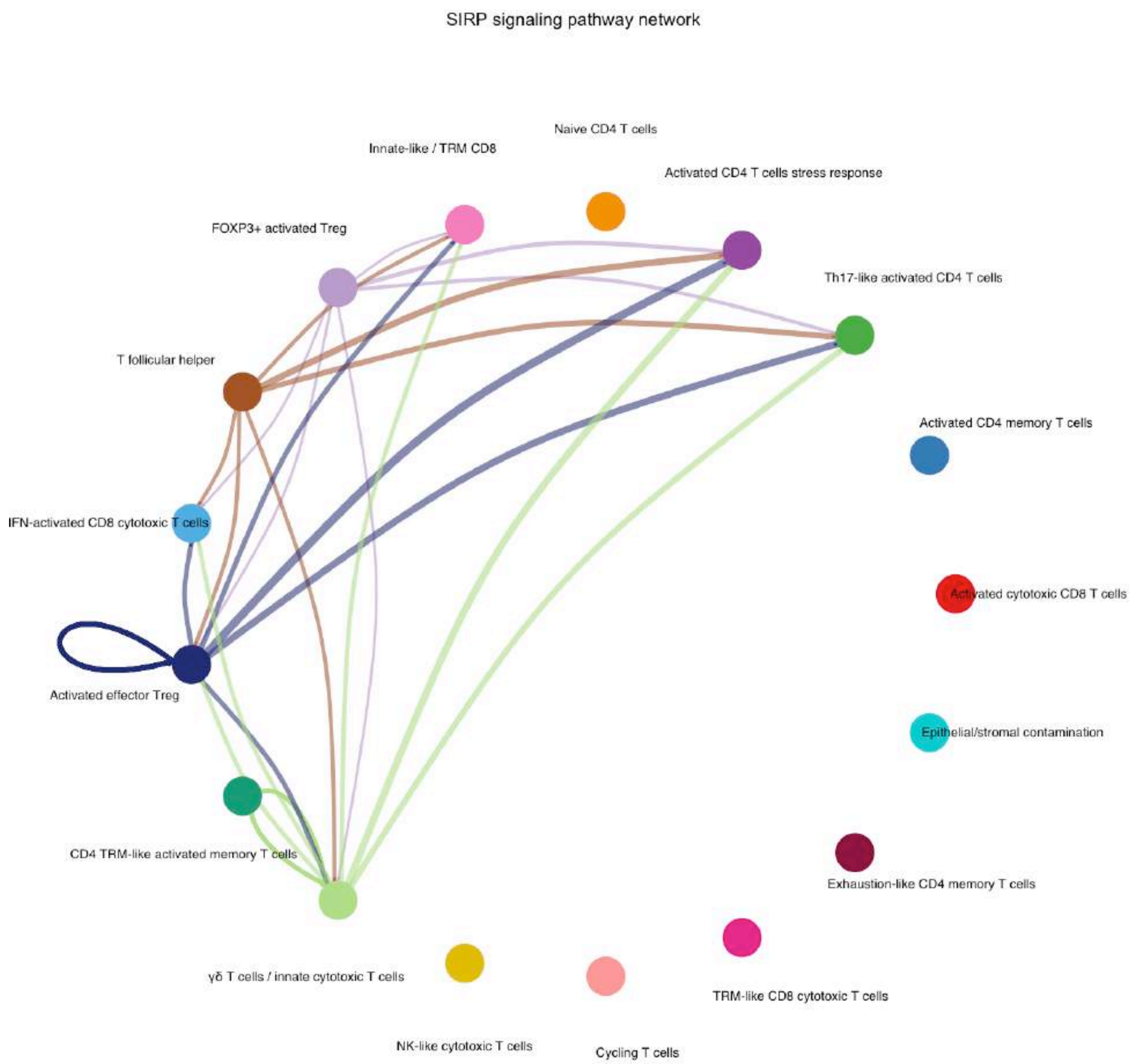
Given the T-cell-dominated composition of the LPL compartment, the observed SIRP pattern is most consistent with altered SIRPG–CD47-associated immune cell communication rather than classical myeloid checkpoint signaling.

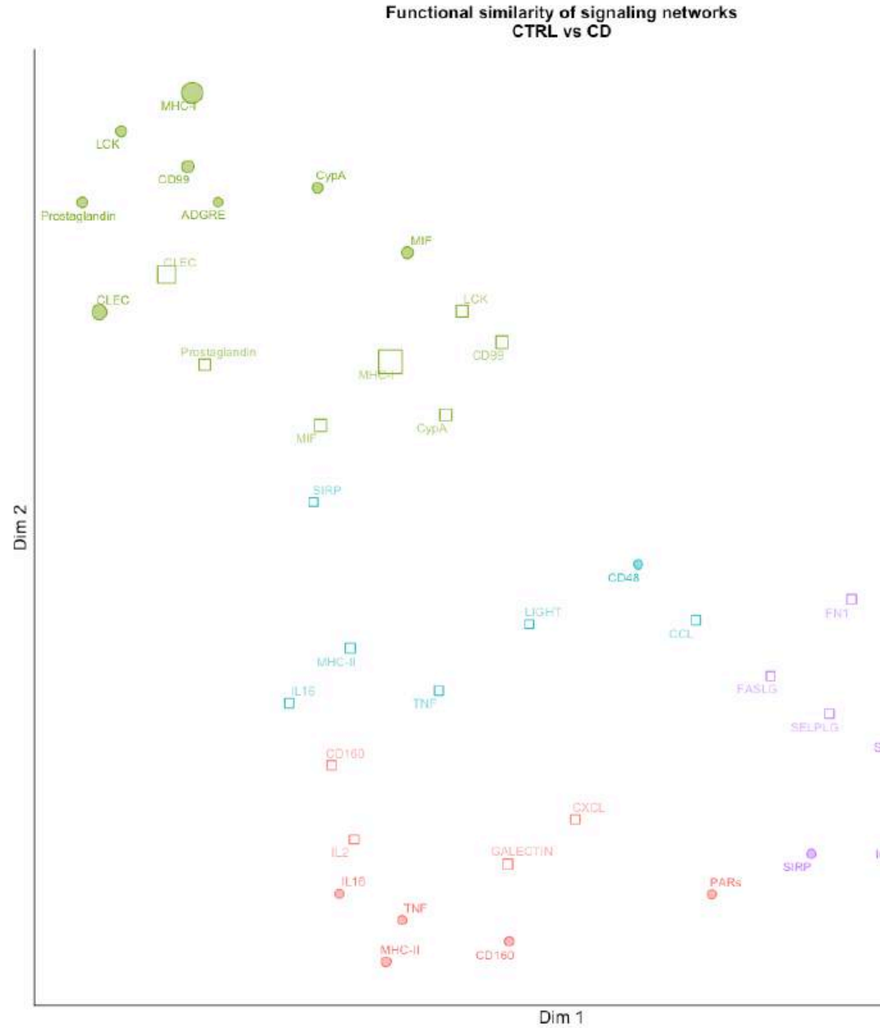
Figure 16. SIRP signaling and pathway-level functional similarity in LPL under Control and Crohn's disease (CD).

6.3.0.1 Circle plots Control

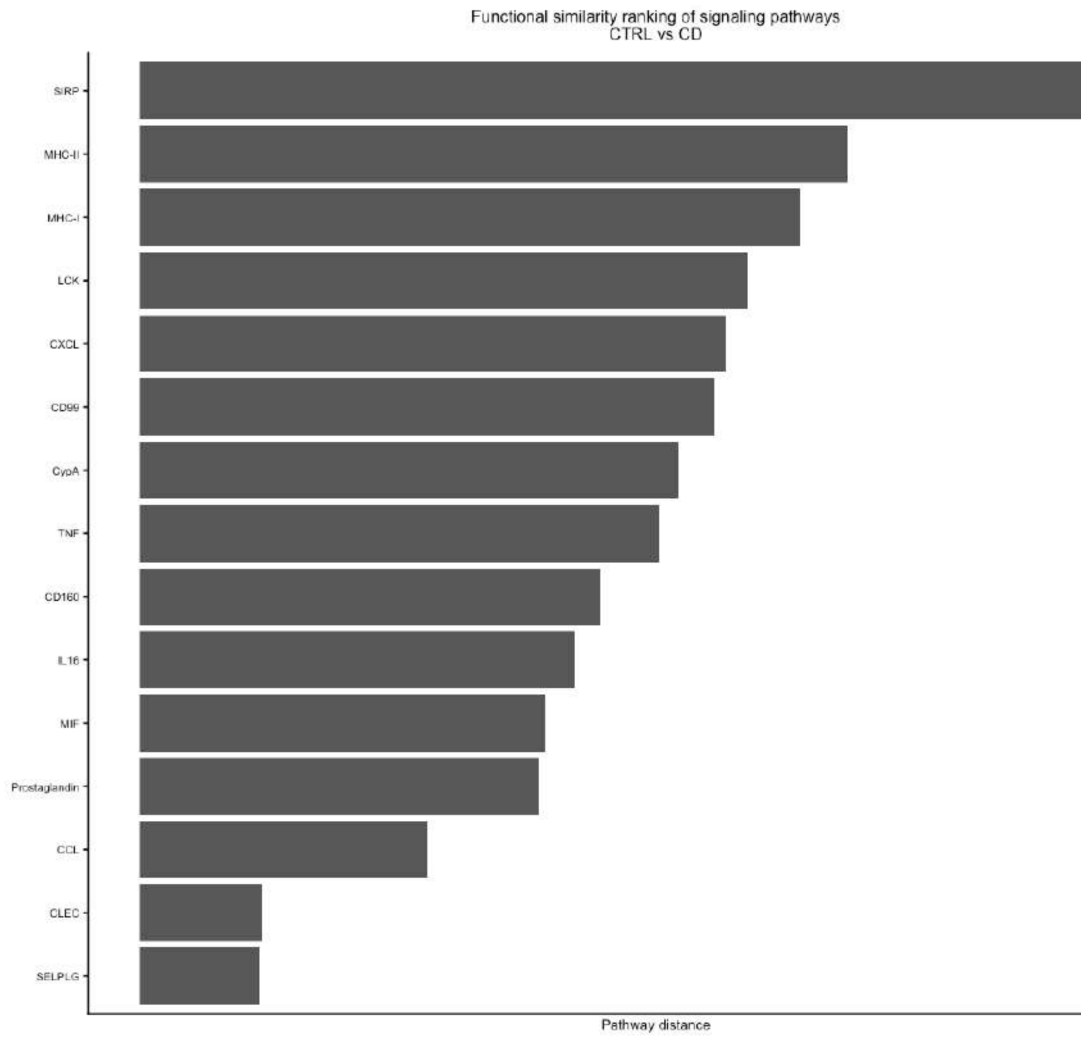


Crohn's disease (CD)


SIRP signaling pathway network
Naive CD4 T cells
Innate-like / TRM CD8
Activated CD4 T cells stress response
FOXP3+ activated Treg
Th17-like activated CD4 T cells
T follicular helper
Activated CD4 memory T cells
IFN-activated CD8 cytotoxic T cells
Activated cytotoxic CD8 T cells
Activated effector Treg
Epithelial/stromal contamination
CD4 TRM-like activated memory T cells
Exhaustion-like CD4 memory T cells
γδ T cells / innate cytotoxic T cells
TRM-like CD8 cytotoxic T cells
NK-like cytotoxic T cells
Cycling T cells



6.3.0.2 Functional similarity embedding



6.3.0.3 Rank similarity

6.4 GALECTIN — *CD-associated, within-CD characterized*

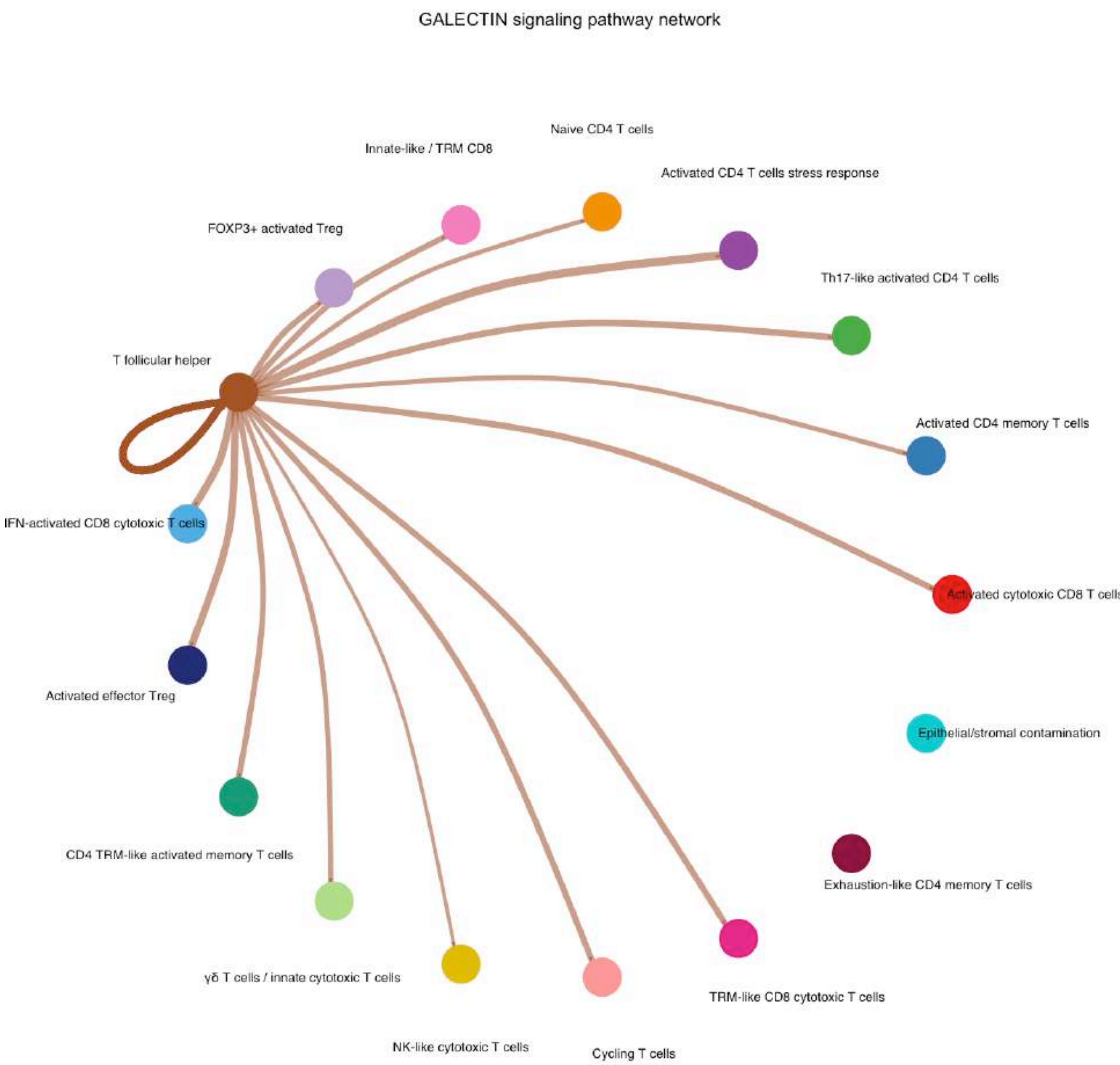
GALECTIN signaling was detected in the CD LPL CellChat network through 38 inferred ligand–receptor interactions, whereas no GALECTIN interactions were detected in Control.

Direct communication inference should be distinguished from network topology: the circle plot highlights T follicular helper cells as a major network hub, including prominent self-signaling, whereas sender assignment should be derived from the underlying ligand–receptor communication table rather than visual centrality alone.

Given the strong condition-dependent cellular composition of LPL and the absence of detected GALECTIN interactions in Control, this result should be interpreted as a CD-associated communication pattern rather than evidence of disease-specific GALECTIN induction.

Figure 17. GALECTIN signaling in LPL under Crohn's disease (CD), showing the inferred CD-exclusive communication pattern and ligand–receptor interactions.

6.4.0.1 Circle plot Crohn's disease (CD)



6.5 IL2 — ***CD-associated, receptor-context dependent***

IL2 signaling was detected only in the CD CellChat network, with 11 inferred ligand–receptor interactions and no detected IL2 interactions in Control. The only detected sender population was Innate-like/TRM CD8 T cells.

Because this population is strongly imbalanced between conditions (2194 CD cells versus 16 Control cells), the result should be interpreted as a CD-associated communication pattern rather than a quantitative increase of IL2 signaling. CellChat supports the presence of an IL2 communication program within the CD cellular landscape but cannot establish disease-specific induction.

The inferred receptor usage was not restricted to a canonical Treg-specific configuration. Predicted IL2 receptor complexes included IL2RA-containing and IL2RB/IL2RG-containing configurations across different receiver populations, suggesting that the inferred IL2 signaling landscape involves multiple cellular contexts rather than a single high-affinity IL2–Treg axis.

Figure 18. IL2 signaling in LPL under Crohn's disease (CD), showing the inferred CD-exclusive communication network.

6.5.0.1 Circle plot Crohn's disease (CD)

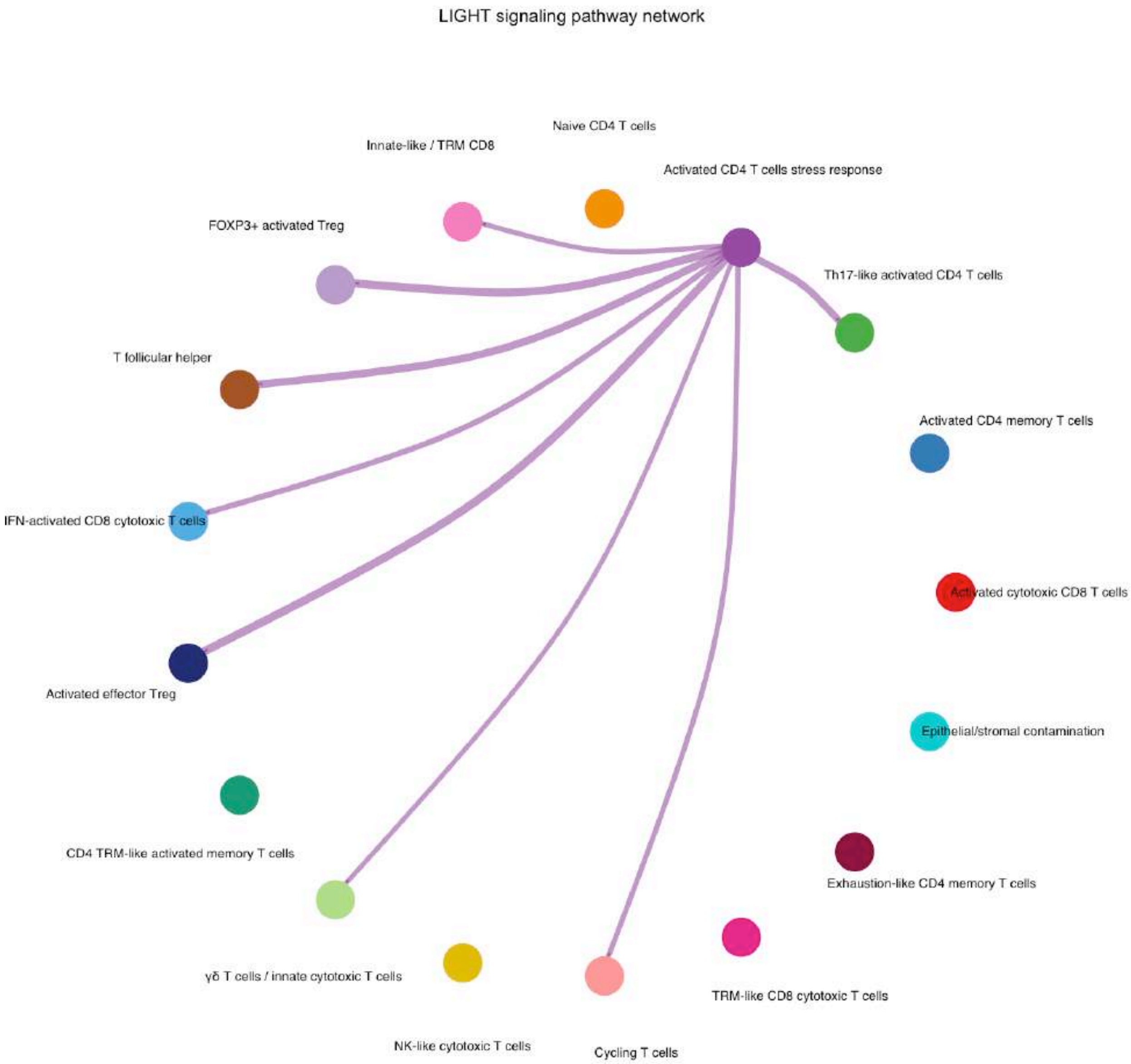
6.6 LIGHT — *fragile, population-limited*

LIGHT signaling was detected only in the CD CellChat network, with the only inferred sender population being Activated CD4 T cells stress response. This population is extremely imbalanced between conditions (15 CD cells versus 2628 Control cells), limiting the interpretation of this interaction.

The only inferred ligand–receptor pair was TNFSF14–TNFRSF14 (HVEM), representing a distinct receptor context from the IEL LIGHT signaling pattern, where TNFSF14–LT R interactions were detected. Given the very small number of CD sender cells and the strong population imbalance, this result should be considered hypothesis-generating rather than evidence of disease-associated LIGHT induction.

Figure 19. LIGHT (TNFSF14) signaling in LPL under Crohn's disease (CD), showing the inferred LIGHT–HVEM communication axis.

6.6.0.1 Circle plot Crohn's disease (CD)



6.7 Communication patterns and cell-type signaling changes

Non-negative matrix factorization (NMF) was used to identify higher-order communication programs in the LPL compartment. Control and CD required different outgoing pattern numbers (Control K=4; CD K=8), whereas incoming communication patterns showed the same decomposition size (K=7 in both conditions).

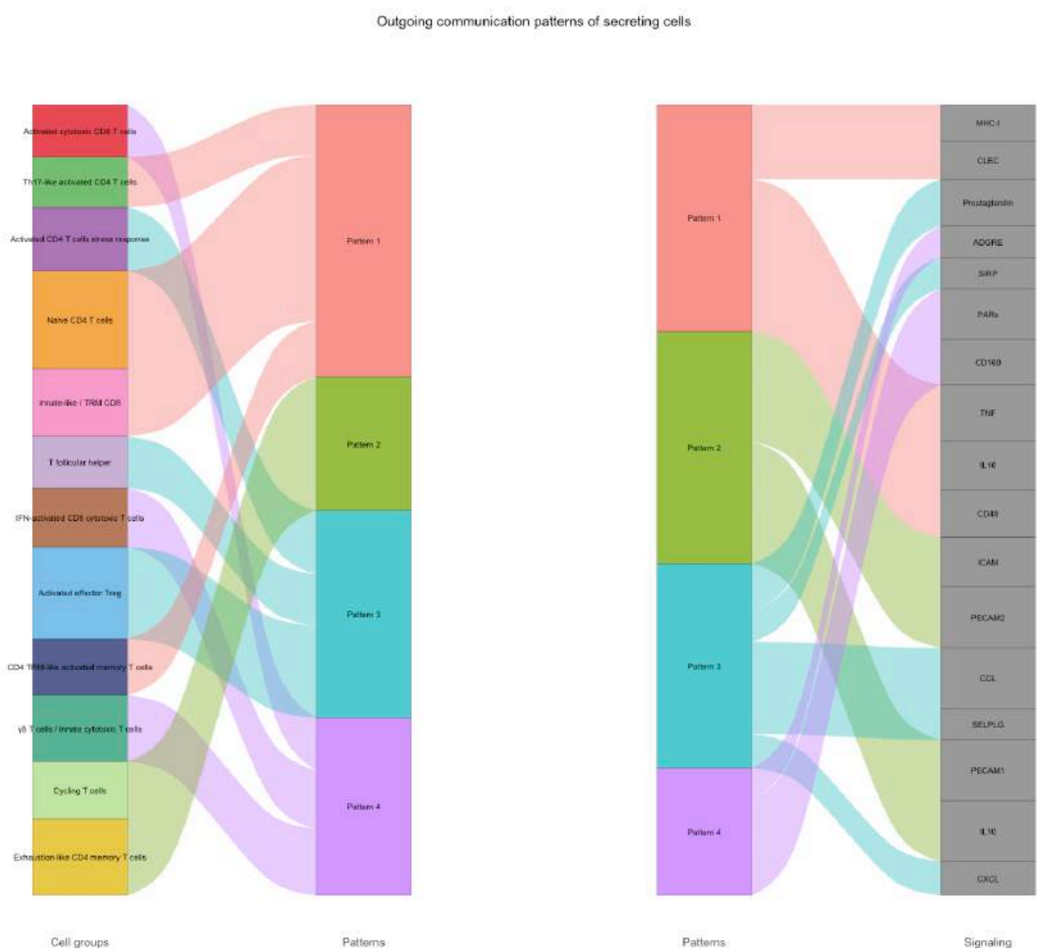
Because the outgoing decomposition was performed with different K values, individual outgoing pattern indices cannot be interpreted as one-to-one equivalents across conditions. The outgoing results should therefore be interpreted as differences in the complexity and organization of inferred signaling programs rather than direct pattern-to-pattern comparisons.

Incoming patterns, where the same K was selected in both conditions, allow a more direct qualitative comparison of the overall organization of receiver-associated signaling programs, while still remaining dependent on the underlying cell-type composition and CellChat inference framework.

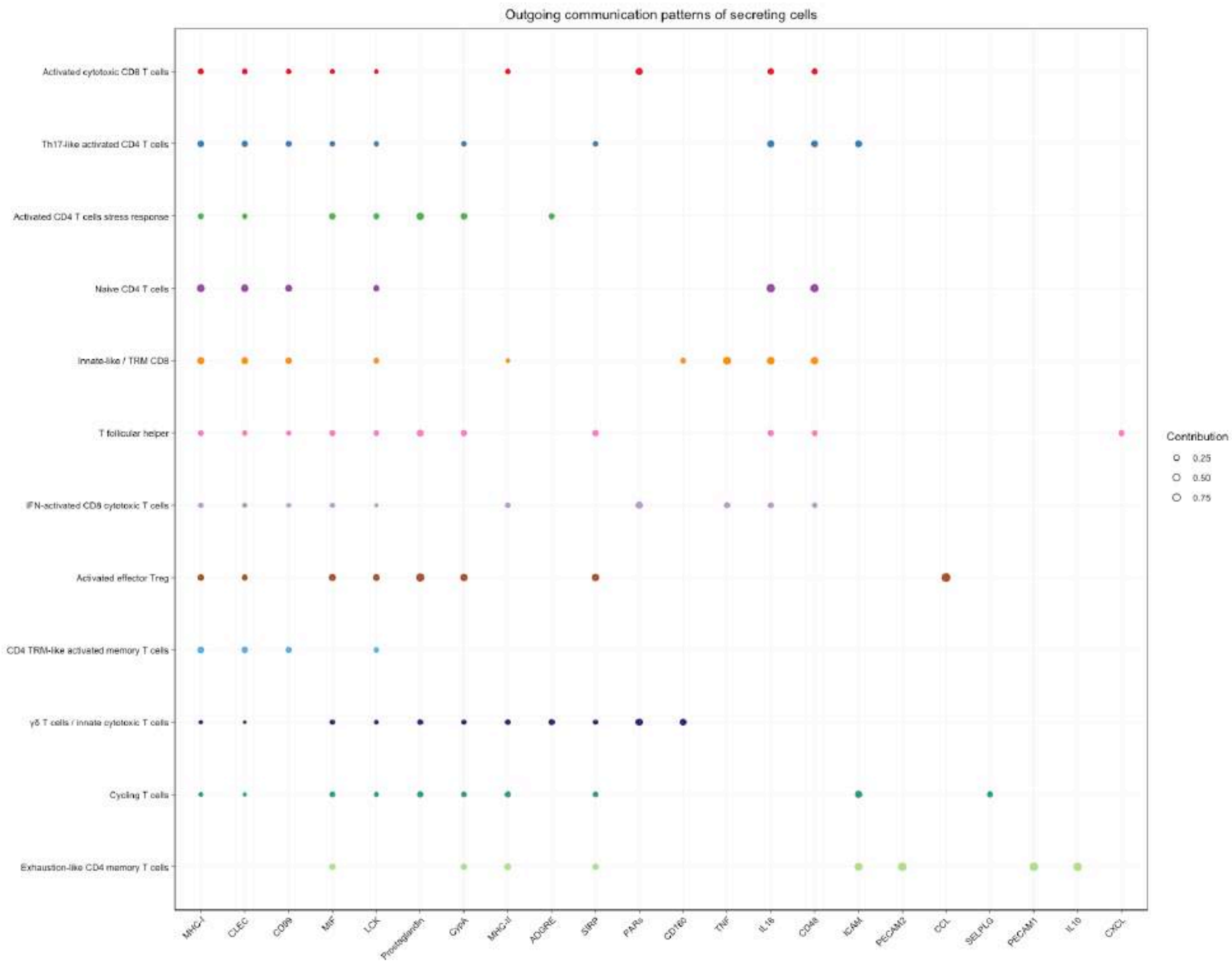
Figure 20. NMF-derived communication patterns in LPL under Control and Crohn's disease (CD). Outgoing communication programs are shown as river plots, while incoming communication programs are summarized using dot plots. Control outgoing and incoming decompositions use K = 4 and K = 7, respectively; CD uses K = 8 and K = 7.

6.7.0.1 Outgoing communication patterns

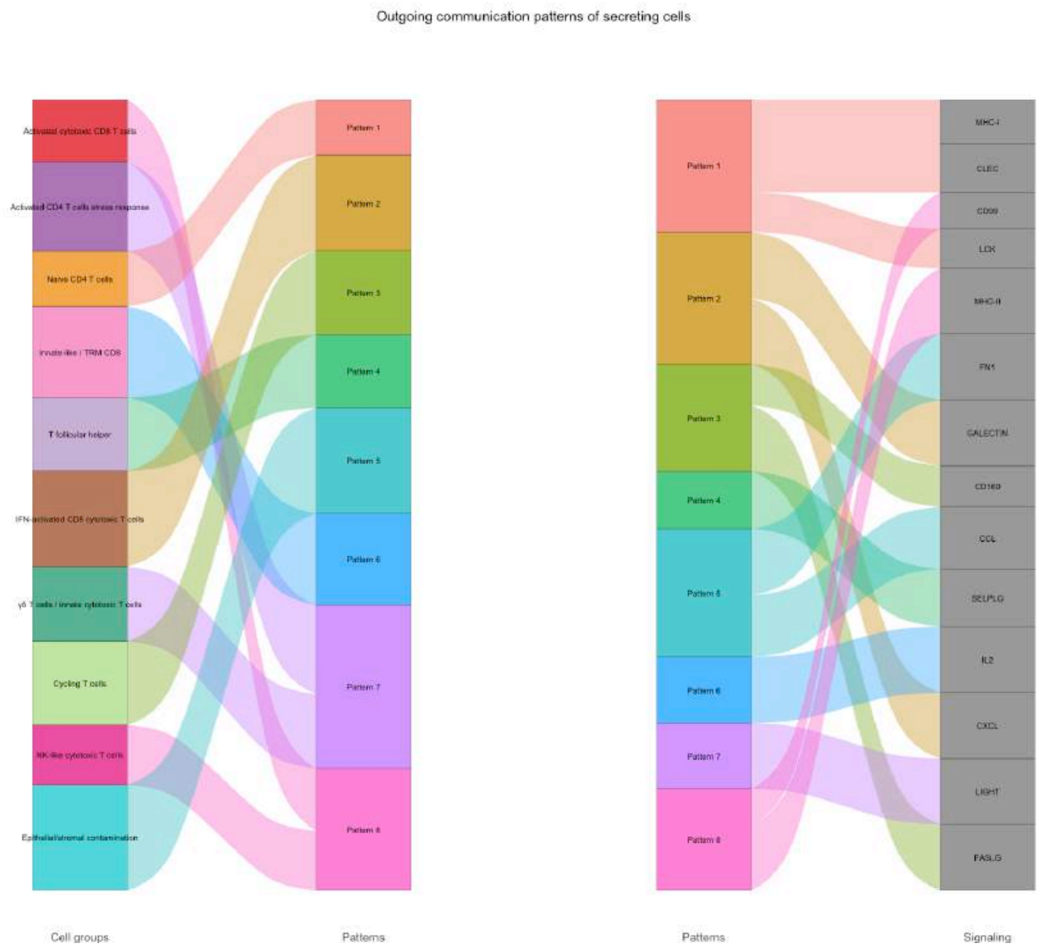
6.7.0.1.1 Control River plot



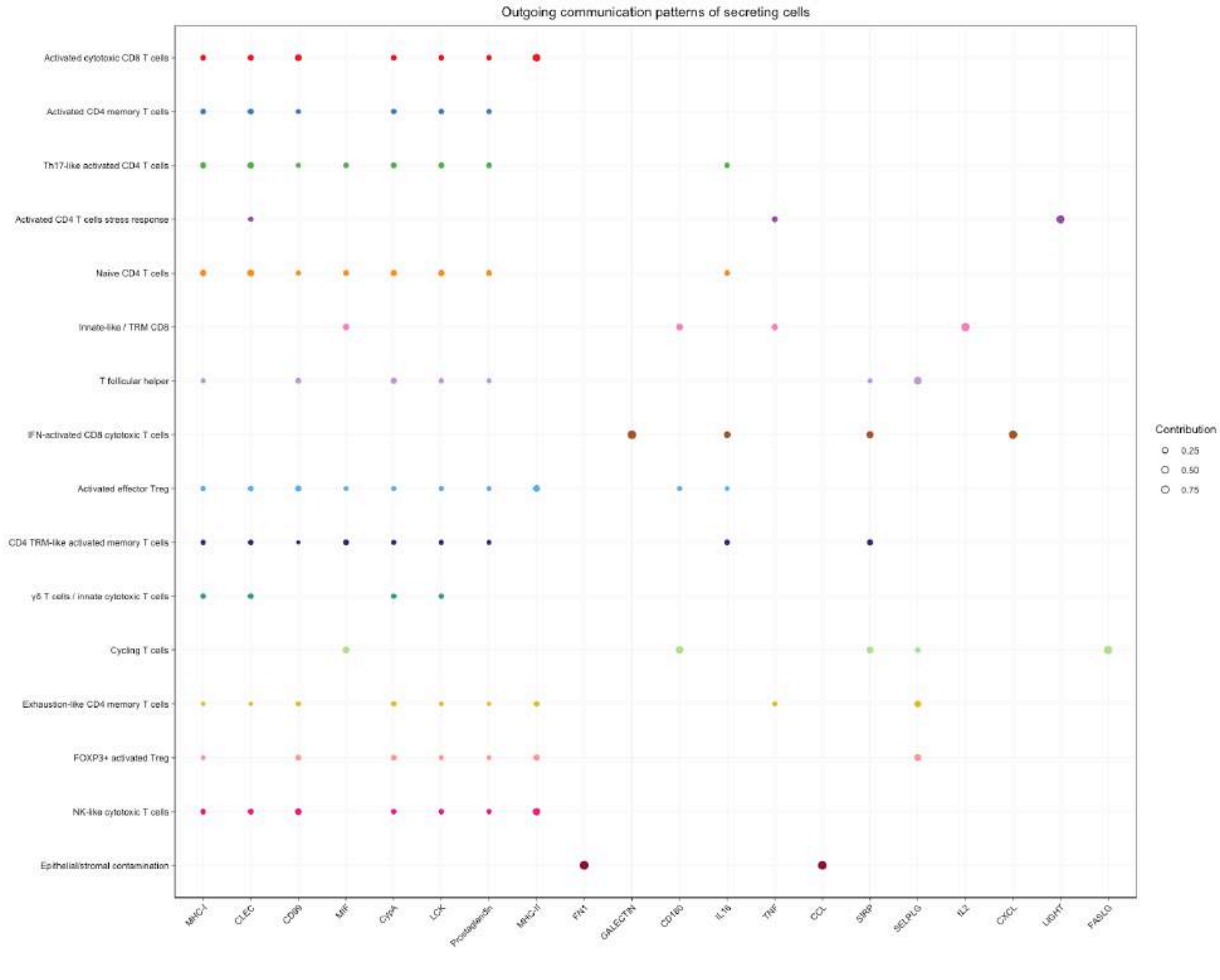
Dot plot



6.7.0.1.2 Crohn's disease (CD) River plot

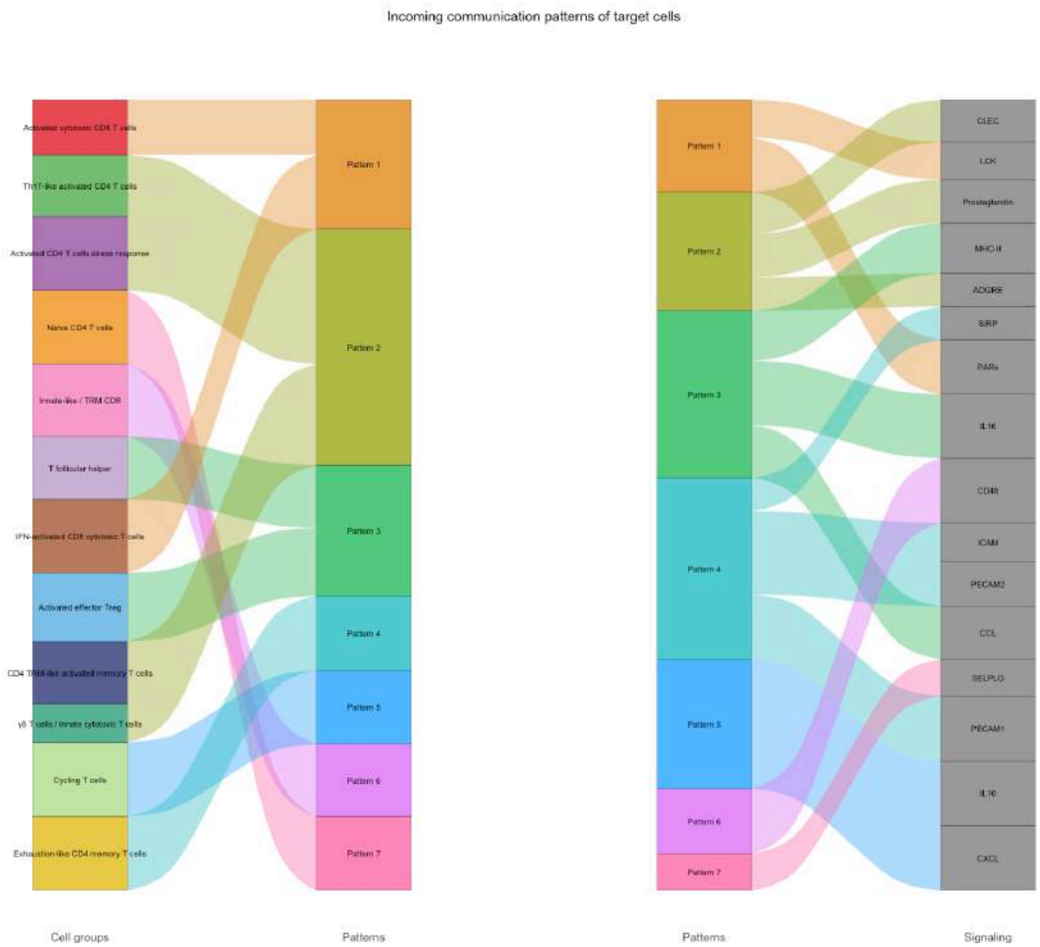

Outgoing communication patterns of secreting cells
Activated cytotoxic CD8 T cells
Activated CD4 T cells stress response
Naive CD4 T cells
Innate-like / TRM CD8
T follicular helper
IFN-activated CD8 cytotoxic T cells
γδ T cells / innate cytotoxic T cells
Cycling T cells
NK-like cytotoxic T cells
Epithelial/stromal contamination
Cell groups
Patterns
MHC-I
CLEC
CD99
LCK
MHC-II
FN1
GALECTIN
CD160
CCL
SELPLG
IL2
CXCL
LIGHT
FASLG
Patterns
Signaling

Dot plot

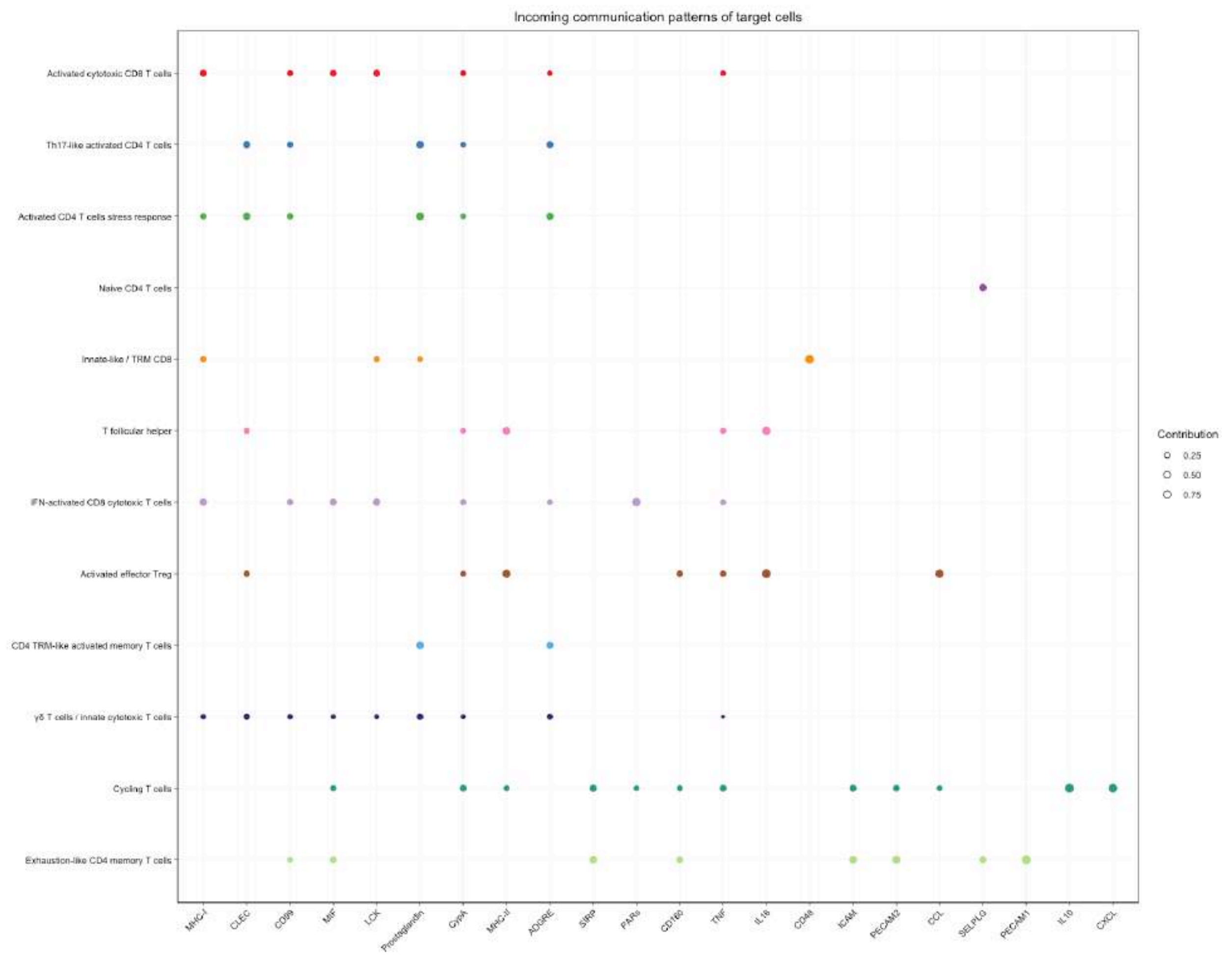


6.7.0.2 Incoming communication patterns

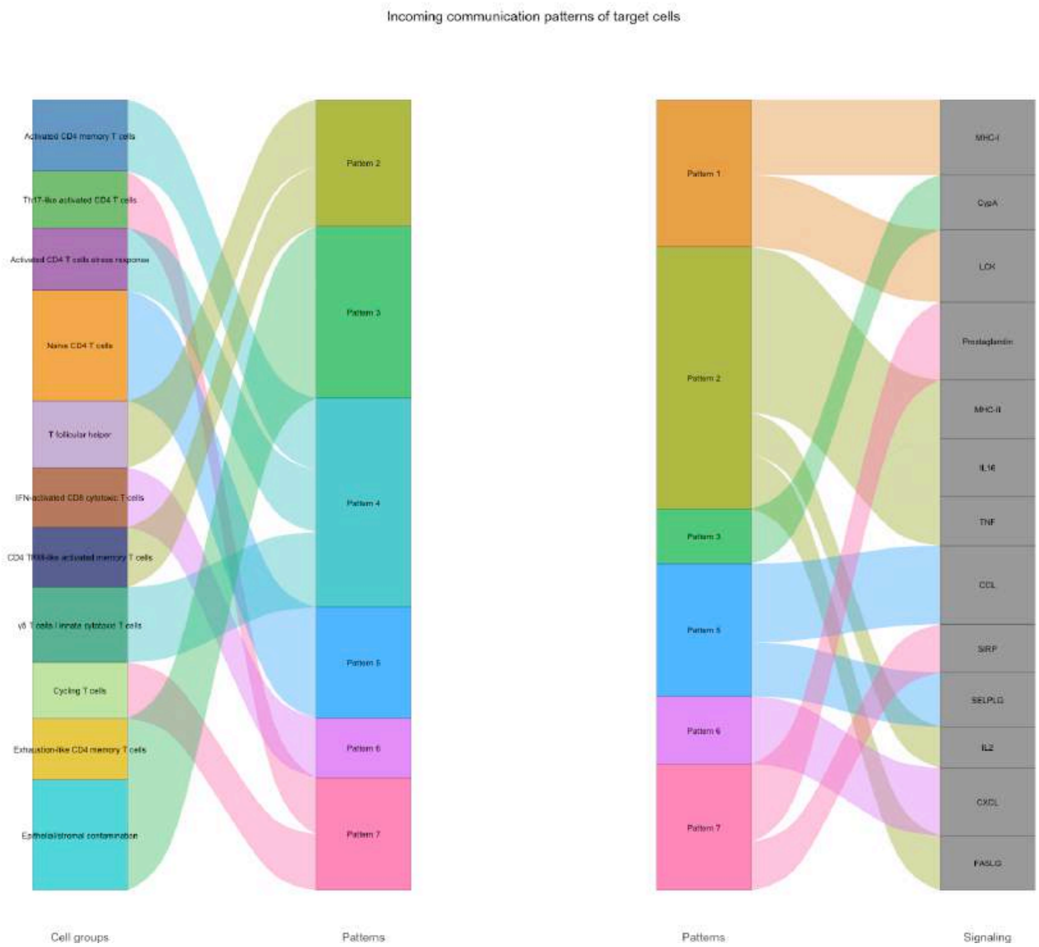
6.7.0.2.1 Control River plot


Incoming communication patterns of target cells
Activated cytotoxic CD8 T cells
Th17-like activated CD4 T cells
Activated CD4 T cells stress response
Naive CD4 T cells
Innate-like / TRM CD8
T follicular helper
IFN-activated CD8 cytotoxic T cells
Activated effector Treg
CD4 TRM-like activated memory T cells
γδ T cells / Innate cytotoxic T cells
Cycling T cells
Exhaustion-like CD4 memory T cells
Pattern 1
Pattern 2
Pattern 3
Pattern 4
Pattern 5
Pattern 6
Pattern 7
Cell groups
Patterns
CLEC
LCK
Prostaglandin
MHC-II
ADGRE
SIRP
PARs
IL16
CD48
ICAM
PECAM2
CCL
SELPLG
PECAM1
IL10
CXCL
Signaling

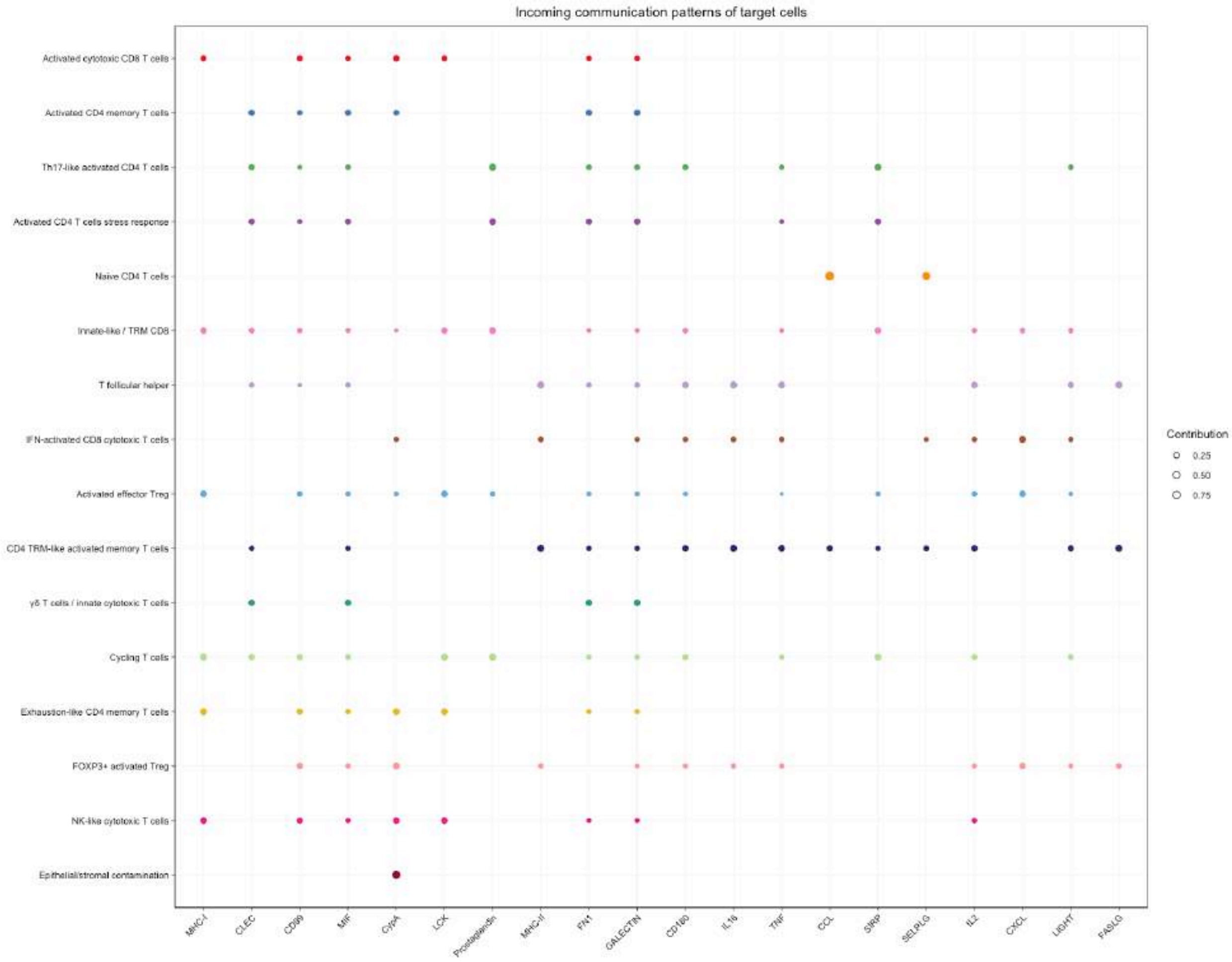
Dot plot



6.7.0.2.2 Crohn's disease (CD) River plot


Incoming communication patterns of target cells
Activated CD4 memory T cells
Th17-like activated CD4 T cells
Activated CD4 T cells stress response
Naive CD4 T cells
T follicular helper
IFN-activated CD8 cytotoxic T cells
CD4 TRM-like activated memory T cells
γδ T cells / innate cytotoxic T cells
Cycling T cells
Exhaustion-like CD4 memory T cells
Epithelial/stromal contamination
Pattern 1
Pattern 2
Pattern 3
Pattern 4
Pattern 5
Pattern 6
Pattern 7
MHC-I
CypA
LCK
Prostaglandin
MHC-II
IL16
TNF
CCL
SIRP
SELPLG
IL2
CXCL
FASLG
Cell groups
Patterns
Signaling

Dot plot



7 Cross-Validation with GSEA

The GSEA finding of **IL2-STAT5 signaling up in CD within LPL-Treg** finds a directional counterpart in CellChat because IL2 is a pathway gained in CD. The convergence is **partial, not clean**: receptor usage is not specifically high-affinity/Treg-selective and the sender is Innate-like/TRM CD8 rather than Treg.

This should therefore be presented as **directional convergence between two independent methods**, not as precise mechanistic validation of the same molecular circuit.

8 Known Technical Issues (CellChat v2) — Methods Note

The following behaviors were diagnosed empirically during the LPL analysis:

1. **`netVisual_bubble()` errors** when a pathway is absent from the second dataset in `comparison`. The comparison index must therefore reflect the dataset in which the pathway is actually present.
2. **`netVisual_bubble()` can silently return an empty plot** when the true sender is excluded by `sources.use` / `targets.use`. This affected LIGHT, IL2, and GALECTIN in LPL; unrestricted plots were therefore regenerated.

3. **Hierarchy plots** require at least two receiver elements. SIRP, IL2, and GALECTIN were excluded from hierarchy visualization where a second receiver could not be assigned without biological arbitrariness. MHC-I and MHC-II retained hierarchy plots.

9 Comparison with External Literature

Finding	Independent literature	Concordance
MHC-II / CD40-axis signaling toward Treg and TH17-like populations	CD40 signaling and intestinal Treg/TH17 biology; Barthels et al. 2017; Wang et al. 2026	Mechanistically consistent
LIGHT–LT R toward Treg/macrophages, IEL	LIGHT association with IBD and LT R biology	Consistent, with dual-role nuance
CDH1(E-cadherin)–CD103, IEL	Established retention mechanism for IEL and CD103 Tregs	Direct mechanistic support
SIRPG–CD47 reorganization, LPL	SIRPG/CD47 T-cell costimulation literature	Mechanistically plausible
IEL T-cell compositional reduction in CD	Jaeger et al. 2021; Song et al. 2024	Source-paper concordance + independent replication
LP CD8 /cytotoxic expansion in CD	Jaeger et al. 2021	Direct source-paper concordance

Correction note

An earlier internal draft cited a separate “Chen et al. (2021, Nat. Commun., PMID 33771991)” as an independent replication cohort. Direct verification showed that PMID 33771991 corresponds to Jaeger et al. (2021) itself. Therefore, those findings are corroborated by the source publication, not by an independent cohort. The genuinely independent replication identified for the -IEL reduction is Song et al. (2024).

10 Cross-Compartment Synthesis

1. **Network complexity is elevated in CD in both compartments, but through different mechanisms.** IEL shows a diffuse expansion (more pathways, lower per-edge strength, partly threshold-driven); LPL shows a proportional expansion (more interactions and more strength together).
2. **MHC-II is the most consistent disease-associated hub across both compartments**, but neither compartment shows a simple intensification of a stable circuit. The dominant receiving population changes.
3. **Regulatory T-cell biology is a recurring node of interest** across MHC-II, GALECTIN, CDH1 in IEL and MHC-I, MHC-II, IL2 in LPL, with centrality, signaling-role and NMF outputs converging on increased involvement of regulatory T-cell-associated populations. picture.
4. **LPL lacks the myeloid/epithelial signaling axes present in IEL**, including the MIF-macrophage and CDH1-epithelial analogues.
5. **TNF/IFNG/IL17/TGFb are largely undetected in CellChat in both compartments** (TNF is detected in LPL only), despite prominence in GSEA/DESeq2. This is most plausibly a detection-threshold issue and must not be interpreted as biological absence.

11 Limitations

1. **Cell-type imbalance** is the dominant methodological constraint and governs the validity tier of every claim.
2. **IEL-Cytotoxic-TRM-like / LPL macropopulation heterogeneity:** the IEL cytotoxic TRM-like population aggregates CD8 and lineages that could not be statistically separated.
3. **Detection-threshold artifacts:** absence of IFN-II/IL17/TGFb (and of TNF in IEL) should not be interpreted as biological absence.
4. **Asymmetry in `celltypes_of_interest`:** the IEL deep-dive set is weighted toward CD-emergent populations, whereas LPL explicitly balances comparable and CD-relevant populations.
5. **LIGHT (LPL)** rests on an extremely small sender population (n=15) and should be treated as hypothesis-generating.
6. **No independent re-derivation in this report:** quantitative values are taken from previously reviewed pipeline outputs.
7. **LPL session continuity:** scripts 17/18 were not re-verified line-by-line in the session that produced the LPL findings log; their description follows the established pipeline architecture. Script 20 was fully verified.

12 Reproducibility

12.1 Environment

Package	Version
R	4.6.0
Seurat	5.5.0
CellChat	v2 (Jin et al., 2024)

12.2 Reference scripts

- `13_CellChat_IEL_CTRL.r`
- `14_CellChat_IEL_CD.r`
- `15_CellChat_IEL_Merge.r`
- `16_CellChat_IEL_Comparison.r`
- `17_CellChat_LPL_CTRL.r`
- `18_CellChat_LPL_CD.r`
- `19_CellChat_LPL_Merge.r`
- `20_CellChat_LPL_Comparison.r`

12.3 Reproducibility settings

Random seed: `1234`, called immediately before each stochastic block (functional similarity embedding, `selectK`, `identifyCommunicationPatterns`).

`netEmbedding(..., umap.method = "uwot")` was used as the R-native UMAP backend, avoiding the Python/reticulate dependency.

13 Deliverables

- Per-condition, per-compartment CellChat objects and lifted/merged comparison objects.

- Global network summary plots.
- RankNet information-flow plots.
- Pathway-specific bubble, circle, gene-expression, chord and hierarchy plots.
- Signaling-role scatter plots.
- Communication-pattern recognition outputs.
- Functional similarity embeddings and `rankSimilarity`.
- This consolidated Quarto report.

13.1 Recommended manuscript figures

The strongest candidates identified in the original report are:

Differential network, both compartments.
RankNet, both compartments.
MHC-II population substitution, IEL and LPL.
CDH1–CD103 axis, IEL.
MHC-I → Activated effector Treg, LPL.
Functional similarity embedding + `rankSimilarity`, LPL.

14 Conclusion

Cell–cell communication analysis of the IEL and LPL compartments provides a network-level complement to pseudobulk/GSEA findings, converging on a small set of well-supported, literature-concordant observations: a denser and structurally more complex communication network in CD in both compartments; a recurring, mechanistically nuanced role for MHC-II and Treg-centered signaling; compartment-specific axes (LIGHT–LT R toward Treg in IEL; SIRPG–CD47 reorganization in LPL); and a transcriptomic support for a CDH1–CD103 interaction consistent with known IEL retention biology.

The explicit, prospectively applied cell-type validity framework is an integral part of the analytical design and should be retained in any future CellChat extension or cross-validation with LIANA.

15 Appendix A: Glossary of Abbreviations

Term	Definition
CD	Crohn's disease
IEL	Intraepithelial lymphocytes
LPL	Lamina propria lymphocytes
TRM	Tissue-resident memory (T cell)
APC	Antigen-presenting cell
NMF	Non-negative matrix factorization
L–R	Ligand–receptor
MHC-I / MHC-II	Major histocompatibility complex, class I / II
GSEA	Gene Set Enrichment Analysis

16 Appendix B: References

1. Jaeger N., et al. (2021). *Single-cell analyses of Crohn's disease tissues reveal intestinal intraepithelial T cells heterogeneity and altered subset distributions.* Nature Communications, 12, 1921.
2. Jin S., Plikus M.V., Nie Q. (2024). CellChat for systematic analysis of cell-cell communication from single-cell transcriptomics. *Nature Protocols*, 20, 180–219. https://doi.org/10.1038/s41596-024-01045-4 communication from single-cell transcriptomics*. bioRxiv 2023.11.05.565674.
3. Barthels C., et al. (2017). *CD40-signalling abrogates induction of ROR t Treg cells by intestinal CD103 DCs and causes fatal colitis.* Nature Communications, 8, 14715.
4. Wang M., et al. (2026). *Fusobacterium nucleatum* drives CD40-mediated dendritic cell activation and Th17/Treg imbalance to exacerbate intestinal inflammation in Crohn's disease. Frontiers in Immunology.
5. Fan et al. (2021). Elevated levels of the cytokine LIGHT in Crohn's disease. medRxiv.
6. Herro R., et al. (2018) and related works on TNFSF14/LT R in intestinal inflammation resolution.
7. Piccio L., et al. (2005). Adhesion of human T cells to antigen-presenting cells through SIRP 2–CD47 interaction costimulates T-cell proliferation. Blood, 105(6), 2421–2427.
8. Yang et al. (2025). Integrin CD103 expression in naive CD8 T cells promotes cytokine-driven acquisition of memory phenotype and effector function. Immunity.
9. Roberts et al. (2021). Integrin E(CD103) 7 in epithelial cancer.
10. Song et al. (2024). Hyperactivation and enhanced cytotoxicity of reduced CD8 T cells in the intestine of patients with Crohn's disease correlates with disease activity. BMC Immunology.
11. Kong et al. (2024). Dysregulation of intraepithelial lymphocytes precedes Crohn's disease-like ileitis. Science Immunology.

17 Appendix C: Figure Index

#	Compartment	Main output	Section
1	Both	Global interaction count/strength	4
2	Both	Differential network / heatmap	4
3	IEL	RankNet	4
4	LPL	Functional similarity + rankSimilarity	4, 6.3
5	LPL	RankNet	4
6	IEL	MHC-I circle + hierarchy	5.1
7	IEL	MHC-II population substitution	5.2
8	IEL	LIGHT–LT R axis	5.3
9	IEL	MIF	5.4
10	IEL	GALECTIN	5.5
11	IEL	CDH1–CD103	5.6
12	IEL	Signaling-role scatter	5.7
13	IEL	NMF communication patterns	5.7
14	LPL	MHC-I / Activated effector Treg	6.1
15	LPL	MHC-II population substitution	6.2
16	LPL	SIRP reorganization	6.3
17	LPL	GALECTIN	6.4
18	LPL	IL2 receptor usage	6.5
19	LPL	LIGHT, n=15 sender cells	6.6
20	LPL	NMF communication patterns	6.7